# EXPLORATORY DATA ANALYSIS


## 23BDS0280

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## COURSE PROJECT

# Exploratory Data Analysis Lab - Telco Customer Churn

Dataset:
https://raw.githubusercontent.com/salemprakash/EDA/main/Data/telco%20customer%20churn.csv

```
%matplotlib inline
```

## Setup - Imports & Dataset URL

```
#Setup - Imports & Dataset URL
import warnings
warnings.filterwarnings("ignore")
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
from scipy.spatial.distance import pdist
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from pandas.plotting import parallel_coordinates
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.arima.model import ARIMA

URL = "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/telco%20customer%20churn.csv"
REG_NO = "23BDS0280"
pd.set_option("display.max_columns", None)
pd.set_option("display.width", 110)


def load_raw():
    return pd.read_csv(URL, lineterminator="\r", na_values=["?", " ", ""])


def load_clean():
    d = load_raw()
    for c in d.select_dtypes(include="number").columns:
        d[c] = d[c].fillna(d[c].median())
    for c in d.select_dtypes(exclude="number").columns:
        d[c] = d[c].fillna(d[c].mode()[0])
    return d.drop_duplicates().reset_index(drop=True)


def r_scale(x):
    return (x - x.mean()) / x.std(ddof=1)
```

# PDF 2 - Data Import, Transform and Basic Statistical Analysis

## Import Dataset - PDF #2 - Question #1

```
#Import Dataset - PDF #2 - Question #1
df = load_raw()
df_raw = df.copy()
```

## First 10 Records - PDF #2 - Question #2

```
#First 10 Records - PDF #2 - Question #2
print(df.head(10))
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
0  7590-VHVEG  Female            0.0     Yes         No     1.0           No  No phone service
1  5575-GNVDE    Male            0.0      No         No    34.0          Yes                No
2  3668-QPYBK    Male            0.0      No         No     2.0          Yes                No
3  7795-CFOCW    Male            0.0      No         No    45.0           No  No phone service
4  9237-HQITU  Female            0.0      No         No     2.0          Yes                No
5  9305-CDSKC  Female            0.0      No         No     8.0          Yes               Yes
6  1452-KIOVK    Male            0.0      No        Yes    22.0          Yes               Yes
7  6713-OKOMC  Female            0.0      No         No    10.0           No  No phone service
8  7892-POOKP  Female            0.0     Yes         No    28.0          Yes               Yes
9  6388-TABGU    Male            0.0      No        Yes    62.0          Yes                No

  InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
0             DSL             No          Yes               No          No          No              No
1             DSL            Yes           No              Yes          No          No              No
2             DSL            Yes          Yes               No          No          No              No
3             DSL            Yes           No              Yes         Yes          No              No
4     Fiber optic             No           No               No          No          No              No
5     Fiber optic             No           No              Yes          No         Yes             Yes
6     Fiber optic             No          Yes               No          No         Yes              No
7             DSL            Yes           No               No          No          No              No
8     Fiber optic             No           No              Yes         Yes         Yes             Yes
9             DSL            Yes          Yes               No          No          No              No

        Contract PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges Churn
0  Month-to-month              Yes           Electronic check           29.85         29.85    No
1        One year               No               Mailed check           56.95       1889.50    No
2  Month-to-month              Yes               Mailed check           53.85        108.15   Yes
3        One year               No  Bank transfer (automatic)           42.30       1840.75    No
4  Month-to-month              Yes           Electronic check           70.70        151.65   Yes
5  Month-to-month              Yes           Electronic check           99.65        820.50   Yes
6  Month-to-month              Yes    Credit card (automatic)           89.10       1949.40    No
7  Month-to-month               No               Mailed check           29.75        301.90    No
8  Month-to-month              Yes           Electronic check          104.80       3046.05   Yes
9        One year               No  Bank transfer (automatic)           56.15       3487.95    No
```

## Last 10 Records - PDF #2 - Question #3

```
#Last 10 Records - PDF #2 - Question #3
print(df.tail(10))
```

```
      customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
7033  9767-FFLEM    Male            0.0      No         No    38.0          Yes                No
7034  0639-TSIQW  Female            0.0      No         No    67.0          Yes               Yes
7035  8456-QDAVC    Male            0.0      No         No    19.0          Yes                No
7036  7750-EYXWZ  Female            0.0      No         No    12.0           No  No phone service
7037  2569-WGERO  Female            0.0      No         No    72.0          Yes                No
7038  6840-RESVB    Male            0.0     Yes        Yes    24.0          Yes               Yes
7039  2234-XADUH  Female            0.0     Yes        Yes    72.0          Yes               Yes
7040  4801-JZAZL  Female            0.0     Yes        Yes    11.0           No  No phone service
7041  8361-LTMKD    Male            1.0     Yes         No     4.0          Yes               Yes
7042  3186-AJIEK    Male            0.0      No         No    66.0          Yes                No

     InternetService       OnlineSecurity         OnlineBackup     DeviceProtection          TechSupport  \
7033     Fiber optic                   No                   No                   No                   No
7034     Fiber optic                  Yes                  Yes                  Yes                   No
7035     Fiber optic                   No                   No                   No                   No
7036             DSL                   No                  Yes                  Yes                  Yes
7037              No  No internet service  No internet service  No internet service  No internet service
7038             DSL                  Yes                   No                  Yes                  Yes
7039     Fiber optic                   No                  Yes                  Yes                   No
7040             DSL                  Yes                   No                   No                   No
7041     Fiber optic                   No                   No                   No                   No
7042     Fiber optic                  Yes                   No                  Yes                  Yes

              StreamingTV      StreamingMovies        Contract PaperlessBilling              PaymentMethod  \
7033                   No                   No  Month-to-month              Yes    Credit card (automatic)
7034                  Yes                   No  Month-to-month              Yes    Credit card (automatic)
7035                  Yes                   No  Month-to-month              Yes  Bank transfer (automatic)
7036                  Yes                  Yes        One year               No           Electronic check
7037  No internet service  No internet service        Two year              Yes  Bank transfer (automatic)
7038                  Yes                  Yes        One year              Yes               Mailed check
7039                  Yes                  Yes        One year              Yes    Credit card (automatic)
7040                   No                   No  Month-to-month              Yes           Electronic check
7041                   No                   No  Month-to-month              Yes               Mailed check
7042                  Yes                  Yes        Two year              Yes  Bank transfer (automatic)

      MonthlyCharges  TotalCharges Churn
7033           69.50       2625.25    No
7034          102.95       6886.25   Yes
7035           78.70       1495.10    No
7036           60.65        743.30    No
7037           21.15       1419.40    No
7038           84.80       1990.50    No
7039          103.20       7362.90    No
7040           29.60        346.45    No
7041           74.40        306.60   Yes
7042          105.65       6844.50    No
```

## Rows and Columns - PDF #2 - Question #4

```
#Rows and Columns - PDF #2 - Question #4
print("Rows:", df.shape[0], "Columns:", df.shape[1])
```

```
Rows: 7043 Columns: 21
```

## Variable Names - PDF #2 - Question #5

```
#Variable Names - PDF #2 - Question #5
print(df.columns.tolist())
```

```
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'I
nternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingM
ovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']
```

## Data Type / Structure - PDF #2 - Question #6

```
#Data Type / Structure - PDF #2 - Question #6
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7035 non-null   float64
 3   Partner           7038 non-null   str
 4   Dependents        7041 non-null   str
 5   tenure            7041 non-null   float64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7042 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7042 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7040 non-null   float64
 19  TotalCharges      7032 non-null   float64
 20  Churn             7040 non-null   str
dtypes: float64(4), str(17)
memory usage: 1.1 MB
```

## Summary Statistics - PDF #2 - Question #7

```
#Summary Statistics - PDF #2 - Question #7
print(df.describe(include="all"))
```

```
        customerID gender  SeniorCitizen Partner Dependents       tenure PhoneService MultipleLines  \
count         7043   7043    7035.000000    7038       7041  7041.000000          7043          7043
unique        7043      2            NaN       2          2          NaN             2             3
top     7590-VHVEG   Male            NaN      No         No          NaN           Yes            No
freq             1   3555            NaN    3640       4931          NaN          6361          3390
mean           NaN    NaN       0.162331     NaN        NaN    32.373242           NaN           NaN
std            NaN    NaN       0.368781     NaN        NaN    24.562077           NaN           NaN
min            NaN    NaN       0.000000     NaN        NaN     0.000000           NaN           NaN
25%            NaN    NaN       0.000000     NaN        NaN     9.000000           NaN           NaN
50%            NaN    NaN       0.000000     NaN        NaN    29.000000           NaN           NaN
75%            NaN    NaN       0.000000     NaN        NaN    55.000000           NaN           NaN
max            NaN    NaN       1.000000     NaN        NaN    72.000000           NaN           NaN

       InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
count             7043           7043         7043             7043        7042        7043            7043
unique               3              3            3                3           3           3               3
top        Fiber optic             No           No               No          No          No              No
freq              3096           3498         3088             3095        3472        2810            2785
mean               NaN            NaN          NaN              NaN         NaN         NaN             NaN
std                NaN            NaN          NaN              NaN         NaN         NaN             NaN
min                NaN            NaN          NaN              NaN         NaN         NaN             NaN
25%                NaN            NaN          NaN              NaN         NaN         NaN             NaN
50%                NaN            NaN          NaN              NaN         NaN         NaN             NaN
75%                NaN            NaN          NaN              NaN         NaN         NaN             NaN
max                NaN            NaN          NaN              NaN         NaN         NaN             NaN

             Contract PaperlessBilling     PaymentMethod  MonthlyCharges  TotalCharges Churn
count            7043             7042              7043     7040.000000   7032.000000  7040
unique              3                2                 4             NaN           NaN     2
top     Month-to-month              Yes  Electronic check             NaN           NaN    No
freq             3875             4170              2365             NaN           NaN  5172
mean              NaN              NaN               NaN       64.761889   2283.300441   NaN
std               NaN              NaN               NaN       30.087384   2266.771362   NaN
min               NaN              NaN               NaN       18.250000     18.800000   NaN
25%               NaN              NaN               NaN       35.500000    401.450000   NaN
50%               NaN              NaN               NaN       70.350000   1397.475000   NaN
75%               NaN              NaN               NaN       89.850000   3794.737500   NaN
max               NaN              NaN               NaN      118.750000   8684.800000   NaN
```

## Missing Values per Column - PDF #2 - Question #8

```
#Missing Values per Column - PDF #2 - Question #8
print(df.isna().sum())
```

```
customerID           0
gender               0
SeniorCitizen        8
Partner              5
Dependents           2
tenure               2
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          1
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     1
PaymentMethod        0
MonthlyCharges       3
TotalCharges        11
Churn                3
dtype: int64
```

## Total Missing Values - PDF #2 - Question #9

```
#Total Missing Values - PDF #2 - Question #9
print("Total missing:", df.isna().sum().sum())
```

```
Total missing: 36
```

## Missing Values per Column - PDF #2 - Question #10

```
#Missing Values per Column - PDF #2 - Question #10
print(df.isna().sum()[df.isna().sum() > 0])
```

```
SeniorCitizen        8
Partner              5
Dependents           2
tenure               2
TechSupport          1
PaperlessBilling     1
MonthlyCharges       3
TotalCharges        11
Churn                3
dtype: int64
```

## Rows with at least one Missing Value - PDF #2 - Question #11

```
#Rows with at least one Missing Value - PDF #2 - Question #11
na_rows = df[df.isna().any(axis=1)]
print("Rows with NA:", len(na_rows))
print(na_rows.head(10))
```

```
Rows with NA: 34
     customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
16   8191-XWSZG  Female            NaN      No         No    52.0          Yes                No
369  3520-FJGCV    Male            NaN     Yes        Yes    72.0          Yes               Yes
370  6563-VRERX    Male            NaN     Yes        Yes    33.0          Yes                No
371  0259-GBZSH    Male            NaN      No         No     2.0          Yes               Yes
372  6122-EFVKN    Male            NaN      No        Yes    24.0           No  No phone service
373  2805-EDJPQ  Female            NaN     Yes        Yes    56.0          Yes               Yes
374  6862-CQUMB    Male            NaN      No         No    37.0          Yes                No
375  7156-MXBJE  Female            NaN      No         No    43.0          Yes               Yes
472  1304-BCCFO    Male            0.0     NaN         No     9.0          Yes                No
473  4104-PVRPS    Male            0.0     NaN         No    24.0          Yes                No

    InternetService       OnlineSecurity         OnlineBackup     DeviceProtection          TechSupport  \
16               No  No internet service  No internet service  No internet service  No internet service
369     Fiber optic                  Yes                  Yes                  Yes                  Yes
370              No  No internet service  No internet service  No internet service  No internet service
371     Fiber optic                   No                   No                   No                   No
372             DSL                  Yes                   No                   No                  Yes
373     Fiber optic                   No                  Yes                  Yes                  Yes
374     Fiber optic                   No                  Yes                   No                   No
375             DSL                   No                  Yes                  Yes                  Yes
472             DSL                   No                   No                  Yes                   No
473             DSL                  Yes                   No                   No                  Yes

             StreamingTV      StreamingMovies        Contract PaperlessBilling            PaymentMethod  \
16   No internet service  No internet service        One year               No             Mailed check
369                  Yes                  Yes        Two year              Yes  Credit card (automatic)
370  No internet service  No internet service        One year               No             Mailed check
371                  Yes                   No  Month-to-month              Yes         Electronic check
372                   No                   No        Two year               No             Mailed check
373                   No                  Yes        One year              Yes  Credit card (automatic)
374                  Yes                  Yes  Month-to-month              Yes  Credit card (automatic)
375                  Yes                  Yes        One year               No  Credit card (automatic)
472                  Yes                  Yes  Month-to-month              Yes             Mailed check
473                   No                   No  Month-to-month               No             Mailed check

     MonthlyCharges  TotalCharges Churn
16            20.65       1022.95    No
369          112.60       8126.65    No
370           20.60        690.25    No
371           85.65        181.50   Yes
372           35.75        830.80    No
373           99.75       5608.40    No
374           96.10       3646.80    No
375           85.10       3662.25    No
472           70.05        564.40    No
473           53.60       1315.35    No
```

## Remove Rows with Missing Values - PDF #2 - Question #12

```
#Remove Rows with Missing Values - PDF #2 - Question #12
df_dropped = df.dropna()
print(df_dropped.shape)
print(df_dropped.head(10))
```

```
(7009, 21)
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
0  7590-VHVEG  Female            0.0     Yes         No     1.0           No  No phone service
1  5575-GNVDE    Male            0.0      No         No    34.0          Yes                No
2  3668-QPYBK    Male            0.0      No         No     2.0          Yes                No
3  7795-CFOCW    Male            0.0      No         No    45.0           No  No phone service
4  9237-HQITU  Female            0.0      No         No     2.0          Yes                No
5  9305-CDSKC  Female            0.0      No         No     8.0          Yes               Yes
6  1452-KIOVK    Male            0.0      No        Yes    22.0          Yes               Yes
7  6713-OKOMC  Female            0.0      No         No    10.0           No  No phone service
8  7892-POOKP  Female            0.0     Yes         No    28.0          Yes               Yes
9  6388-TABGU    Male            0.0      No        Yes    62.0          Yes                No

  InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
0             DSL             No          Yes               No          No          No              No
1             DSL            Yes           No              Yes          No          No              No
2             DSL            Yes          Yes               No          No          No              No
3             DSL            Yes           No              Yes         Yes          No              No
4     Fiber optic             No           No               No          No          No              No
5     Fiber optic             No           No              Yes          No         Yes             Yes
6     Fiber optic             No          Yes               No          No         Yes              No
7             DSL            Yes           No               No          No          No              No
8     Fiber optic             No           No              Yes         Yes         Yes             Yes
9             DSL            Yes          Yes               No          No          No              No

        Contract PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges Churn
0  Month-to-month              Yes           Electronic check           29.85         29.85    No
1        One year               No               Mailed check           56.95       1889.50    No
2  Month-to-month              Yes               Mailed check           53.85        108.15   Yes
3        One year               No  Bank transfer (automatic)           42.30       1840.75    No
4  Month-to-month              Yes           Electronic check           70.70        151.65   Yes
5  Month-to-month              Yes           Electronic check           99.65        820.50   Yes
6  Month-to-month              Yes    Credit card (automatic)           89.10       1949.40    No
7  Month-to-month               No               Mailed check           29.75        301.90    No
8  Month-to-month              Yes           Electronic check          104.80       3046.05   Yes
9        One year               No  Bank transfer (automatic)           56.15       3487.95    No
```

## Mean Imputation (MonthlyCharges) - PDF #2 - Question #13

```
#Mean Imputation (MonthlyCharges) - PDF #2 - Question #13
df["MonthlyCharges"] = df["MonthlyCharges"].fillna(df["MonthlyCharges"].mean())
print(df["MonthlyCharges"].isna().sum())
```

```
0
```

## Median Imputation (TotalCharges) - PDF #2 - Question #14

```
#Median Imputation (TotalCharges) - PDF #2 - Question #14
df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())
print(df["TotalCharges"].isna().sum())
```

```
0
```

## Mode Imputation (Partner) - PDF #2 - Question #15

```
#Mode Imputation (Partner) - PDF #2 - Question #15
df["Partner"] = df["Partner"].fillna(df["Partner"].mode()[0])
print(df["Partner"].isna().sum())
```

```
0
```

## Forward Fill - PDF #2 - Question #16

```
#Forward Fill - PDF #2 - Question #16
df_ffill = df_raw.ffill()
print(df_ffill.isna().sum().sum())
```

```
0
```

## Backward Fill - PDF #2 - Question #17

```
#Backward Fill - PDF #2 - Question #17
df_bfill = df_raw.bfill()
print(df_bfill.isna().sum().sum())
```

```
0
```

## Constant Fill (0 / "Unknown") - PDF #2 - Question #18

```
#Constant Fill (0 / "Unknown") - PDF #2 - Question #18
num_cols = df.select_dtypes(include="number").columns
cat_cols = df.select_dtypes(exclude="number").columns
df[num_cols] = df[num_cols].fillna(0)
df[cat_cols] = df[cat_cols].fillna("Unknown")
print(df.isna().sum().sum())
```

```
0
```

## Remove Duplicates - PDF #2 - Question #19

```
#Remove Duplicates - PDF #2 - Question #19
print("Duplicates:", df.duplicated().sum())
df = df.drop_duplicates()
print(df.shape)
```

```
Duplicates: 0
(7043, 21)
```

## Convert to Factors (category) - PDF #2 - Question #20

```
#Convert to Factors (category) - PDF #2 - Question #20
factor_cols = [c for c in cat_cols if c != "customerID"] + ["SeniorCitizen"]
df[factor_cols] = df[factor_cols].astype("category")
print(df.dtypes)
```

```
customerID            str
gender             category
SeniorCitizen      category
Partner            category
Dependents         category
tenure              float64
PhoneService       category
MultipleLines      category
InternetService    category
OnlineSecurity     category
OnlineBackup       category
DeviceProtection   category
TechSupport        category
StreamingTV        category
StreamingMovies    category
Contract           category
PaperlessBilling   category
PaymentMethod      category
MonthlyCharges      float64
TotalCharges        float64
Churn              category
dtype: object
```

## Rename Columns - PDF #2 - Question #21

```
#Rename Columns - PDF #2 - Question #21
df = df.rename(columns={"tenure": "Tenure", "gender": "Gender"})
print(df.columns.tolist())
```

```
['customerID', 'Gender', 'SeniorCitizen', 'Partner', 'Dependents', 'Tenure', 'PhoneService', 'MultipleLines', 'I
nternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingM
ovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']
```

## Derived Column - PDF #2 - Question #22

```
#Derived Column - PDF #2 - Question #22
df["AvgChargePerMonth"] = np.where(df["Tenure"] > 0, df["TotalCharges"] / df["Tenure"],
df["MonthlyCharges"])
print(df[["Tenure", "TotalCharges", "AvgChargePerMonth"]].head(10))
```

```
   Tenure  TotalCharges  AvgChargePerMonth
0     1.0         29.85          29.850000
1    34.0       1889.50          55.573529
2     2.0        108.15          54.075000
3    45.0       1840.75          40.905556
4     2.0        151.65          75.825000
5     8.0        820.50         102.562500
6    22.0       1949.40          88.609091
7    10.0        301.90          30.190000
8    28.0       3046.05         108.787500
9    62.0       3487.95          56.257258
```

## Summary Statistics - PDF #2 - Question #23

```
#Summary Statistics - PDF #2 - Question #23
print(df.describe(include="all"))
```

```
        customerID Gender  SeniorCitizen Partner Dependents       Tenure PhoneService MultipleLines  \
count         7043   7043         7043.0    7043       7043  7043.000000         7043          7043
unique        7043      2            2.0       2          3          NaN            2             3
top     7590-VHVEG   Male            0.0      No         No          NaN          Yes            No
freq             1   3555         5901.0    3645       4931          NaN         6361          3390
mean           NaN    NaN            NaN     NaN        NaN    32.364049          NaN           NaN
std            NaN    NaN            NaN     NaN        NaN    24.564646          NaN           NaN
min            NaN    NaN            NaN     NaN        NaN     0.000000          NaN           NaN
25%            NaN    NaN            NaN     NaN        NaN     9.000000          NaN           NaN
50%            NaN    NaN            NaN     NaN        NaN    29.000000          NaN           NaN
75%            NaN    NaN            NaN     NaN        NaN    55.000000          NaN           NaN
max            NaN    NaN            NaN     NaN        NaN    72.000000          NaN           NaN

       InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
count             7043           7043         7043             7043        7043        7043            7043
unique               3              3            3                3           4           3               3
top        Fiber optic             No           No               No          No          No              No
freq              3096           3498         3088             3095        3472        2810            2785
mean               NaN            NaN          NaN              NaN         NaN         NaN             NaN
std                NaN            NaN          NaN              NaN         NaN         NaN             NaN
min                NaN            NaN          NaN              NaN         NaN         NaN             NaN
25%                NaN            NaN          NaN              NaN         NaN         NaN             NaN
50%                NaN            NaN          NaN              NaN         NaN         NaN             NaN
75%                NaN            NaN          NaN              NaN         NaN         NaN             NaN
max                NaN            NaN          NaN              NaN         NaN         NaN             NaN

             Contract PaperlessBilling     PaymentMethod  MonthlyCharges  TotalCharges Churn  \
count            7043             7043              7043     7043.000000   7043.000000  7043
unique              3                3                 4             NaN           NaN     3
top     Month-to-month              Yes  Electronic check             NaN           NaN    No
freq             3875             4170              2365             NaN           NaN  5172
mean              NaN              NaN               NaN       64.761889   2281.916928   NaN
std               NaN              NaN               NaN       30.080975   2265.270398   NaN
min               NaN              NaN               NaN       18.250000     18.800000   NaN
25%               NaN              NaN               NaN       35.525000    402.225000   NaN
50%               NaN              NaN               NaN       70.350000   1397.475000   NaN
75%               NaN              NaN               NaN       89.850000   3786.600000   NaN
max               NaN              NaN               NaN      118.750000   8684.800000   NaN

       AvgChargePerMonth
count        7043.000000
unique               NaN
top                  NaN
freq                 NaN
mean           64.762895
std            30.189874
min            13.775000
25%            35.935156
50%            70.337500
75%            90.174158
max           121.400000
```

## Minimum and Maximum - PDF #2 - Question #24

```
#Minimum and Maximum - PDF #2 - Question #24
print(df.select_dtypes(include="number").agg(["min", "max"]))
```

```
     Tenure  MonthlyCharges  TotalCharges  AvgChargePerMonth
min     0.0           18.25          18.8             13.775
max    72.0          118.75        8684.8            121.400
```

## Range - PDF #2 - Question #25

```
#Range - PDF #2 - Question #25
print("Range of MonthlyCharges:", df["MonthlyCharges"].max() - df["MonthlyCharges"].min())
```

```
Range of MonthlyCharges: 100.5
```

## Variance - PDF #2 - Question #26

```
#Variance - PDF #2 - Question #26
print("Variance of MonthlyCharges:", df["MonthlyCharges"].var())
```

```
Variance of MonthlyCharges: 904.865031933198
```

## Standard Deviation - PDF #2 - Question #27

```
#Standard Deviation - PDF #2 - Question #27
print("SD of MonthlyCharges:", df["MonthlyCharges"].std())
```

```
SD of MonthlyCharges: 30.080974584165286
```

## Interquartile Range (IQR) - PDF #2 - Question #28

```
#Interquartile Range (IQR) - PDF #2 - Question #28
print("IQR of MonthlyCharges:", df["MonthlyCharges"].quantile(0.75) -
df["MonthlyCharges"].quantile(0.25))
```

```
IQR of MonthlyCharges: 54.324999999999996
```

# PDF 3 - Data Cleaning and Univariate Analysis

## Import Dataset - PDF #3 - Question #1

```
#Import Dataset - PDF #3 - Question #1
df = load_raw()
```

## First and Last 6 Records - PDF #3 - Question #2

```
#First and Last 6 Records - PDF #3 - Question #2
print(df.head(6))
print(df.tail(6))
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
0  7590-VHVEG  Female            0.0     Yes         No     1.0           No  No phone service
1  5575-GNVDE    Male            0.0      No         No    34.0          Yes                No
2  3668-QPYBK    Male            0.0      No         No     2.0          Yes                No
3  7795-CFOCW    Male            0.0      No         No    45.0           No  No phone service
4  9237-HQITU  Female            0.0      No         No     2.0          Yes                No
5  9305-CDSKC  Female            0.0      No         No     8.0          Yes               Yes

  InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
0             DSL             No          Yes               No          No          No              No
1             DSL            Yes           No              Yes          No          No              No
2             DSL            Yes          Yes               No          No          No              No
3             DSL            Yes           No              Yes         Yes          No              No
4     Fiber optic             No           No               No          No          No              No
5     Fiber optic             No           No              Yes          No         Yes             Yes

        Contract PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges Churn
0  Month-to-month              Yes           Electronic check           29.85         29.85    No
1        One year               No               Mailed check           56.95       1889.50    No
2  Month-to-month              Yes               Mailed check           53.85        108.15   Yes
3        One year               No  Bank transfer (automatic)           42.30       1840.75    No
4  Month-to-month              Yes           Electronic check           70.70        151.65   Yes
5  Month-to-month              Yes           Electronic check           99.65        820.50   Yes
      customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
7037  2569-WGERO  Female            0.0      No         No    72.0          Yes                No
7038  6840-RESVB    Male            0.0     Yes        Yes    24.0          Yes               Yes
7039  2234-XADUH  Female            0.0     Yes        Yes    72.0          Yes               Yes
7040  4801-JZAZL  Female            0.0     Yes        Yes    11.0           No  No phone service
7041  8361-LTMKD    Male            1.0     Yes         No     4.0          Yes               Yes
7042  3186-AJIEK    Male            0.0      No         No    66.0          Yes                No

     InternetService       OnlineSecurity         OnlineBackup     DeviceProtection          TechSupport  \
7037              No  No internet service  No internet service  No internet service  No internet service
7038             DSL                  Yes                   No                  Yes                 Yes
7039     Fiber optic                   No                  Yes                  Yes                  No
7040             DSL                  Yes                   No                   No                  No
7041     Fiber optic                   No                   No                   No                  No
7042     Fiber optic                  Yes                   No                  Yes                 Yes

              StreamingTV      StreamingMovies        Contract PaperlessBilling              PaymentMethod  \
7037  No internet service  No internet service        Two year              Yes  Bank transfer (automatic)
7038                  Yes                  Yes        One year              Yes               Mailed check
7039                  Yes                  Yes        One year              Yes    Credit card (automatic)
7040                   No                   No  Month-to-month              Yes           Electronic check
7041                   No                   No  Month-to-month              Yes               Mailed check
7042                  Yes                  Yes        Two year              Yes  Bank transfer (automatic)

      MonthlyCharges  TotalCharges Churn
7037           21.15       1419.40    No
7038           84.80       1990.50    No
7039          103.20       7362.90    No
7040           29.60        346.45    No
7041           74.40        306.60   Yes
7042          105.65       6844.50    No
```

## Total Missing Values - PDF #3 - Question #3

```
#Total Missing Values - PDF #3 - Question #3
print("Total missing:", df.isna().sum().sum())
```

```
Total missing: 36
```

## Records with at least one Missing Value - PDF #3 - Question #4

```
#Records with at least one Missing Value - PDF #3 - Question #4
print(df[df.isna().any(axis=1)])
```

| | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines |
|---|---|---|---|---|---|---|---|---|
| 16 | 8191-XWSZG | Female | NaN | No | No | 52.0 | Yes | No |
| 369 | 3520-FJGCV | Male | NaN | Yes | Yes | 72.0 | Yes | Yes |
| 370 | 6563-VRERX | Male | NaN | Yes | Yes | 33.0 | Yes | No |
| 371 | 0259-GBZSH | Male | NaN | No | No | 2.0 | Yes | Yes |
| 372 | 6122-EFVKN | Male | NaN | No | Yes | 24.0 | No | No phone service |
| 373 | 2805-EDJPQ | Female | NaN | Yes | Yes | 56.0 | Yes | Yes |
| 374 | 6862-CQUMB | Male | NaN | No | No | 37.0 | Yes | No |
| 375 | 7156-MXBJE | Female | NaN | No | No | 43.0 | Yes | Yes |
| 472 | 1304-BCCFO | Male | 0.0 | NaN | No | 9.0 | Yes | No |
| 473 | 4104-PVRPS | Male | 0.0 | NaN | No | 24.0 | Yes | No |
| 474 | 9399-APLBT | Female | 0.0 | NaN | Yes | 1.0 | Yes | No |
| 475 | 2359-KMGLI | Male | 0.0 | NaN | No | 24.0 | Yes | Yes |
| 476 | 3780-DDGSE | Male | 1.0 | NaN | Yes | 35.0 | Yes | No |
| 483 | 5168-MQQCA | Female | 0.0 | Yes | NaN | 72.0 | Yes | Yes |
| 488 | 4472-LVYGI | Female | 0.0 | Yes | Yes | 0.0 | No | No phone service |
| 538 | 8905-IAZPF | Female | 0.0 | Yes | No | 69.0 | Yes | No |
| 539 | 5394-MEITZ | Female | 0.0 | Yes | Yes | 60.0 | Yes | No |
| 597 | 5172-MIGPM | Male | 0.0 | No | No | 4.0 | Yes | Yes |
| 753 | 3115-CZMZD | Male | 0.0 | No | Yes | 0.0 | Yes | No |
| 936 | 5709-LVOEQ | Female | 0.0 | Yes | Yes | 0.0 | Yes | No |
| 1082 | 4367-NUYAO | Male | 0.0 | Yes | Yes | 0.0 | Yes | Yes |
| 1340 | 1371-DWPAZ | Female | 0.0 | Yes | Yes | 0.0 | No | No phone service |
| 1546 | 7114-AEOZE | Female | 0.0 | No | No | 60.0 | Yes | No |
| 1547 | 2886-KEFUM | Female | 0.0 | Yes | No | 63.0 | Yes | Yes |
| 1849 | 0621-TWIEM | Male | 0.0 | No | No | 10.0 | Yes | No |
| 3308 | 2706-QZIHY | Female | 0.0 | Yes | No | NaN | Yes | Yes |
| 3331 | 7644-OMVMY | Male | 0.0 | Yes | Yes | 0.0 | Yes | No |
| 3738 | 4950-BDEUX | Male | 0.0 | No | No | NaN | No | No phone service |
| 3826 | 3213-VVOLG | Male | 0.0 | Yes | Yes | 0.0 | Yes | Yes |
| 3845 | 0004-TLHLJ | Male | 0.0 | No | NaN | 4.0 | Yes | No |
| 4380 | 2520-SGTTA | Female | 0.0 | Yes | Yes | 0.0 | Yes | No |
| 5218 | 2923-ARZLG | Male | 0.0 | Yes | Yes | 0.0 | Yes | No |
| 6670 | 4075-WKNIU | Female | 0.0 | Yes | Yes | 0.0 | Yes | Yes |
| 6754 | 2775-SEFEE | Male | 0.0 | No | Yes | 0.0 | Yes | Yes |

| | InternetService | OnlineSecurity | OnlineBackup | DeviceProtection | TechSupport |
|---|---|---|---|---|---|
| 16 | No | No internet service | No internet service | No internet service | No internet service |
| 369 | Fiber optic | Yes | Yes | Yes | Yes |
| 370 | No | No internet service | No internet service | No internet service | No internet service |
| 371 | Fiber optic | No | No | No | No |
| 372 | DSL | Yes | No | No | Yes |
| 373 | Fiber optic | No | Yes | Yes | Yes |
| 374 | Fiber optic | No | Yes | No | No |
| 375 | DSL | No | Yes | Yes | Yes |
| 472 | DSL | No | No | Yes | No |
| 473 | DSL | Yes | No | No | Yes |
| 474 | Fiber optic | No | Yes | No | NaN |
| 475 | Fiber optic | No | Yes | No | No |
| 476 | DSL | No | Yes | No | Yes |
| 483 | Fiber optic | No | Yes | Yes | Yes |
| 488 | DSL | Yes | No | Yes | Yes |
| 538 | Fiber optic | No | Yes | No | No |
| 539 | DSL | No | Yes | Yes | Yes |
| 597 | DSL | No | No | No | Yes |
| 753 | No | No internet service | No internet service | No internet service | No internet service |
| 936 | DSL | Yes | Yes | Yes | No |
| 1082 | No | No internet service | No internet service | No internet service | No internet service |
| 1340 | DSL | Yes | Yes | Yes | Yes |
| 1546 | No | No internet service | No internet service | No internet service | No internet service |
| 1547 | Fiber optic | No | Yes | Yes | No |
| 1849 | DSL | No | No | No | No |
| 3308 | No | No internet service | No internet service | No internet service | No internet service |
| 3331 | No | No internet service | No internet service | No internet service | No internet service |
| 3738 | DSL | No | No | Yes | No |
| 3826 | No | No internet service | No internet service | No internet service | No internet service |
| 3845 | Fiber optic | No | No | Yes | No |
| 4380 | No | No internet service | No internet service | No internet service | No internet service |
| 5218 | No | No internet service | No internet service | No internet service | No internet service |
| 6670 | DSL | No | Yes | Yes | Yes |
| 6754 | DSL | Yes | Yes | No | Yes |

| | StreamingTV | StreamingMovies | Contract | PaperlessBilling | PaymentMethod |
|---|---|---|---|---|---|
| 16 | No internet service | No internet service | One year | No | Mailed check |
| 369 | Yes | Yes | Two year | Yes | Credit card (automatic) |
| 370 | No internet service | No internet service | One year | No | Mailed check |
| 371 | Yes | No | Month-to-month | Yes | Electronic check |
| 372 | No | No | Two year | No | Mailed check |
| 373 | No | Yes | One year | Yes | Credit card (automatic) |
| 374 | Yes | Yes | Month-to-month | Yes | Credit card (automatic) |
| 375 | Yes | Yes | One year | No | Credit card (automatic) |

```
472                 Yes                 Yes  Month-to-month              Yes              Mailed check
473                  No                  No  Month-to-month               No              Mailed check
474                  No                  No  Month-to-month              Yes          Electronic check
475                  No                  No  Month-to-month              Yes          Electronic check
476                 Yes                 Yes  Month-to-month              Yes          Electronic check
483                 Yes                 Yes        One year              Yes  Bank transfer (automatic)
488                 Yes                  No        Two year              NaN  Bank transfer (automatic)
538                 Yes                 Yes        One year              Yes    Credit card (automatic)
539                 Yes                 Yes        Two year              Yes  Bank transfer (automatic)
597                  No                 Yes  Month-to-month               No              Mailed check
753  No internet service  No internet service        Two year               No              Mailed check
936                 Yes                 Yes        Two year               No              Mailed check
1082 No internet service  No internet service        Two year               No              Mailed check
1340                Yes                  No        Two year               No    Credit card (automatic)
1546 No internet service  No internet service        Two year               No  Bank transfer (automatic)
1547                Yes                 Yes        One year              Yes          Electronic check
1849                 No                  No  Month-to-month              Yes          Electronic check
3308 No internet service  No internet service        Two year               No              Mailed check
3331 No internet service  No internet service        Two year               No              Mailed check
3738                Yes                 Yes  Month-to-month               No          Electronic check
3826 No internet service  No internet service        Two year               No              Mailed check
3845                 No                  No  Month-to-month              Yes          Electronic check
4380 No internet service  No internet service        Two year               No              Mailed check
5218 No internet service  No internet service        One year              Yes              Mailed check
6670                Yes                  No        Two year               No              Mailed check
6754                 No                  No        Two year              Yes  Bank transfer (automatic)

      MonthlyCharges  TotalCharges Churn
16             20.65       1022.95    No
369           112.60       8126.65    No
370            20.60        690.25    No
371            85.65        181.50   Yes
372            35.75        830.80    No
373            99.75       5608.40    No
374            96.10       3646.80    No
375            85.10       3662.25    No
472            70.05        564.40    No
473            53.60       1315.35    No
474            74.70         74.70   Yes
475            80.25       1861.50   Yes
476            76.05       2747.20    No
483           108.50       8003.80    No
488            52.55           NaN    No
538            95.65       6744.20   NaN
539            80.60       4946.70   NaN
597              NaN        237.20    No
753            20.25           NaN    No
936            80.85           NaN    No
1082           25.75           NaN    No
1340           56.05           NaN    No
1546             NaN       1128.10    No
1547             NaN       6873.75   Yes
1849           45.55        402.60   NaN
3308           25.20        387.90    No
3331           19.85           NaN    No
3738           49.20       1701.65    No
3826           25.35           NaN    No
3845           73.90        280.85   Yes
4380           20.00           NaN    No
5218           19.70           NaN    No
6670           73.35           NaN    No
6754           61.90           NaN    No
```

## Remove Columns with >70% Missing - PDF #3 - Question #5

```
#Remove Columns with >70% Missing - PDF #3 - Question #5
miss_pct = df.isna().mean() * 100
print(miss_pct.round(3))
df = df.loc[:, miss_pct <= 70]
print(df.shape)
```

```
customerID          0.000
gender              0.000
SeniorCitizen       0.114
Partner             0.071
Dependents          0.028
tenure              0.028
PhoneService        0.000
MultipleLines       0.000
InternetService     0.000
OnlineSecurity      0.000
OnlineBackup        0.000
DeviceProtection    0.000
TechSupport         0.014
StreamingTV         0.000
StreamingMovies     0.000
Contract            0.000
PaperlessBilling    0.014
PaymentMethod       0.000
MonthlyCharges      0.043
TotalCharges        0.156
Churn               0.043
dtype: float64
(7043, 21)
```

## Mean Imputation (Numerical) - PDF #3 - Question #6

```
#Mean Imputation (Numerical) - PDF #3 - Question #6
num_cols = df.select_dtypes(include="number").columns
df_mean = df.copy()
df_mean[num_cols] = df_mean[num_cols].fillna(df_mean[num_cols].mean())
print(df_mean[num_cols].isna().sum())
```

```
SeniorCitizen     0
tenure            0
MonthlyCharges    0
TotalCharges      0
dtype: int64
```

## Median Imputation (Numerical) - PDF #3 - Question #7

```
#Median Imputation (Numerical) - PDF #3 - Question #7
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
print(df[num_cols].isna().sum())
```

```
SeniorCitizen     0
tenure            0
MonthlyCharges    0
TotalCharges      0
dtype: int64
```

## Mode Imputation (Categorical) - PDF #3 - Question #8

```
#Mode Imputation (Categorical) - PDF #3 - Question #8
cat_cols = df.select_dtypes(exclude="number").columns
for c in cat_cols:
    df[c] = df[c].fillna(df[c].mode()[0])
print(df.isna().sum().sum())
```

```
0
```

## Identify Duplicate Rows - PDF #3 - Question #9

```
#Identify Duplicate Rows - PDF #3 - Question #9
print("Duplicate rows:", df.duplicated().sum())
print(df[df.duplicated()])
```

```
Duplicate rows: 0
Empty DataFrame
Columns: [customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetS
ervice, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, Pap
erlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Churn]
Index: []
```

## Remove Duplicate Rows - PDF #3 - Question #10

```
#Remove Duplicate Rows - PDF #3 - Question #10
df = df.drop_duplicates().reset_index(drop=True)
print(df.shape)
```

```
(7043, 21)
```

## Identify Incorrect Data Types - PDF #3 - Question #11

```
#Identify Incorrect Data Types - PDF #3 - Question #11
raw_str = pd.read_csv(URL, lineterminator="\r", dtype=str)
for c in raw_str.columns:
    if pd.to_numeric(raw_str[c], errors="coerce").notna().mean() > 0.95 and
raw_str[c].str.contains(r"[?\s]", na=False).any():
        print(f"{c}: stored as text, should be numeric")
for c in num_cols:
    if df[c].nunique() <= 2:
        print(f"{c}: stored as numeric, should be categorical")
```

```
tenure: stored as text, should be numeric
TotalCharges: stored as text, should be numeric
SeniorCitizen: stored as numeric, should be categorical
```

## Convert Character Variables to Factors - PDF #3 - Question #12

```
#Convert Character Variables to Factors - PDF #3 - Question #12
factor_cols = [c for c in cat_cols if c != "customerID"] + ["SeniorCitizen"]
df[factor_cols] = df[factor_cols].astype("category")
print(df.dtypes)
```

```
customerID             str
gender            category
SeniorCitizen     category
Partner           category
Dependents        category
tenure             float64
PhoneService      category
MultipleLines     category
InternetService   category
OnlineSecurity    category
OnlineBackup      category
DeviceProtection  category
TechSupport       category
StreamingTV       category
StreamingMovies   category
Contract          category
PaperlessBilling  category
PaymentMethod     category
MonthlyCharges     float64
TotalCharges       float64
Churn             category
dtype: object
```

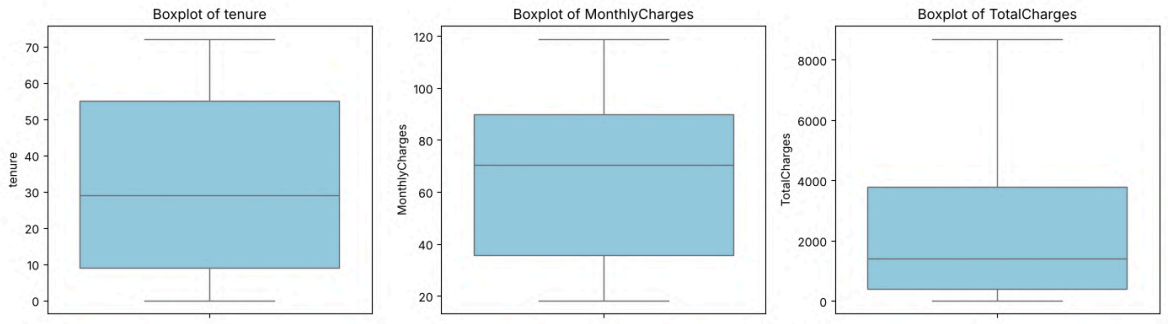
## Lowercase / Uppercase - PDF #3 - Question #13

```
#Lowercase / Uppercase - PDF #3 - Question #13
df["gender"] = df["gender"].astype(str).str.lower().astype("category")
df["Contract"] = df["Contract"].astype(str).str.upper().astype("category")
df["customerID"] = df["customerID"].str.lower()
print(df[["customerID", "gender", "Contract"]].head())
```

```
   customerID  gender        Contract
0  7590-vhveg  female  MONTH-TO-MONTH
1  5575-gnvde    male        ONE YEAR
2  3668-qpybk    male  MONTH-TO-MONTH
3  7795-cfocw    male        ONE YEAR
4  9237-hqitu  female  MONTH-TO-MONTH
```

## Detect Outliers - Boxplots - PDF #3 - Question #14

```
#Detect Outliers - Boxplots - PDF #3 - Question #14
cont_cols = ["tenure", "MonthlyCharges", "TotalCharges"]
fig, axes = plt.subplots(1, 3, figsize=(14, 4))
for ax, c in zip(axes, cont_cols):
    sns.boxplot(y=df[c], ax=ax, color="skyblue")
    ax.set_title(f"Boxplot of {c}")
plt.tight_layout()
plt.show()
```




## Identify Outliers - IQR Method - PDF #3 - Question #15

```
#Identify Outliers - IQR Method - PDF #3 - Question #15
for c in cont_cols:
    q1, q3 = df[c].quantile([0.25, 0.75])
    iqr = q3 - q1
    lo, hi = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    print(f"{c}: bounds=({lo:.2f}, {hi:.2f}), outliers={((df[c] < lo) | (df[c] > hi)).sum()}")
```

```
tenure: bounds=(-60.00, 124.00), outliers=0
MonthlyCharges: bounds=(-45.96, 171.34), outliers=0
TotalCharges: bounds=(-4674.34, 8863.16), outliers=0
```

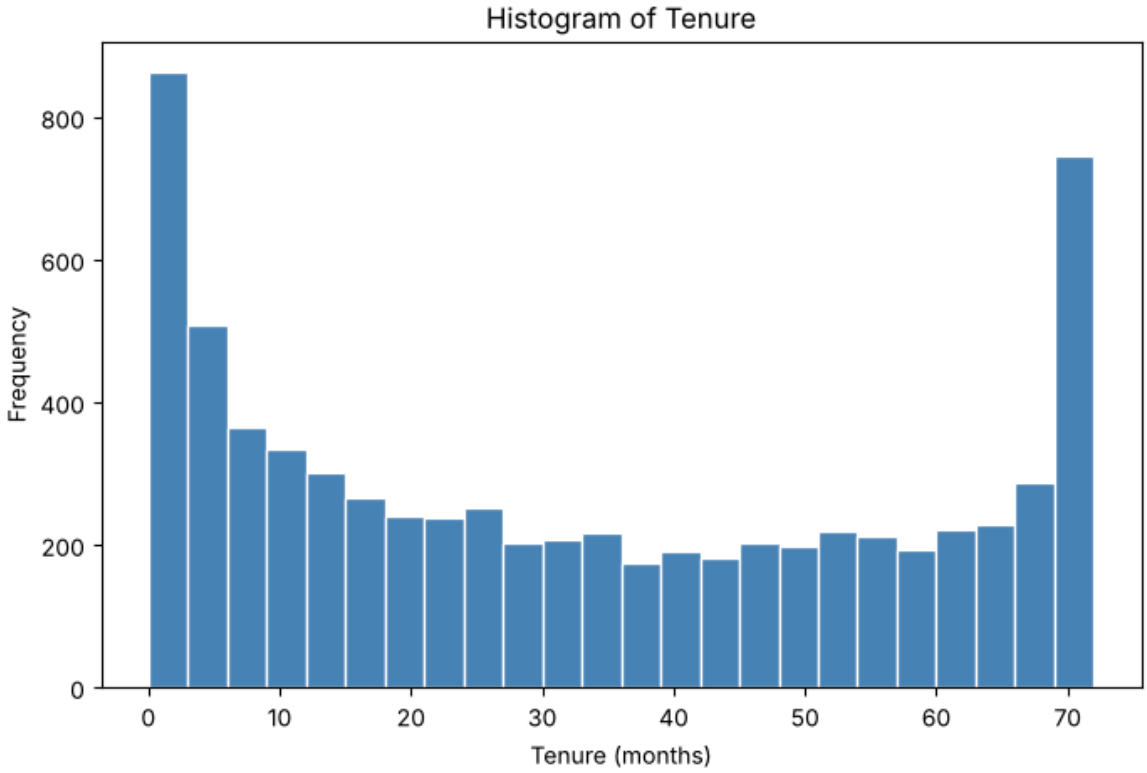
## Remove Extreme Outliers - PDF #3 - Question #16

```
#Remove Extreme Outliers - PDF #3 - Question #16
mask = pd.Series(True, index=df.index)
for c in cont_cols:
    q1, q3 = df[c].quantile([0.25, 0.75])
    iqr = q3 - q1
    mask &= df[c].between(q1 - 3 * iqr, q3 + 3 * iqr)
print("Before:", df.shape)
df = df[mask].reset_index(drop=True)
print("After:", df.shape)
```

```
Before: (7043, 21)
After: (7043, 21)
```

## Histogram + Interpretation - PDF #3 - Question #17

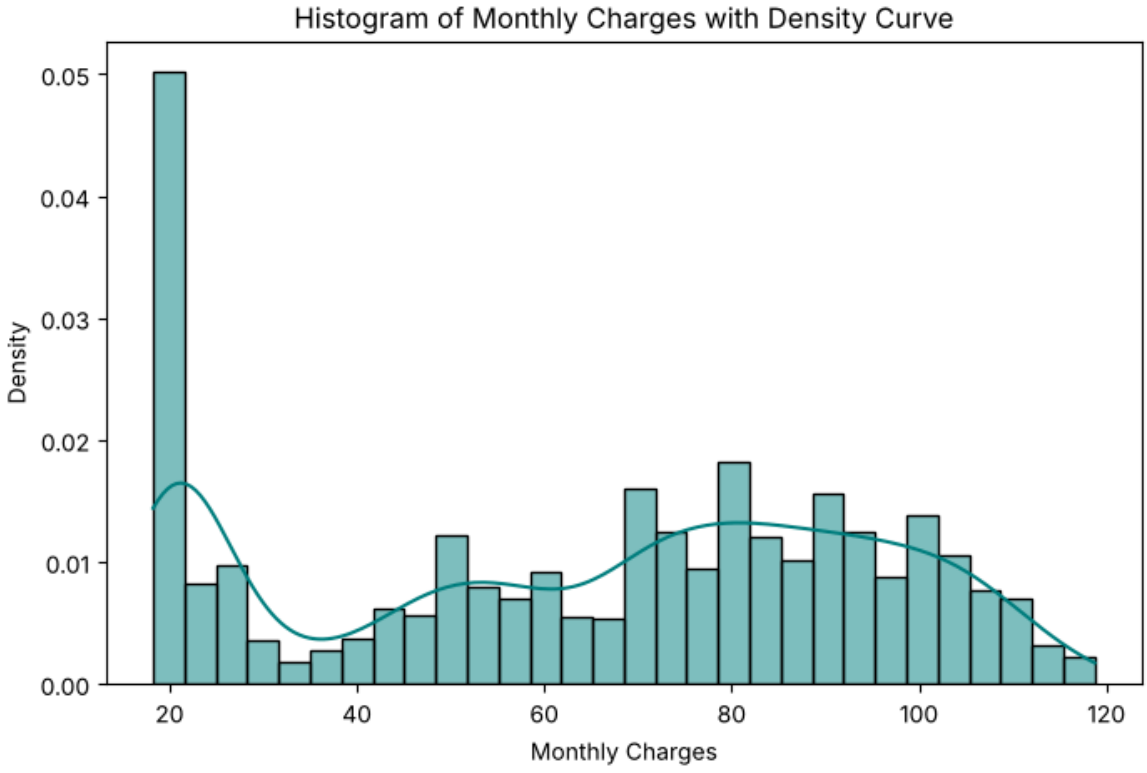
```
#Histogram + Interpretation - PDF #3 - Question #17
plt.figure(figsize=(8, 5))
plt.hist(df["tenure"], bins=24, color="steelblue", edgecolor="white")
plt.title("Histogram of Tenure")
plt.xlabel("Tenure (months)")
plt.ylabel("Frequency")
plt.show()
sk = df["tenure"].skew()
print(f"Skewness={sk:.3f} ->", "right-skewed" if sk > 0.5 else "left-skewed" if sk < -0.5 else
"approximately symmetric",
      "| peaks at very new (<6 mo) and long-term (~72 mo) customers -> bimodal")
```




```
Skewness=0.239 -> approximately symmetric | peaks at very new (<6 mo) and long-term (~72 mo) customers -> bimoda
l
```

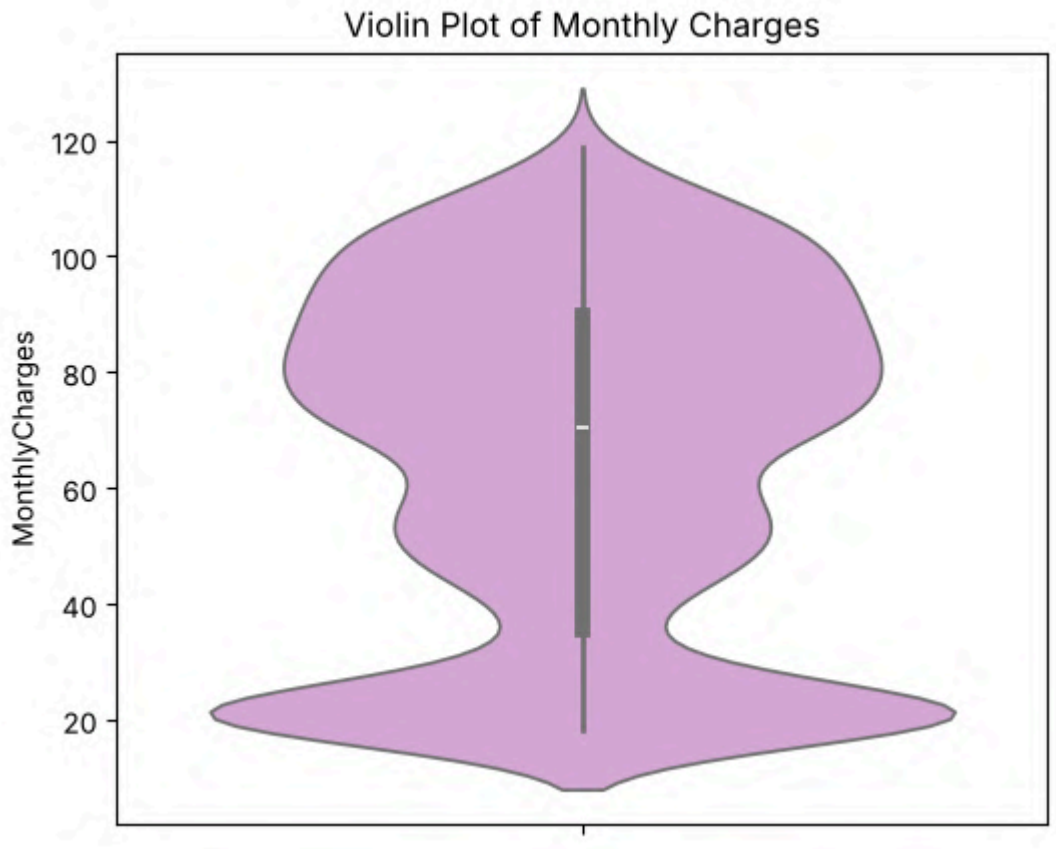
## Histogram with Density Curve - PDF #3 - Question #18

```python
#Histogram with Density Curve - PDF #3 - Question #18
plt.figure(figsize=(8, 5))
sns.histplot(df["MonthlyCharges"], bins=30, kde=True, stat="density", color="teal")
plt.title("Histogram of Monthly Charges with Density Curve")
plt.xlabel("Monthly Charges")
plt.show()
```



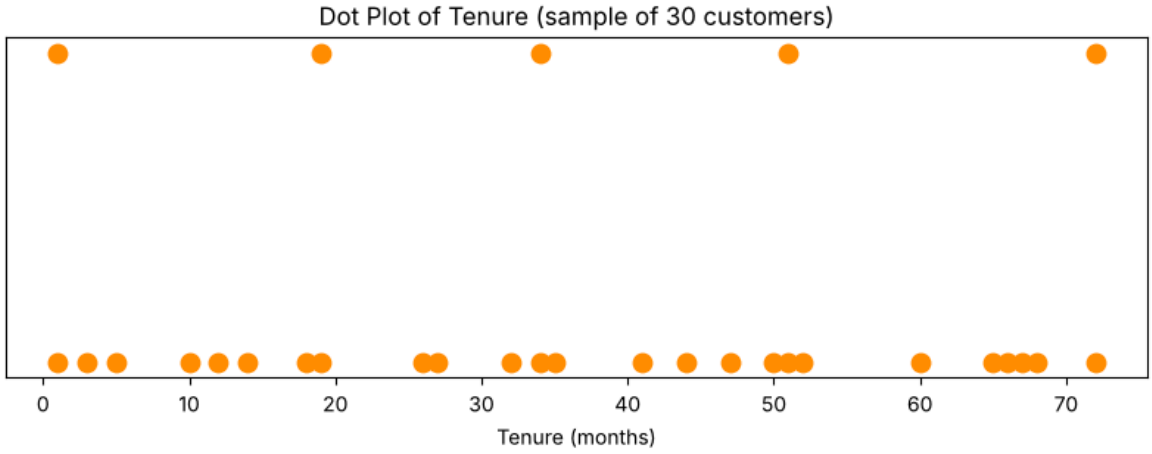
## Violin Plot - PDF #3 - Question #19

```
#Violin Plot - PDF #3 - Question #19
plt.figure(figsize=(6, 5))
sns.violinplot(y=df["MonthlyCharges"], color="plum")
plt.title("Violin Plot of Monthly Charges")
plt.show()
```




## Dot Plot (small sample) - PDF #3 - Question #20

```
#Dot Plot (small sample) - PDF #3 - Question #20
small = df["tenure"].sample(30, random_state=1).astype(int)
counts = {}
xs, ys = [], []
for v in sorted(small):
    counts[v] = counts.get(v, 0) + 1
    xs.append(v)
    ys.append(counts[v])
plt.figure(figsize=(10, 3))
plt.scatter(xs, ys, s=80, color="darkorange")
plt.yticks([])
plt.title("Dot Plot of Tenure (sample of 30 customers)")
plt.xlabel("Tenure (months)")
plt.show()
```




## Frequency Table - PDF #3 - Question #21

```
#Frequency Table - PDF #3 - Question #21
freq = df["PaymentMethod"].value_counts()
print(freq)
```

```
PaymentMethod
Electronic check             2365
Mailed check                 1612
Bank transfer (automatic)    1544
Credit card (automatic)      1522
Name: count, dtype: int64
```

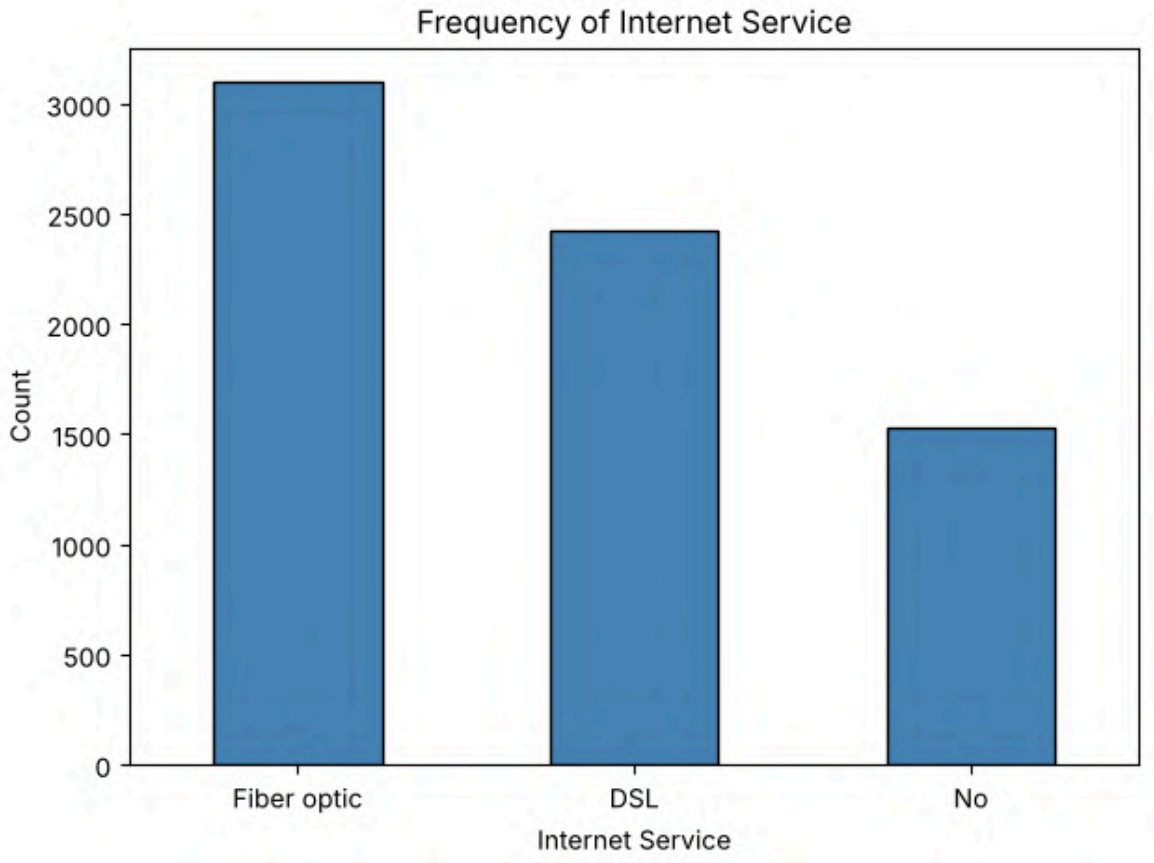
## Proportion and Percentage - PDF #3 - Question #22

```
#Proportion and Percentage - PDF #3 - Question #22
print(pd.DataFrame({"Frequency": freq, "Proportion": (freq / freq.sum()).round(4),
                    "Percentage": (freq / freq.sum() * 100).round(2)}))
```

```
                           Frequency  Proportion  Percentage
PaymentMethod
Electronic check                2365      0.3358       33.58
Mailed check                    1612      0.2289       22.89
Bank transfer (automatic)       1544      0.2192       21.92
Credit card (automatic)         1522      0.2161       21.61
```

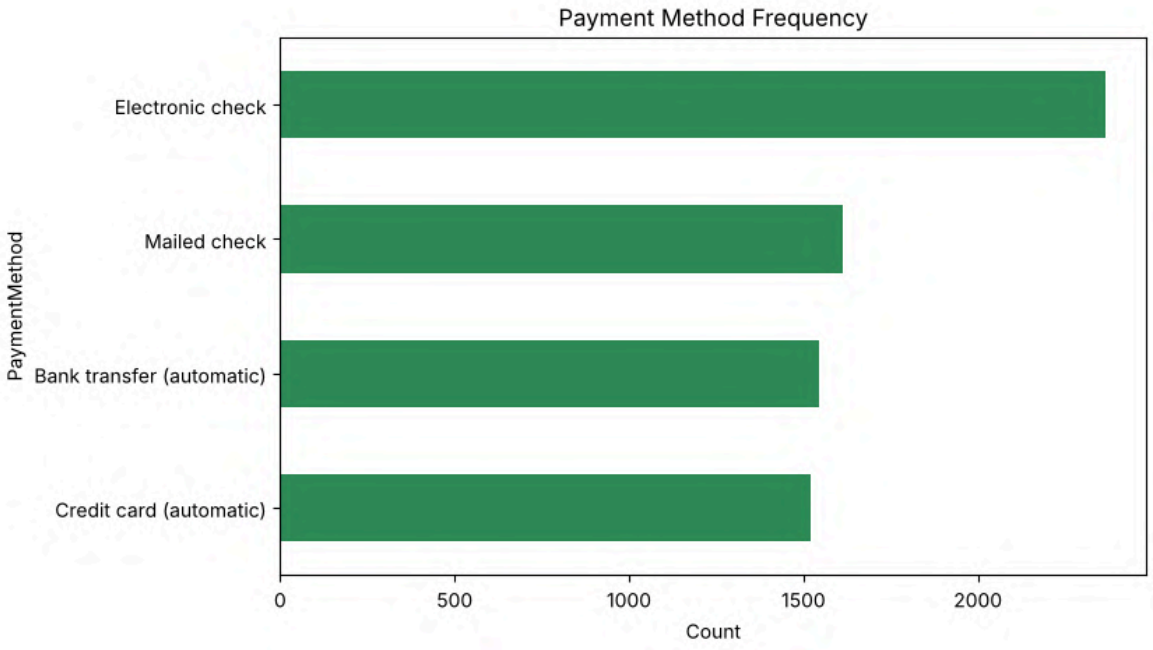
## Bar Chart - PDF #3 - Question #23

```
#Bar Chart - PDF #3 - Question #23
plt.figure(figsize=(7, 5))
df["InternetService"].value_counts().plot(kind="bar", color="steelblue", edgecolor="black")
plt.title("Frequency of Internet Service")
plt.xlabel("Internet Service")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```



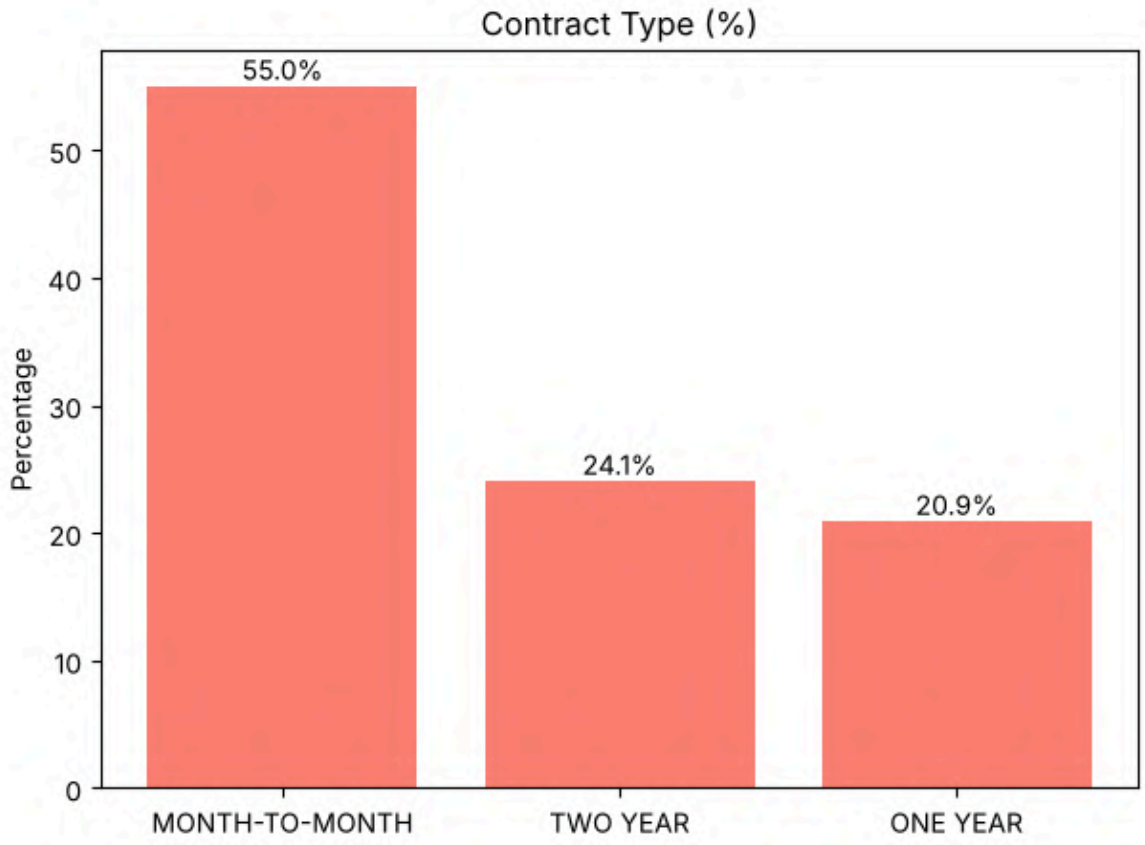

## Horizontal Bar Chart - PDF #3 - Question #24

```
#Horizontal Bar Chart - PDF #3 - Question #24
plt.figure(figsize=(8, 5))
freq.sort_values().plot(kind="barh", color="seagreen")
plt.title("Payment Method Frequency")
plt.xlabel("Count")
plt.show()
```



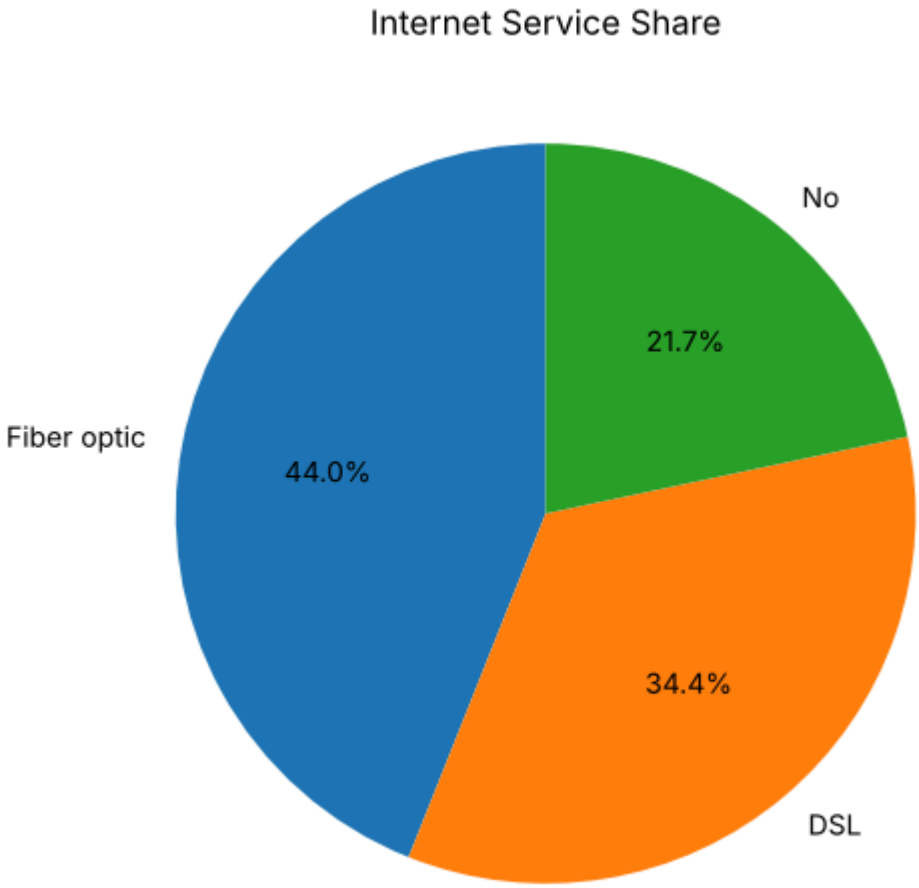
## Percentage Bar Chart - PDF #3 - Question #25

```
#Percentage Bar Chart - PDF #3 - Question #25
pct = df["Contract"].value_counts(normalize=True) * 100
plt.figure(figsize=(7, 5))
bars = plt.bar(pct.index.astype(str), pct.values, color="salmon")
plt.bar_label(bars, fmt="%.1f%%")
plt.title("Contract Type (%)")
plt.ylabel("Percentage")
plt.show()
```



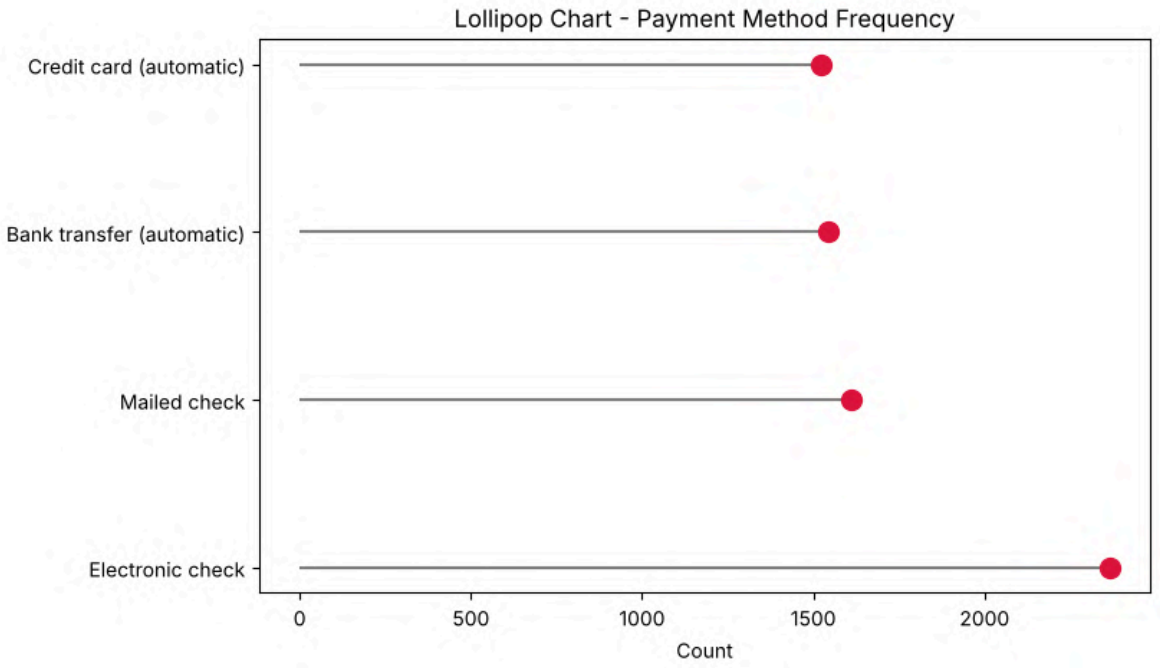

## Pie Chart - PDF #3 - Question #26

```
#Pie Chart - PDF #3 - Question #26
plt.figure(figsize=(6, 6))
df["InternetService"].value_counts().plot(kind="pie", autopct="%1.1f%%", startangle=90)
plt.title("Internet Service Share")
plt.ylabel("")
plt.show()
```



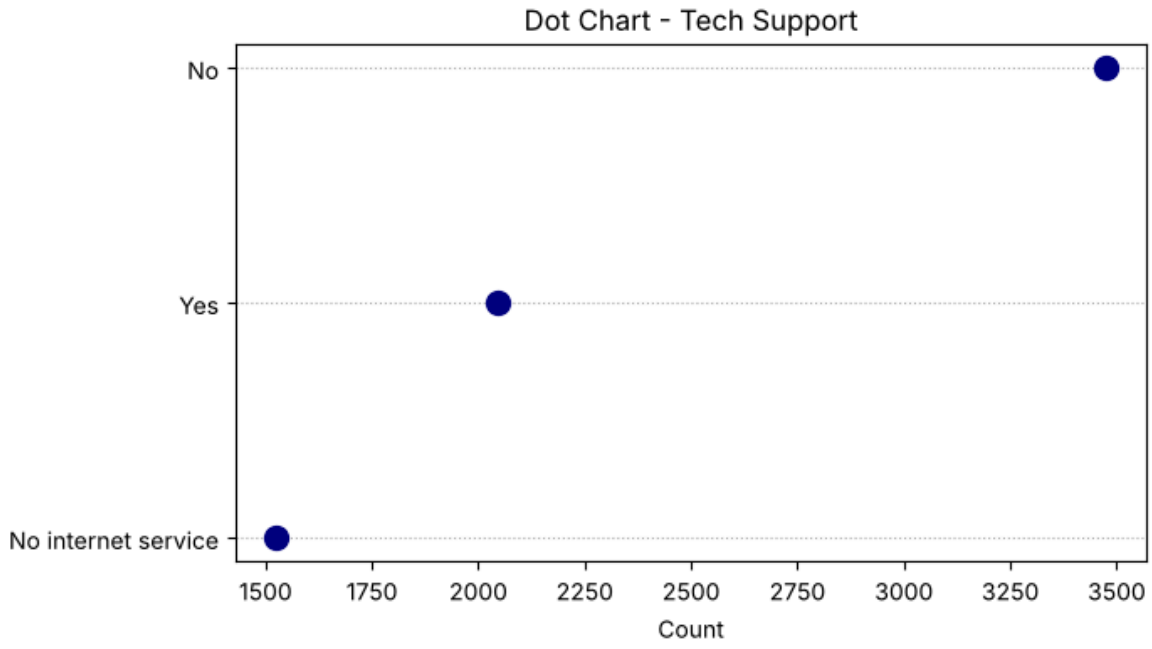

## Lollipop Chart - PDF #3 - Question #27

```
#Lollipop Chart - PDF #3 - Question #27
plt.figure(figsize=(8, 5))
plt.hlines(y=freq.index, xmin=0, xmax=freq.values, color="grey")
plt.plot(freq.values, freq.index, "o", markersize=10, color="crimson")
plt.title("Lollipop Chart - Payment Method Frequency")
plt.xlabel("Count")
plt.show()
```




## Dot Chart - PDF #3 - Question #28

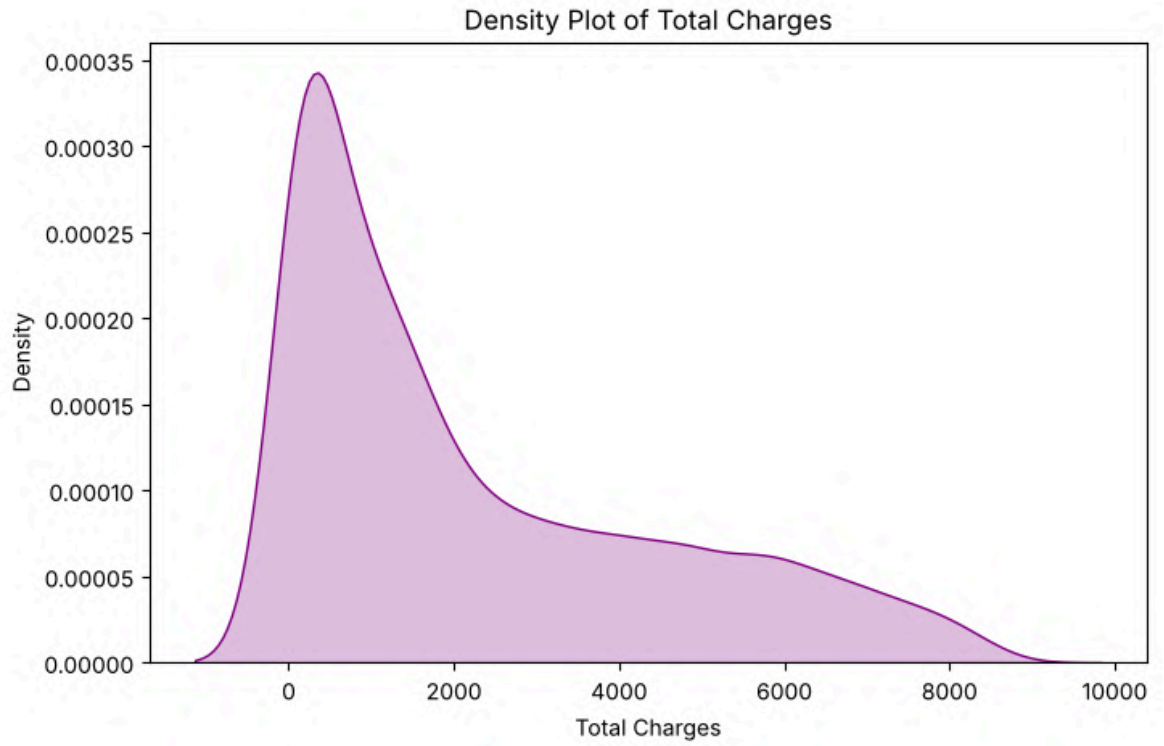
```
#Dot Chart - PDF #3 - Question #28
tech = df["TechSupport"].value_counts().sort_values()
plt.figure(figsize=(7, 4))
plt.plot(tech.values, tech.index.astype(str), "o", markersize=10, color="navy")
plt.grid(axis="y", linestyle=":")
plt.title("Dot Chart - Tech Support")
plt.xlabel("Count")
plt.show()
```

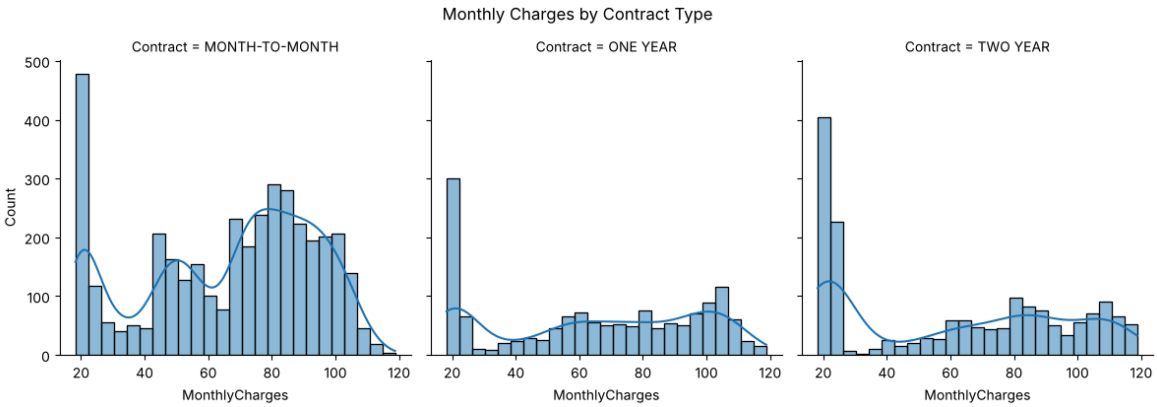


# Density Plot - PDF #3 - Question #29

```
#Density Plot - PDF #3 - Question #29
plt.figure(figsize=(8, 5))
sns.kdeplot(df["TotalCharges"], fill=True, color="purple")
plt.title("Density Plot of Total Charges")
plt.xlabel("Total Charges")
plt.show()

#facet_wrap Distribution by Group - PDF #3 - Question #30
g = sns.displot(data=df, x="MonthlyCharges", col="Contract", bins=25, kde=True, height=4)
g.figure.suptitle("Monthly Charges by Contract Type", y=1.05)
plt.show()
for grp, sub in df.groupby("Contract", observed=True)["MonthlyCharges"]:
    print(f"{grp}: mean={sub.mean():.2f}, median={sub.median():.2f}, sd={sub.std():.2f}, skew=
{sub.skew():.2f}")
```




```
MONTH-TO-MONTH: mean=66.40, median=73.25, sd=26.93, skew=-0.41
ONE YEAR: mean=65.02, median=68.75, sd=31.82, skew=-0.17
TWO YEAR: mean=60.80, median=64.35, sd=34.67, skew=0.08
```

# PDF 4 - Bivariate Analysis

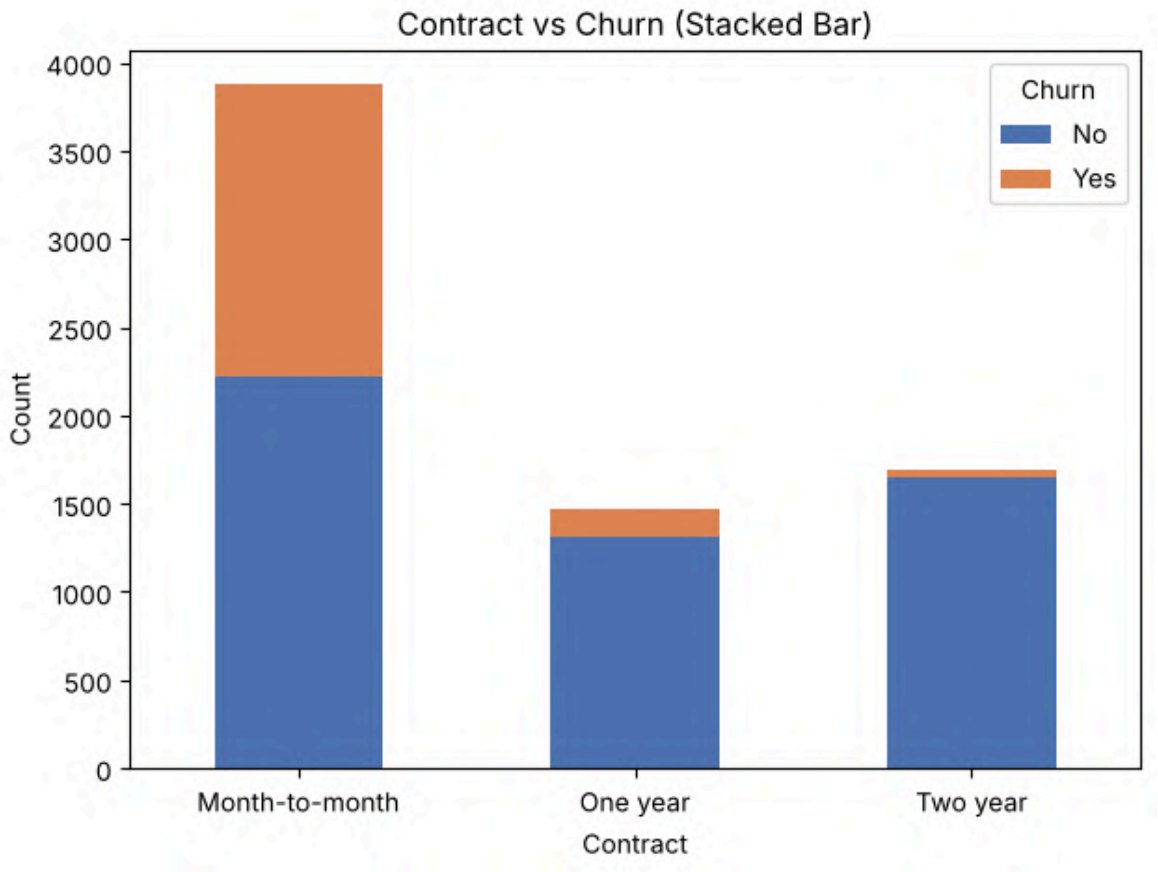
## Load Dataset - PDF #4 - Question #1

```
#Load Dataset - PDF #4 - Question #1
df = load_clean()
```

## Contingency Table + Stacked Bar (Base) - PDF #4 - Question #2

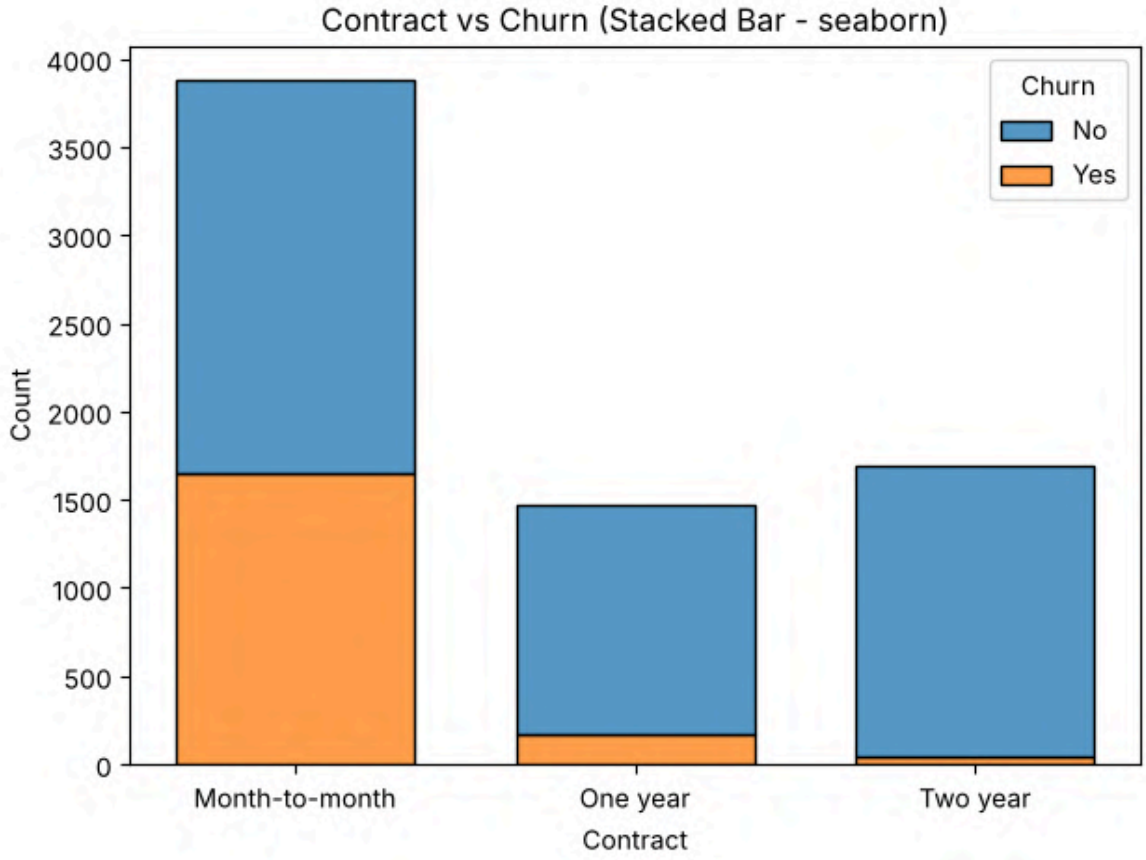
```
#Contingency Table + Stacked Bar (Base) - PDF #4 - Question #2
ct = pd.crosstab(df["Contract"], df["Churn"])
print(ct)
ct.plot(kind="bar", stacked=True, color=["#4C72B0", "#DD8452"], figsize=(7, 5))
plt.title("Contract vs Churn (Stacked Bar)")
plt.xlabel("Contract")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.legend(title="Churn")
plt.show()
```

```
Churn             No   Yes
Contract
Month-to-month  2221  1654
One year        1307   166
Two year        1647    48
```



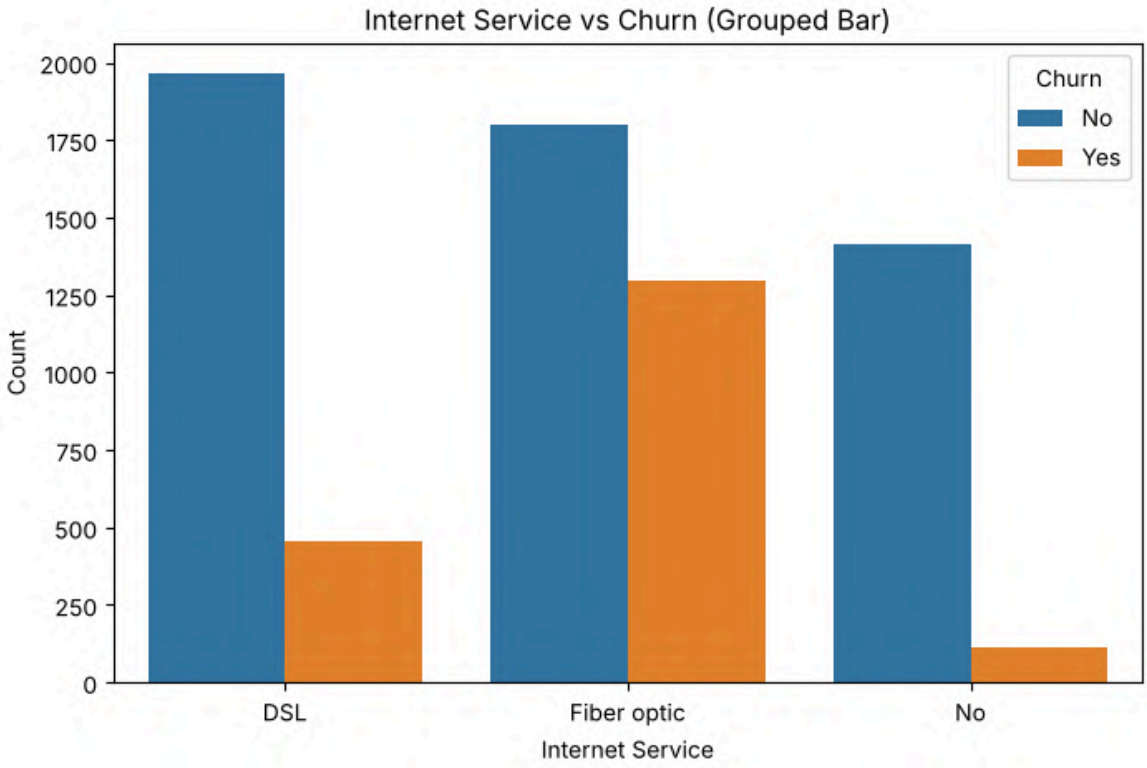
## Contingency Table + Stacked Bar (seaborn / ggplot style) - PDF #4 - Question #2

```
#Contingency Table + Stacked Bar (seaborn / ggplot style) - PDF #4 - Question #2
plt.figure(figsize=(7, 5))
sns.histplot(data=df, x="Contract", hue="Churn", multiple="stack", shrink=0.7, discrete=True)
plt.title("Contract vs Churn (Stacked Bar - seaborn)")
plt.ylabel("Count")
plt.show()
```



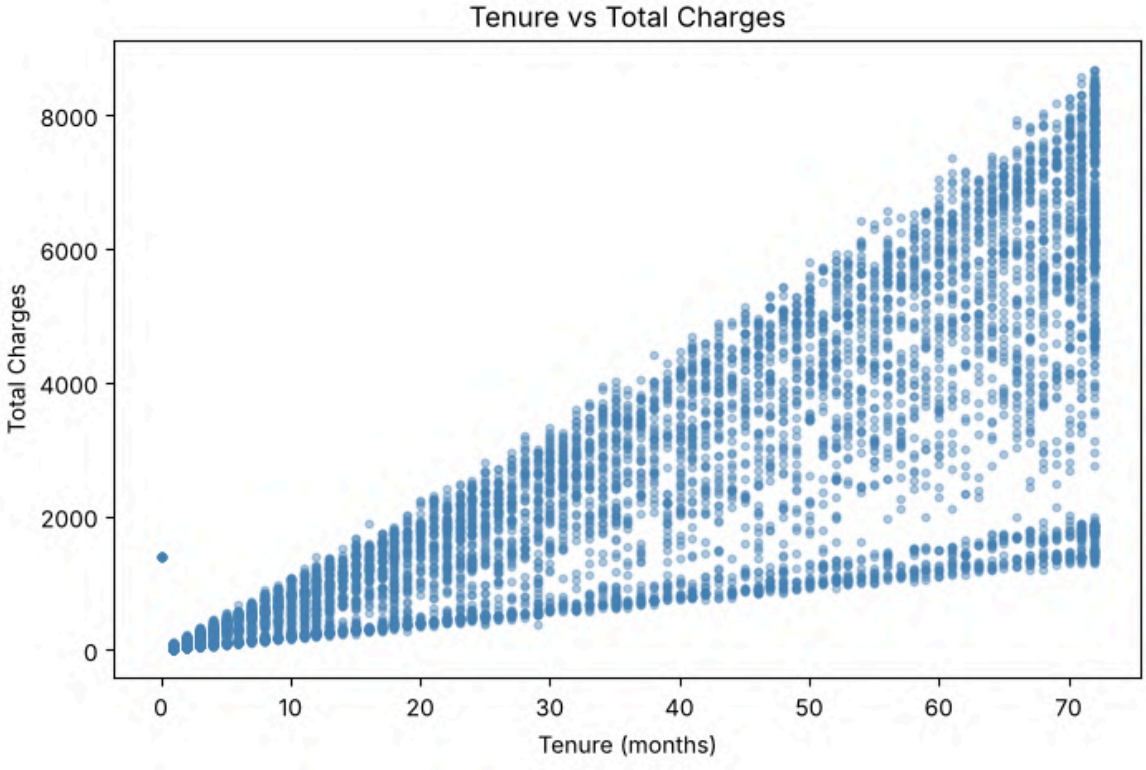

## Grouped Bar Plot - PDF #4 - Question #3

```
#Grouped Bar Plot - PDF #4 - Question #3
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x="InternetService", hue="Churn")
plt.title("Internet Service vs Churn (Grouped Bar)")
plt.xlabel("Internet Service")
plt.ylabel("Count")
plt.show()
```



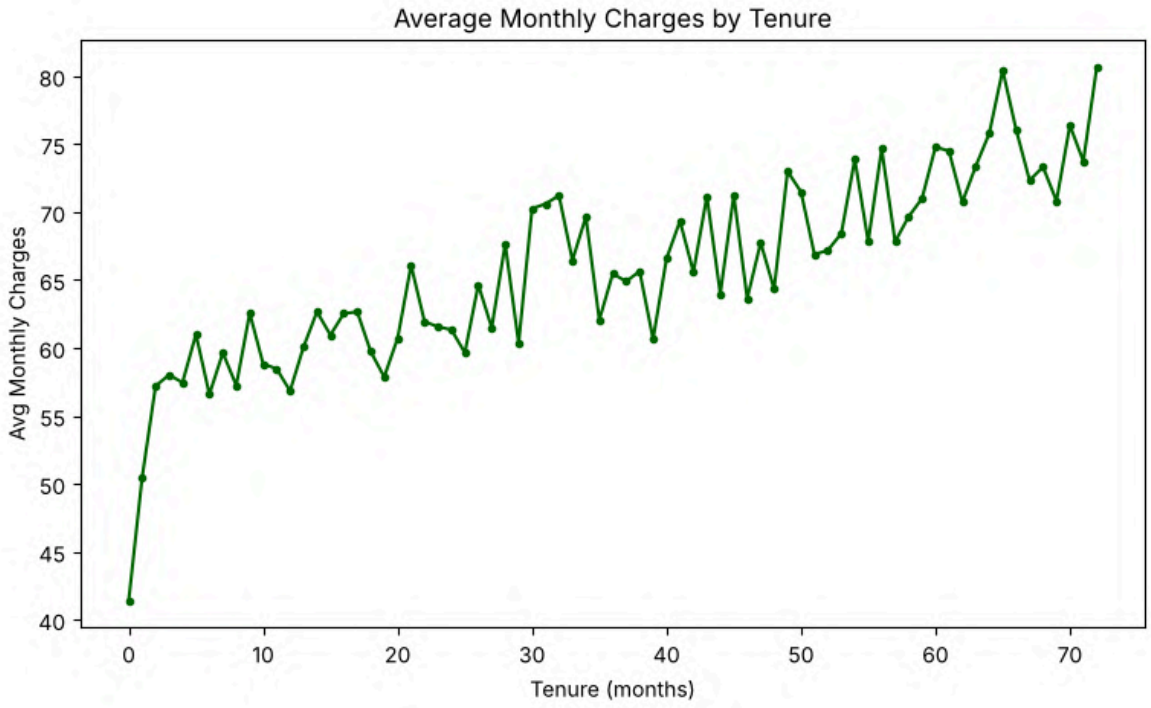

## Scatter Plot - PDF #4 - Question #4

```
#Scatter Plot - PDF #4 - Question #4
plt.figure(figsize=(8, 5))
plt.scatter(df["tenure"], df["TotalCharges"], alpha=0.4, s=10, color="steelblue")
plt.title("Tenure vs Total Charges")
plt.xlabel("Tenure (months)")
plt.ylabel("Total Charges")
plt.show()
```



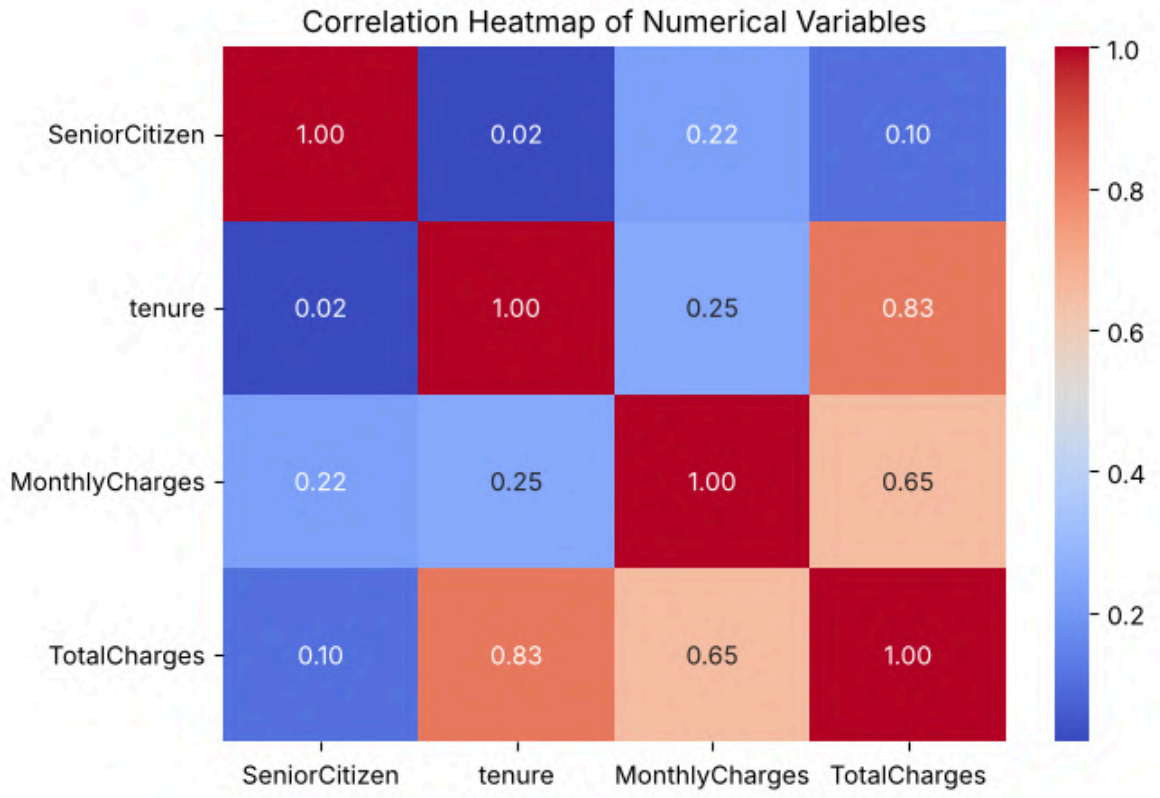
## Line Plot - PDF #4 - Question #5

```
#Line Plot - PDF #4 - Question #5
trend = df.groupby("tenure")["MonthlyCharges"].mean()
plt.figure(figsize=(9, 5))
plt.plot(trend.index, trend.values, marker="o", markersize=3, color="darkgreen")
plt.title("Average Monthly Charges by Tenure")
plt.xlabel("Tenure (months)")
plt.ylabel("Avg Monthly Charges")
plt.show()
```



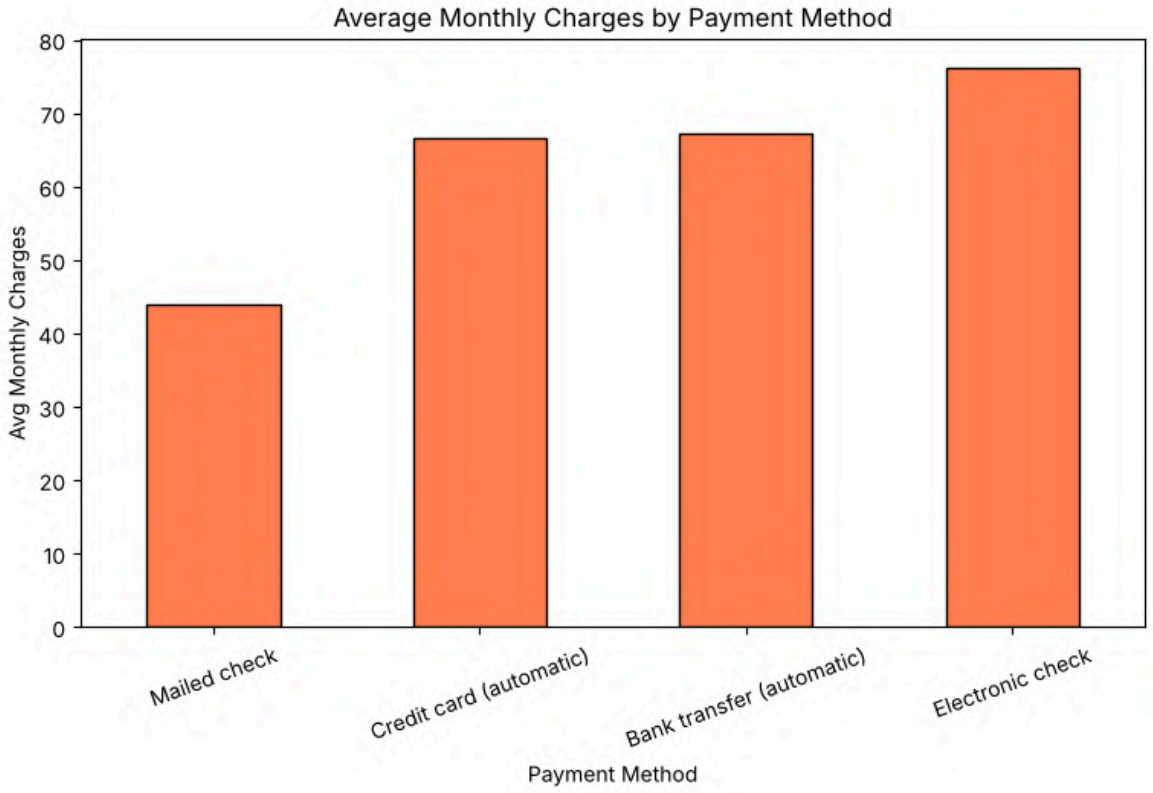

## Heatmap - PDF #4 - Question #6

```
#Heatmap - PDF #4 - Question #6
num = df[["SeniorCitizen", "tenure", "MonthlyCharges", "TotalCharges"]]
plt.figure(figsize=(7, 5))
sns.heatmap(num.corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap of Numerical Variables")
plt.show()
```



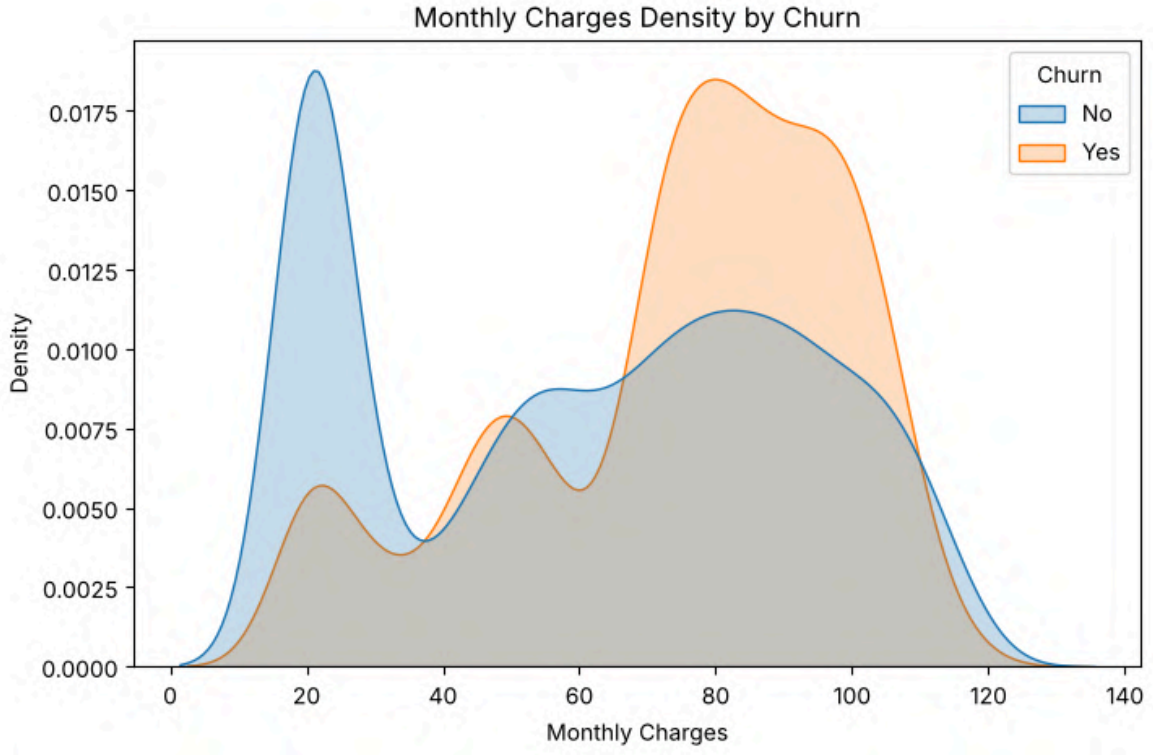

## Bar Chart Comparing Two Variables - PDF #4 - Question #7

```
#Bar Chart Comparing Two Variables - PDF #4 - Question #7
avg_pm = df.groupby("PaymentMethod")["MonthlyCharges"].mean().sort_values()
plt.figure(figsize=(9, 5))
avg_pm.plot(kind="bar", color="coral", edgecolor="black")
plt.title("Average Monthly Charges by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Avg Monthly Charges")
plt.xticks(rotation=20)
plt.show()
```



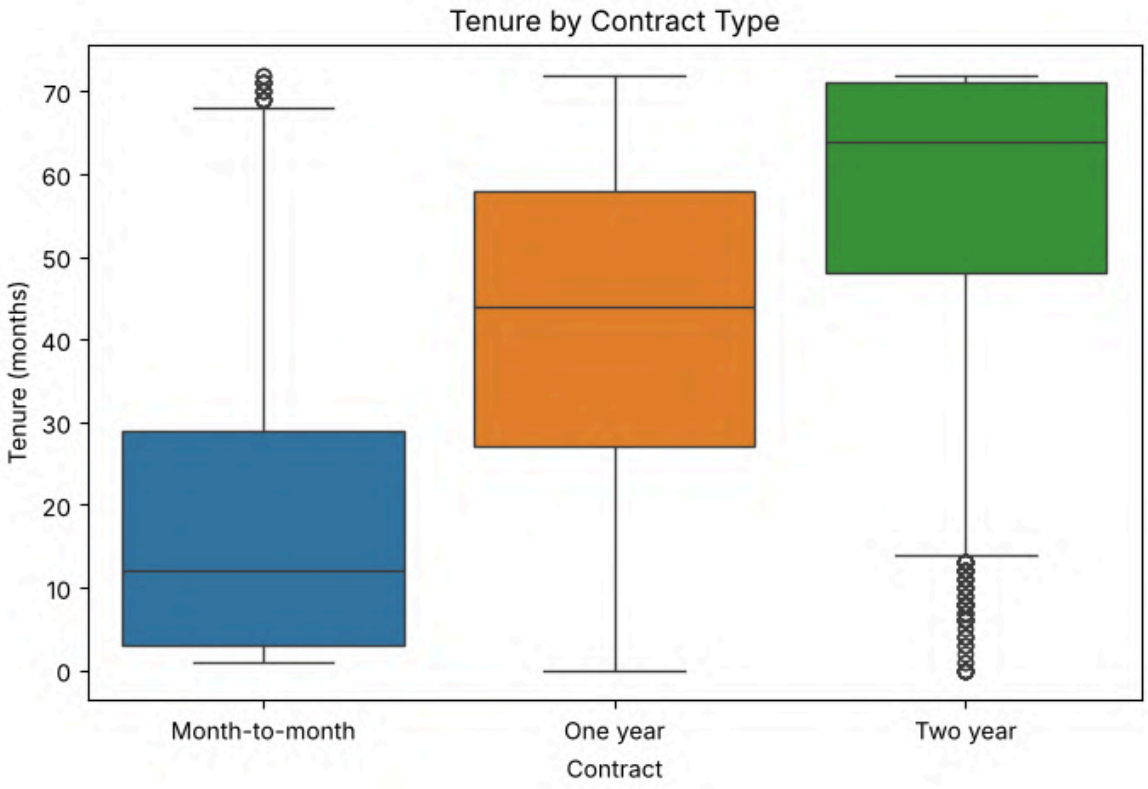

## Density Plot by Category - PDF #4 - Question #8

```
#Density Plot by Category - PDF #4 - Question #8
plt.figure(figsize=(8, 5))
sns.kdeplot(data=df, x="MonthlyCharges", hue="Churn", fill=True, common_norm=False)
plt.title("Monthly Charges Density by Churn")
plt.xlabel("Monthly Charges")
plt.show()
```



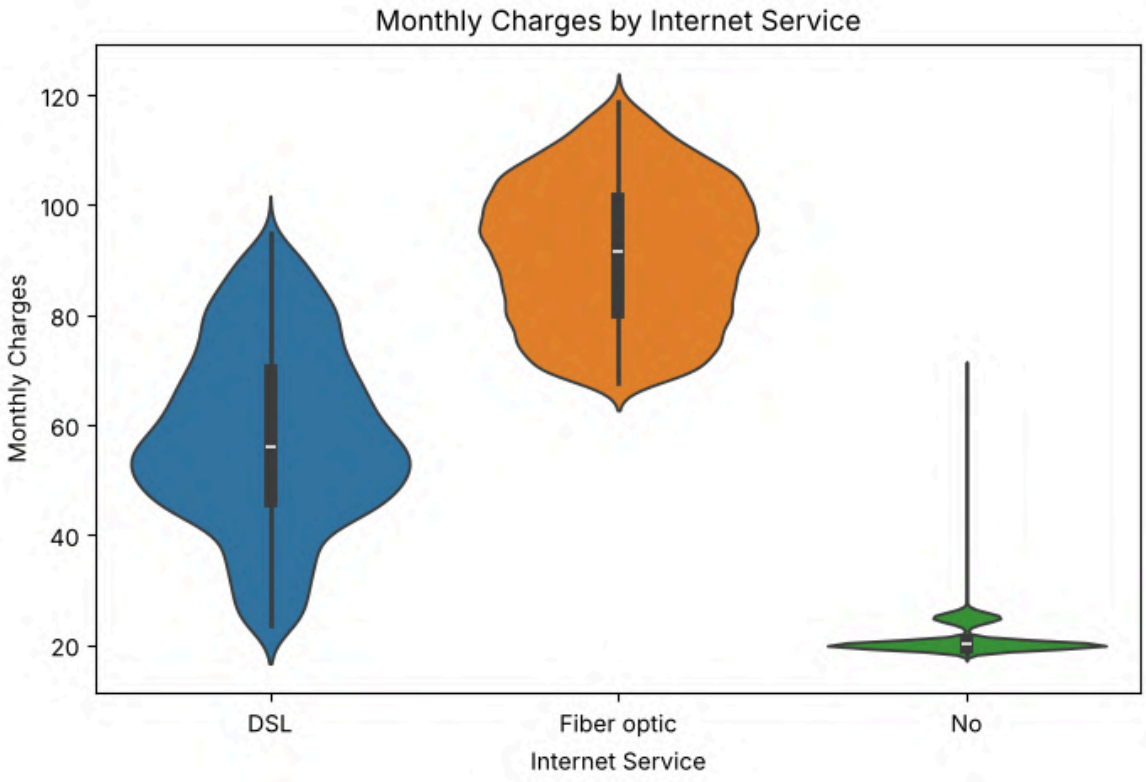
## Box Plot - PDF #4 - Question #9

```
#Box Plot - PDF #4 - Question #9
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="Contract", y="tenure", hue="Contract", legend=False)
plt.title("Tenure by Contract Type")
plt.xlabel("Contract")
plt.ylabel("Tenure (months)")
plt.show()
```



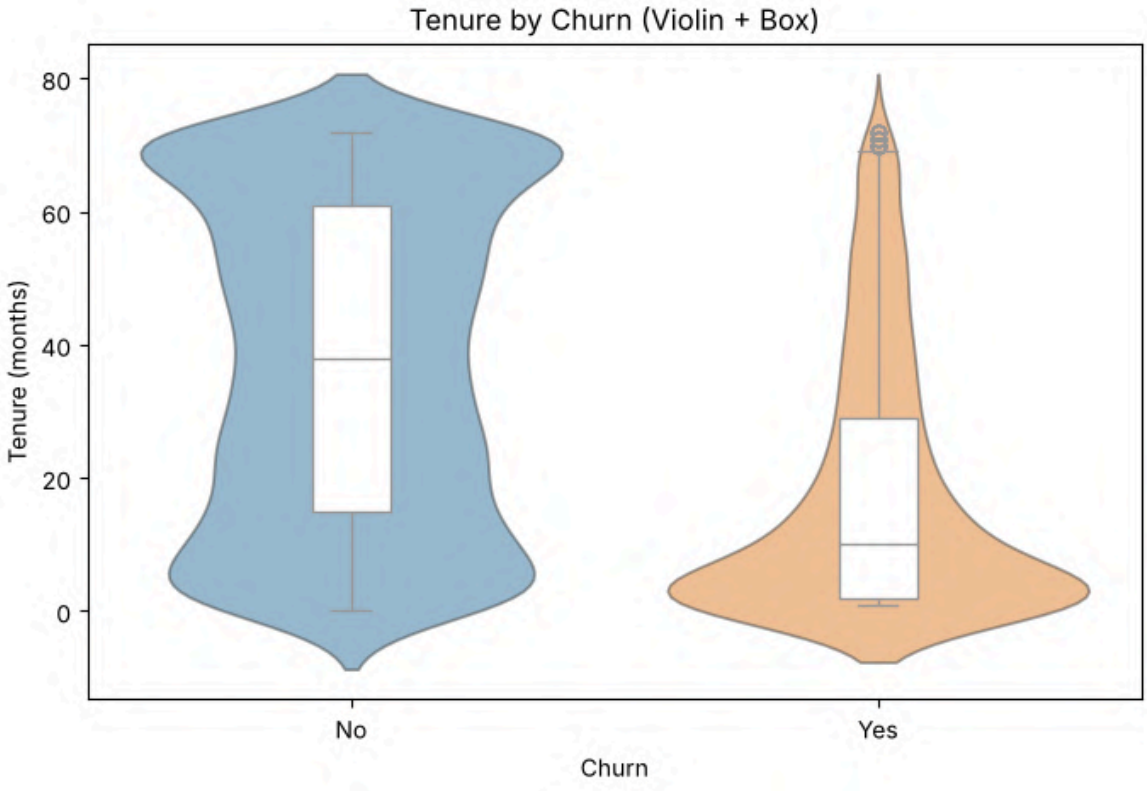

## Violin Plot - PDF #4 - Question #10

```
#Violin Plot - PDF #4 - Question #10
plt.figure(figsize=(8, 5))
sns.violinplot(data=df, x="InternetService", y="MonthlyCharges", hue="InternetService",
legend=False)
plt.title("Monthly Charges by Internet Service")
plt.xlabel("Internet Service")
plt.ylabel("Monthly Charges")
plt.show()
```




## Violin + Box Plot Combined - PDF #4 - Question #11

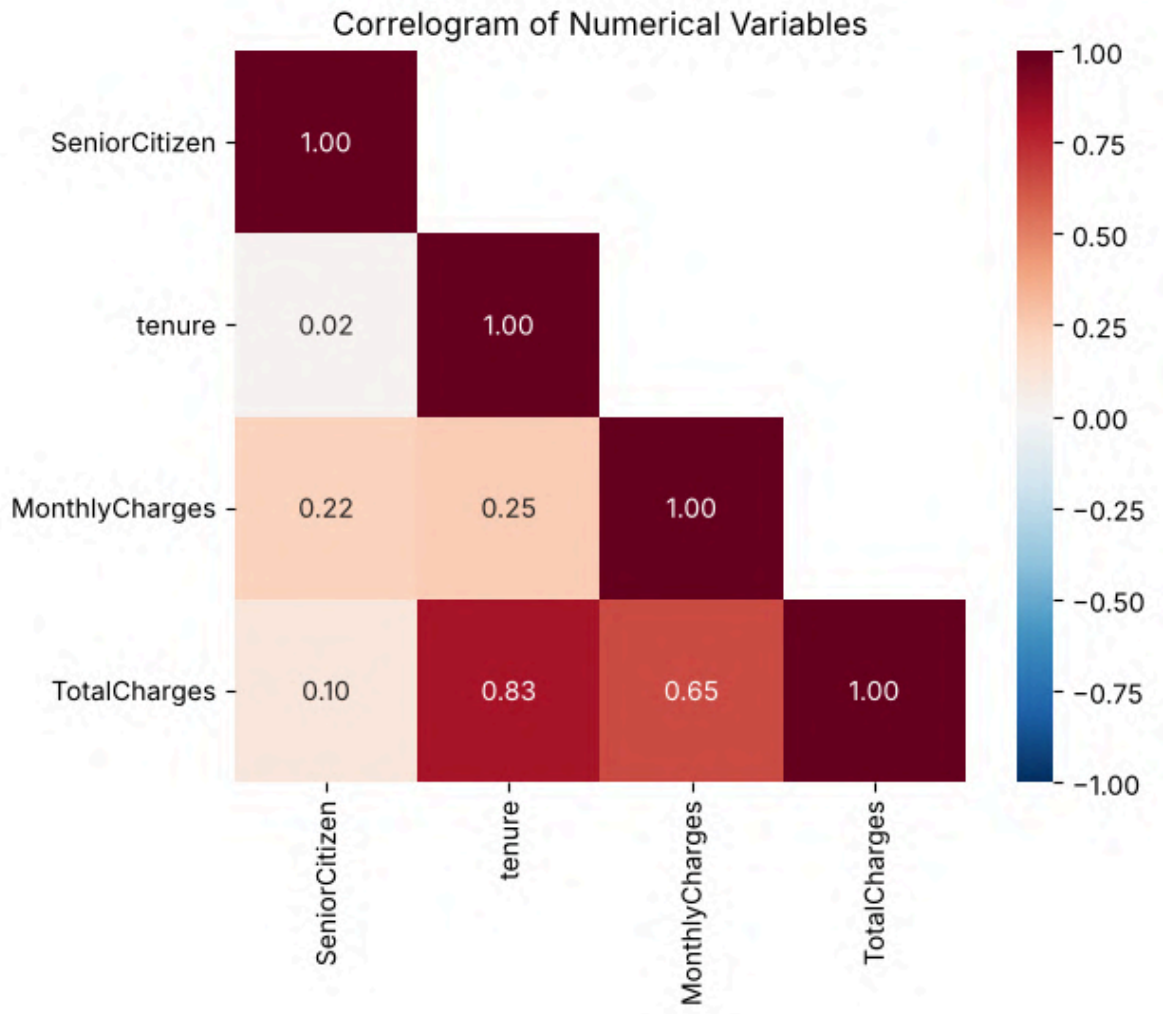
```
#Violin + Box Plot Combined - PDF #4 - Question #11
plt.figure(figsize=(8, 5))
sns.violinplot(data=df, x="Churn", y="tenure", hue="Churn", inner=None, legend=False, alpha=0.5)
sns.boxplot(data=df, x="Churn", y="tenure", width=0.15, color="white")
plt.title("Tenure by Churn (Violin + Box)")
plt.xlabel("Churn")
plt.ylabel("Tenure (months)")
plt.show()
```



## Correlation Matrix + Correlogram - PDF #4 - Question #12

```
#Correlation Matrix + Correlogram - PDF #4 - Question #12
corr = num.corr()
print(corr)
mask = np.triu(np.ones_like(corr, dtype=bool), k=1)
plt.figure(figsize=(7, 5))
sns.heatmap(corr, mask=mask, annot=True, cmap="RdBu_r", vmin=-1, vmax=1, square=True, fmt=".2f")
plt.title("Correlogram of Numerical Variables")
plt.show()
```

```
                SeniorCitizen    tenure  MonthlyCharges  TotalCharges
SeniorCitizen        1.000000  0.016547        0.220200      0.102652
tenure               0.016547  1.000000        0.247916      0.825433
MonthlyCharges       0.220200  0.247916        1.000000      0.650558
TotalCharges         0.102652  0.825433        0.650558      1.000000
```




## Titles, Labels, Legends - PDF #4 - Question #13

```
#Titles, Labels, Legends - PDF #4 - Question #13
# applied to every plot above
```

# PDF 5 - Multivariate Analysis

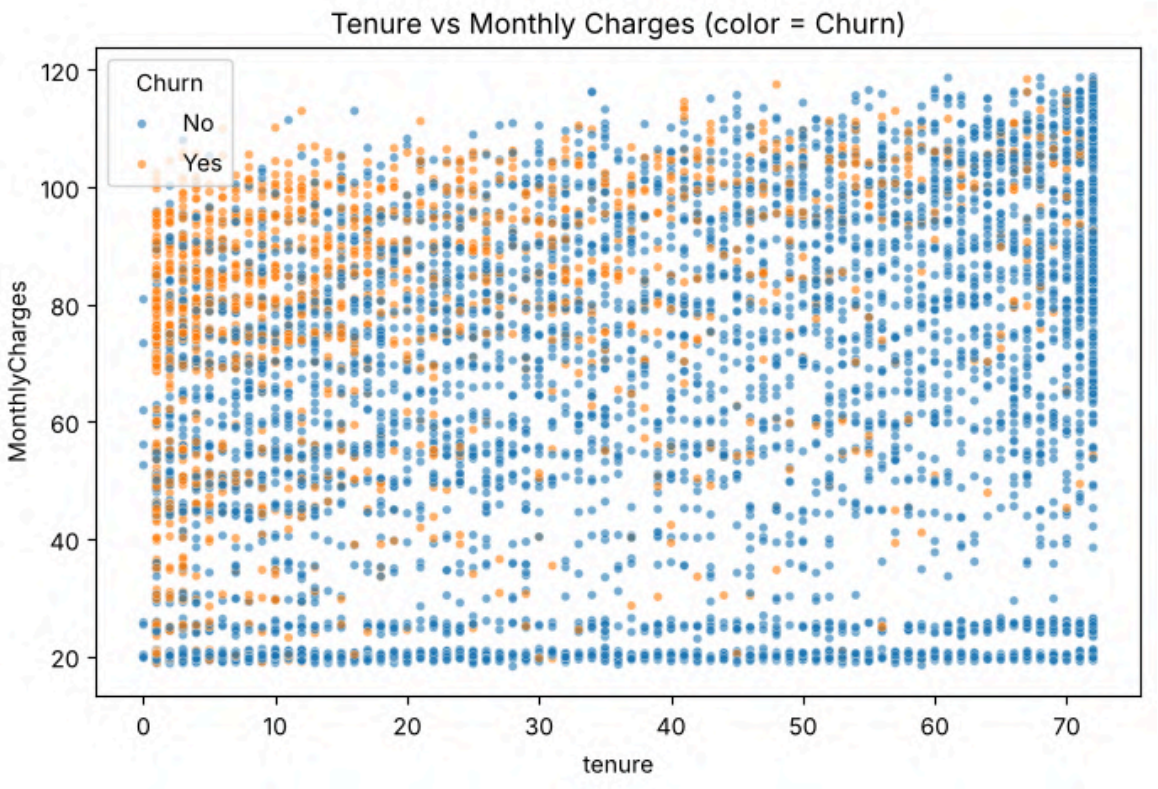
## Load Dataset + Identify Variable Types - PDF #5 - Question #1

```
#Load Dataset + Identify Variable Types - PDF #5 - Question #1
df = load_clean()
num_cols = df.select_dtypes(include="number").columns.tolist()
cat_cols = df.select_dtypes(exclude="number").columns.tolist()
print("Numerical:", num_cols)
print("Categorical:", cat_cols)
```

```
Numerical: ['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges']
Categorical: ['customerID', 'gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetServic
e', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Cont
ract', 'PaperlessBilling', 'PaymentMethod', 'Churn']
```

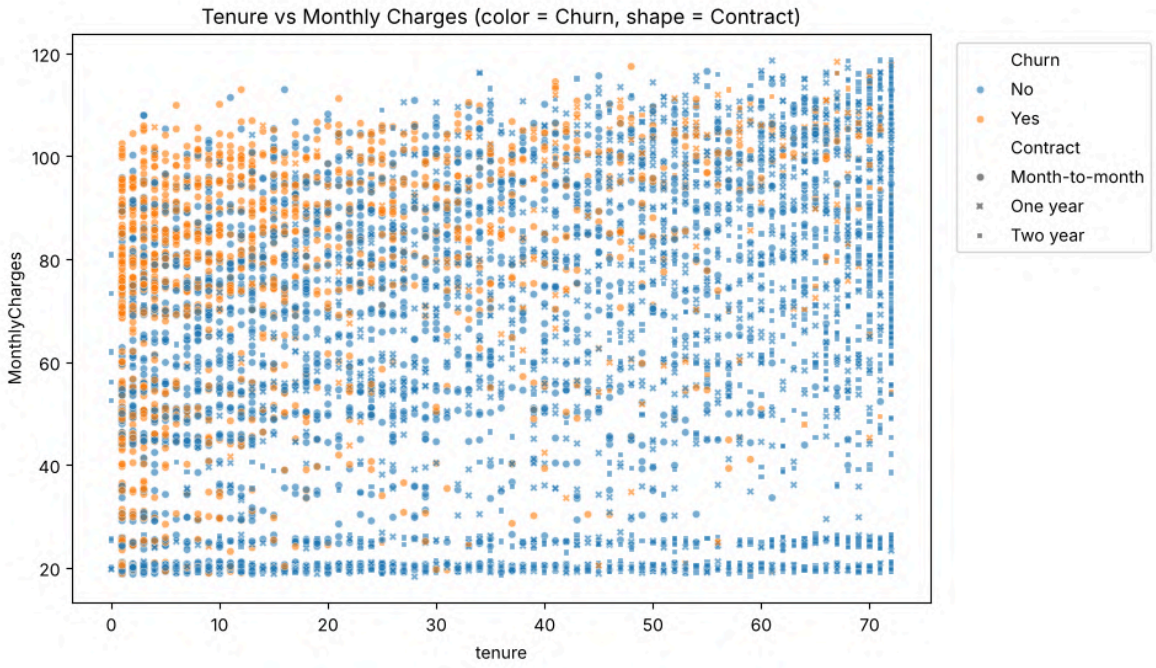
## Scatter - 3 Variables (color) - PDF #5 - Question #2

```
#Scatter - 3 Variables (color) - PDF #5 - Question #2
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="tenure", y="MonthlyCharges", hue="Churn", alpha=0.6, s=15)
plt.title("Tenure vs Monthly Charges (color = Churn)")
plt.show()
```



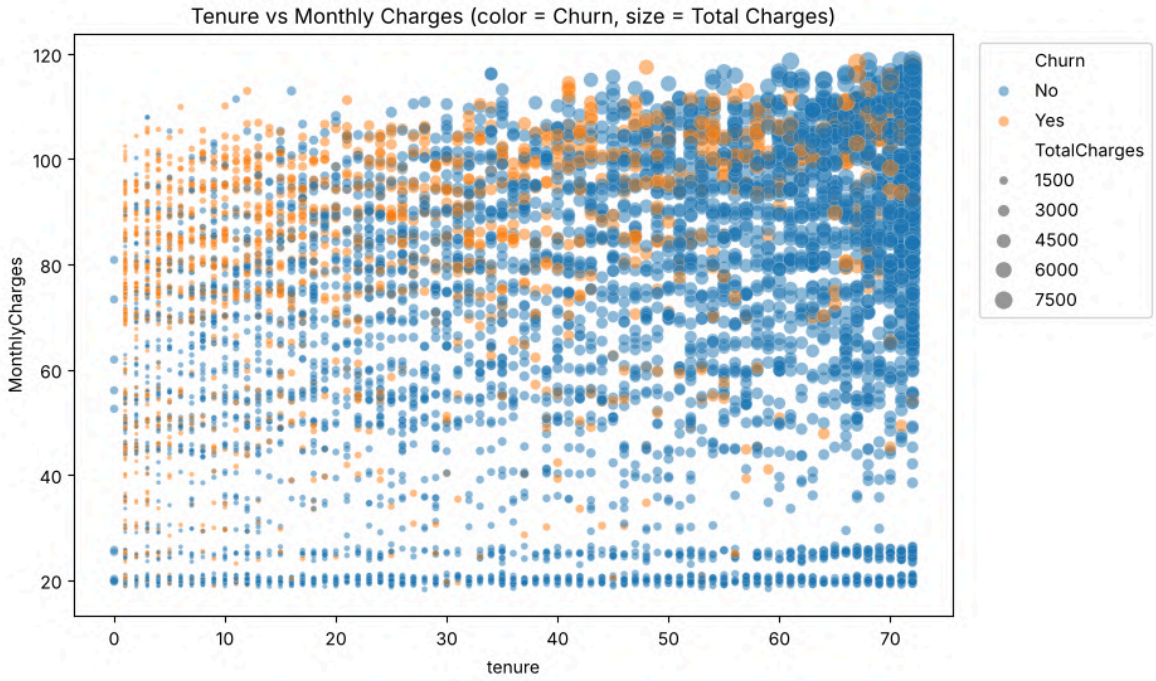

## Scatter - 4 Variables (color + shape) - PDF #5 - Question #3

```
#Scatter - 4 Variables (color + shape) - PDF #5 - Question #3
plt.figure(figsize=(9, 6))
sns.scatterplot(data=df, x="tenure", y="MonthlyCharges", hue="Churn", style="Contract", alpha=0.6,
s=20)
plt.title("Tenure vs Monthly Charges (color = Churn, shape = Contract)")
plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left")
plt.show()
```




## Scatter - 4 Variables (color + size) - PDF #5 - Question #4

```
#Scatter - 4 Variables (color + size) - PDF #5 - Question #4
plt.figure(figsize=(9, 6))
sns.scatterplot(data=df, x="tenure", y="MonthlyCharges", hue="Churn", size="TotalCharges", sizes=
(5, 120), alpha=0.5)
plt.title("Tenure vs Monthly Charges (color = Churn, size = Total Charges)")
plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left")
plt.show()
```

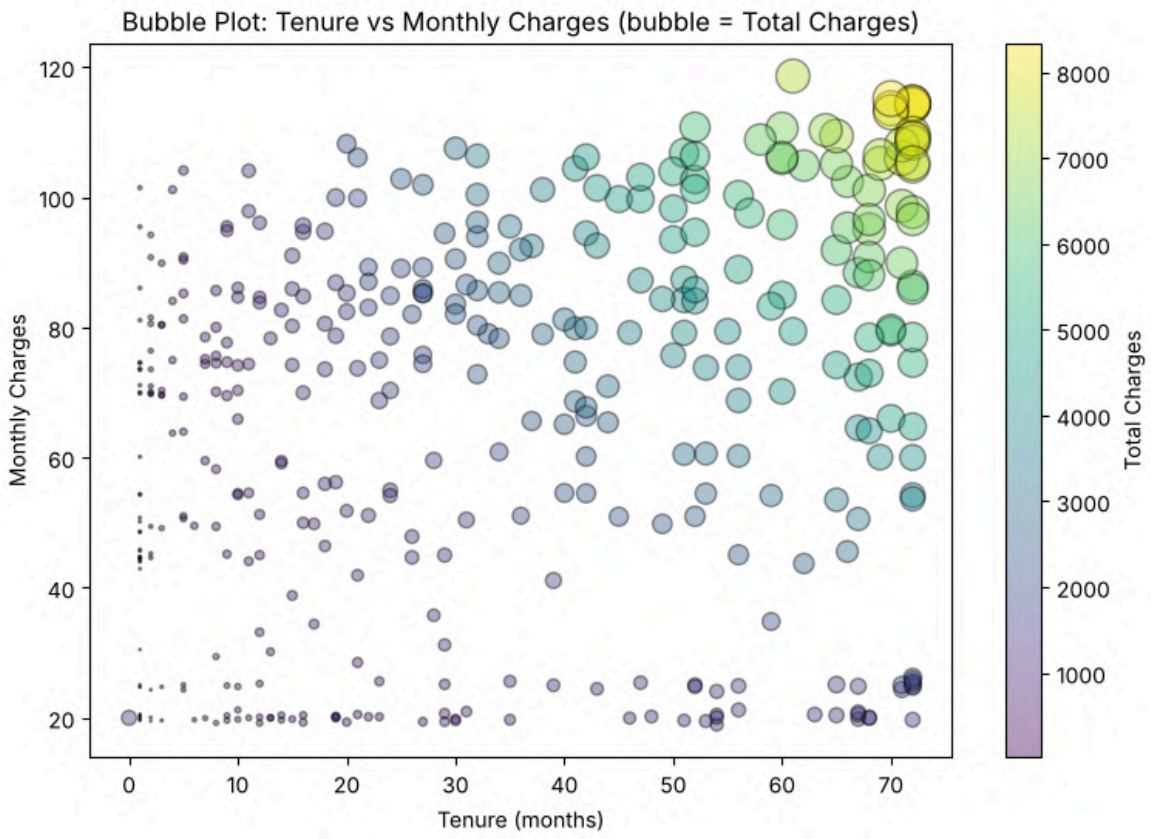


# Bubble Plot - PDF #5 - Question #5

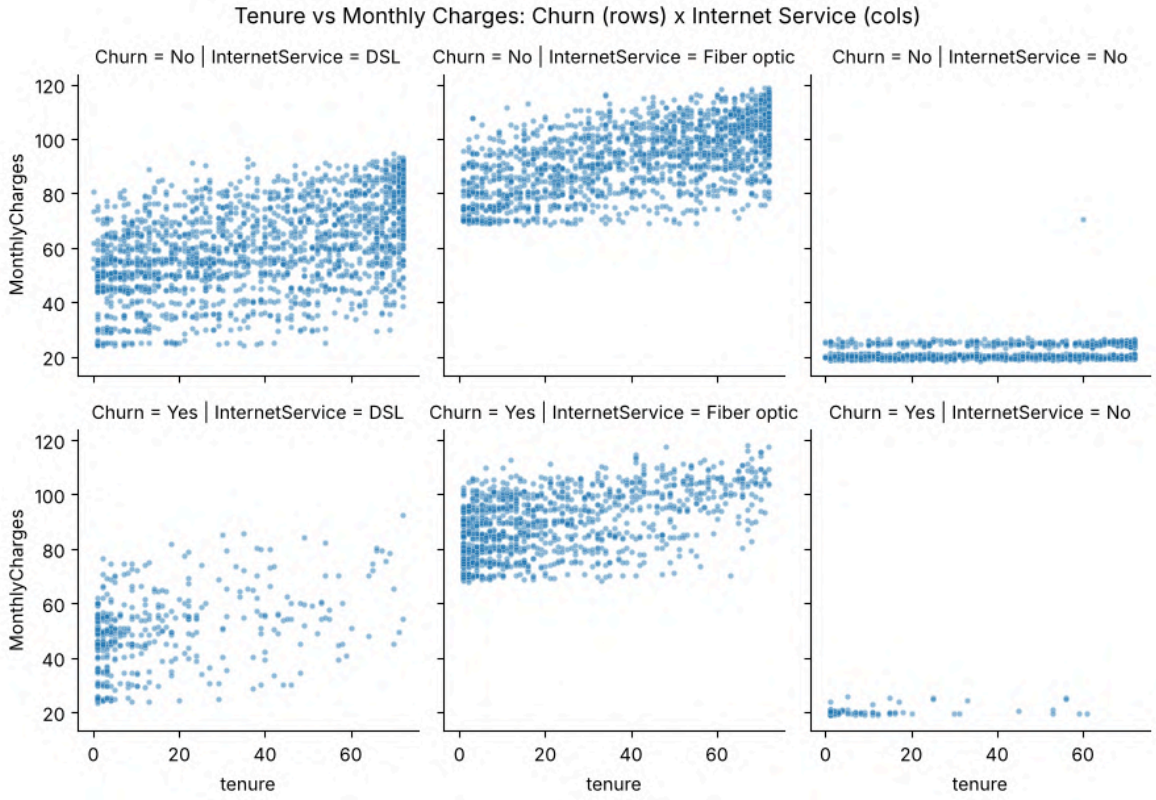
```
#Bubble Plot - PDF #5 - Question #5
samp = df.sample(400, random_state=1)
plt.figure(figsize=(9, 6))
plt.scatter(samp["tenure"], samp["MonthlyCharges"], s=samp["TotalCharges"] / 30, alpha=0.4,
            c=samp["TotalCharges"], cmap="viridis", edgecolors="black")
plt.colorbar(label="Total Charges")
plt.title("Bubble Plot: Tenure vs Monthly Charges (bubble = Total Charges)")
plt.xlabel("Tenure (months)")
plt.ylabel("Monthly Charges")
plt.show()

#facet_wrap Subplots - PDF #5 - Question #6
g = sns.relplot(data=df, x="tenure", y="MonthlyCharges", col="PaymentMethod", col_wrap=2,
hue="Churn", s=10, alpha=0.5, height=3.5)
g.figure.suptitle("Tenure vs Monthly Charges by Payment Method", y=1.03)
plt.show()

#facet_grid Subplots - PDF #5 - Question #7
g = sns.relplot(data=df, x="tenure", y="MonthlyCharges", row="Churn", col="InternetService", s=10,
alpha=0.5, height=3)
g.figure.suptitle("Tenure vs Monthly Charges: Churn (rows) x Internet Service (cols)", y=1.03)
plt.show()
```

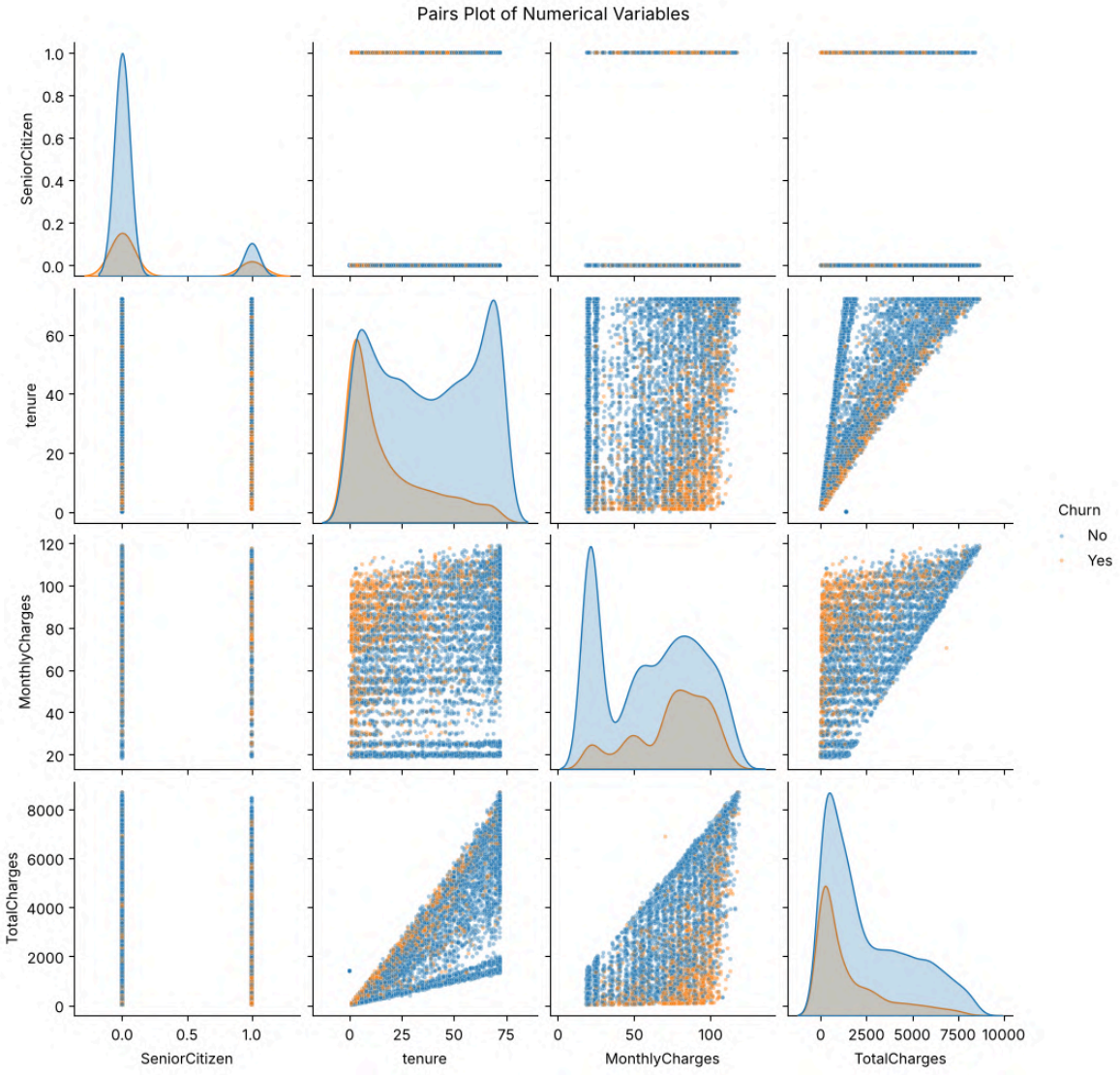





## Pairs Plot - PDF #5 - Question #8

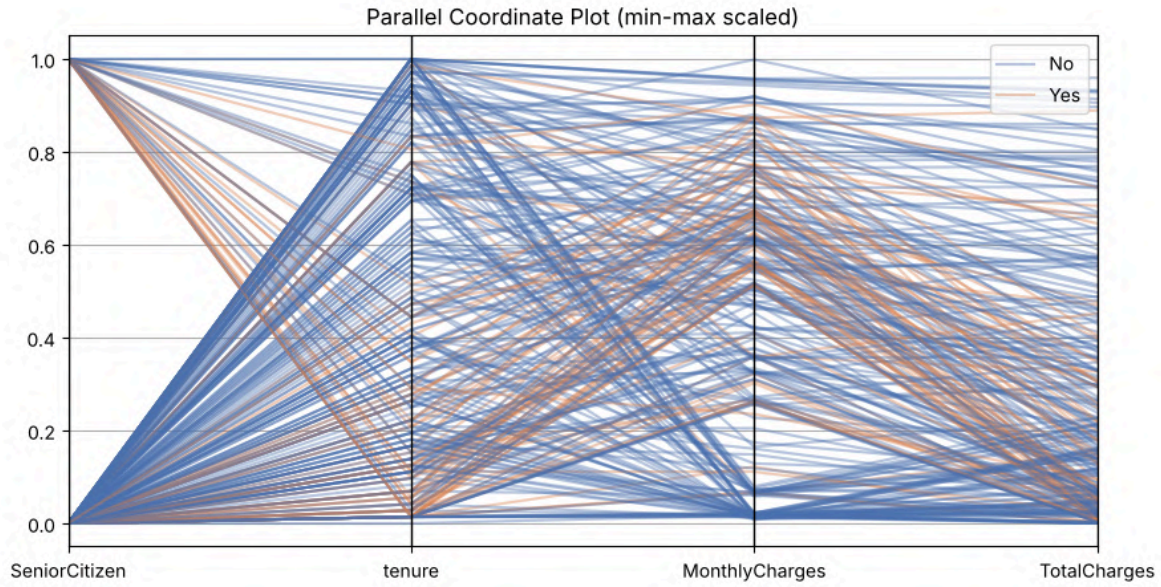
```
#Pairs Plot - PDF #5 - Question #8
pair_cols = ["SeniorCitizen", "tenure", "MonthlyCharges", "TotalCharges"]
sns.pairplot(df[pair_cols + ["Churn"]], hue="Churn", plot_kws={"s": 8, "alpha": 0.4})
plt.suptitle("Pairs Plot of Numerical Variables", y=1.02)
plt.show()
print(df[pair_cols].corr())
```




| | SeniorCitizen | tenure | MonthlyCharges | TotalCharges |
|---|---|---|---|---|
| SeniorCitizen | 1.000000 | 0.016547 | 0.220200 | 0.102652 |
| tenure | 0.016547 | 1.000000 | 0.247916 | 0.825433 |
| MonthlyCharges | 0.220200 | 0.247916 | 1.000000 | 0.650558 |
| TotalCharges | 0.102652 | 0.825433 | 0.650558 | 1.000000 |

## Parallel Coordinate Plot - PDF #5 - Question #9

```
#Parallel Coordinate Plot - PDF #5 - Question #9
pc = df[pair_cols].apply(lambda x: (x - x.min()) / (x.max() - x.min()))
pc["Churn"] = df["Churn"]
plt.figure(figsize=(10, 5))
parallel_coordinates(pc.sample(300, random_state=1), "Churn", color=["#4C72B0", "#DD8452"],
alpha=0.4)
plt.title("Parallel Coordinate Plot (min-max scaled)")
plt.show()
```


Parallel Coordinate Plot (min-max scaled)
No
Yes
1.0
0.8
0.6
0.4
0.2
0.0
SeniorCitizen
tenure
MonthlyCharges
TotalCharges

# PDF 6 - Time Series Analysis

## Load Dataset (series = avg MonthlyCharges by tenure month) - PDF #6 - Question #1

```
#Load Dataset (series = avg MonthlyCharges by tenure month) - PDF #6 - Question #1
df = load_clean()
s = df[df["tenure"] > 0].groupby("tenure")["MonthlyCharges"].mean()
ts = pd.Series(s.values, index=pd.period_range("2019-01", periods=len(s), freq="M"),
name="AvgMonthlyCharges")
```

## First 5 and Last 5 Observations - PDF #6 - Question #2

```
#First 5 and Last 5 Observations - PDF #6 - Question #2
print(ts.head(5))
print(ts.tail(5))
```

```
2019-01    50.485808
2019-02    57.206303
2019-03    58.015000
2019-04    57.459943
2019-05    61.003759
Freq: M, Name: AvgMonthlyCharges, dtype: float64
2024-08    73.321000
2024-09    70.823158
2024-10    76.378992
2024-11    73.735588
2024-12    80.695856
Freq: M, Name: AvgMonthlyCharges, dtype: float64
```

## Start and End - PDF #6 - Question #3

```
#Start and End - PDF #6 - Question #3
print("Start:", ts.index[0], "End:", ts.index[-1])
```

```
Start: 2019-01 End: 2024-12
```

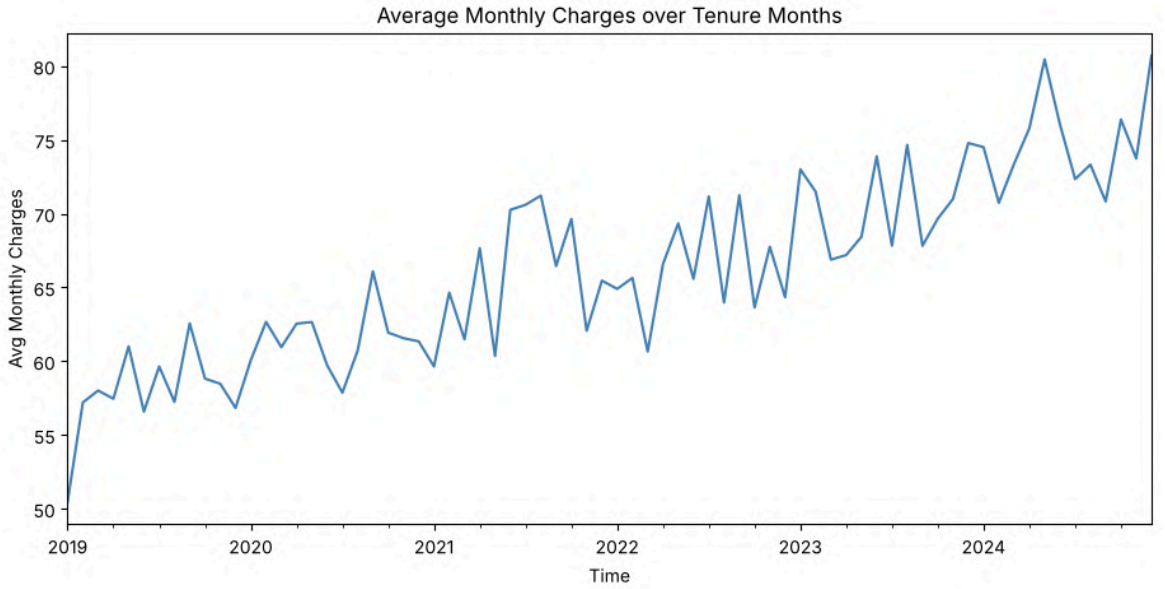
## Frequency and Class - PDF #6 - Question #4

```
#Frequency and Class - PDF #6 - Question #4
print("Frequency:", ts.index.freqstr, "(12 per year)")
print("Class:", type(ts), "| Length:", len(ts))
```

```
Frequency: M (12 per year)
Class: <class 'pandas.Series'> | Length: 72
```

## Plot Trend - PDF #6 - Question #5

```
#Plot Trend - PDF #6 - Question #5
plt.figure(figsize=(11, 5))
ts.plot(color="steelblue")
plt.title("Average Monthly Charges over Tenure Months")
plt.xlabel("Time")
plt.ylabel("Avg Monthly Charges")
plt.show()
```



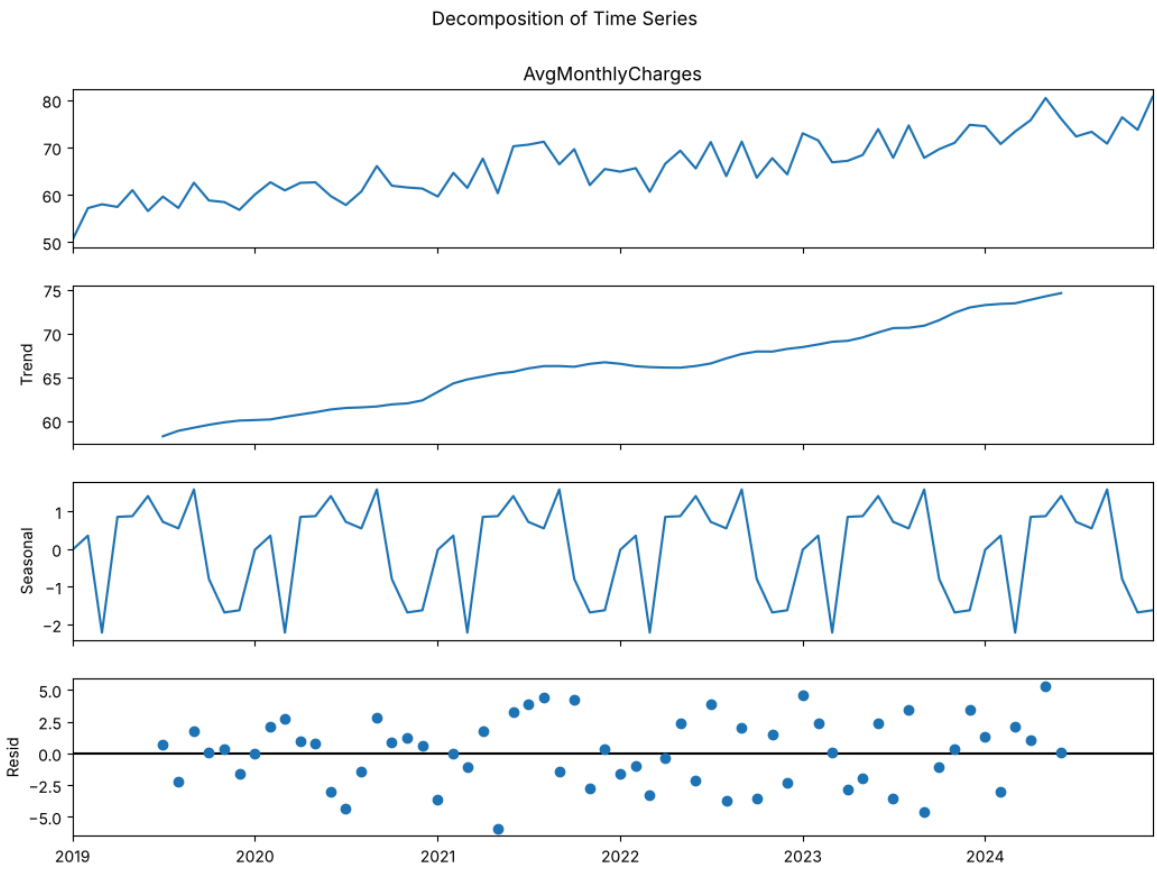

## Decompose - PDF #6 - Question #6

```
#Decompose - PDF #6 - Question #6
decomp = seasonal_decompose(ts.to_timestamp(), model="additive", period=12)
print(decomp.trend.dropna().head())
print(decomp.seasonal.head(12))
```

```
2019-07-01    58.261511
2019-08-01    58.888358
2019-09-01    59.238277
2019-10-01    59.572621
2019-11-01    59.853333
Freq: MS, Name: trend, dtype: float64
2019-01-01   -0.018933
2019-02-01    0.354226
2019-03-01   -2.212974
2019-04-01    0.852390
2019-05-01    0.871557
2019-06-01    1.403521
2019-07-01    0.719835
2019-08-01    0.547577
2019-09-01    1.579536
2019-10-01   -0.794108
2019-11-01   -1.680060
2019-12-01   -1.622567
Freq: MS, Name: seasonal, dtype: float64
```

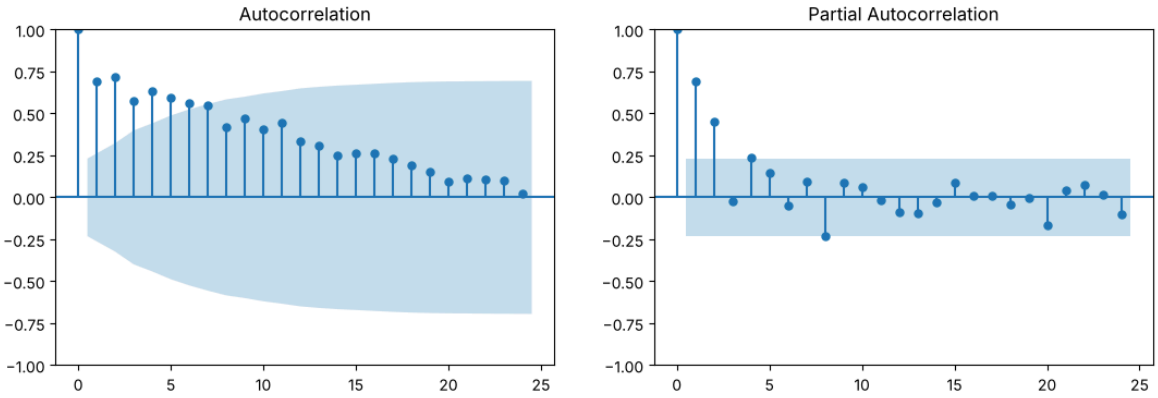
## Plot Decomposed Components - PDF #6 - Question #7

```
#Plot Decomposed Components - PDF #6 - Question #7
fig = decomp.plot()
fig.set_size_inches(11, 8)
fig.suptitle("Decomposition of Time Series", y=1.01)
plt.show()
```




## ACF and PACF - PDF #6 - Question #8

```
#ACF and PACF - PDF #6 - Question #8
fig, axes = plt.subplots(1, 2, figsize=(13, 4))
plot_acf(ts.values, lags=24, ax=axes[0])
plot_pacf(ts.values, lags=24, ax=axes[1], method="ywm")
plt.show()
```




## Fit ARIMA - PDF #6 - Question #9

```
#Fit ARIMA - PDF #6 - Question #9
model = ARIMA(ts, order=(1, 1, 1)).fit()
```

## ARIMA Summary - PDF #6 - Question #10

```
#ARIMA Summary - PDF #6 - Question #10
print(model.summary())
print("AIC:", model.aic, "| BIC:", model.bic)
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:      AvgMonthlyCharges   No. Observations:                   72
Model:                 ARIMA(1, 1, 1)   Log Likelihood                -188.754
Date:                Fri, 16 Oct 2026   AIC                            383.507
Time:                        15:49:54   BIC                            390.295
Sample:                  01-31-2019   HQIC                           386.207
                       - 12-31-2024
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.1844      0.178     -1.036      0.300      -0.533       0.165
ma.L1         -0.5926      0.157     -3.774      0.000      -0.900      -0.285
sigma2        11.8178      2.540      4.654      0.000       6.840      16.795
===================================================================================
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):                 2.92
Prob(Q):                              0.92   Prob(JB):                         0.23
Heteroskedasticity (H):               1.72   Skew:                             0.11
Prob(H) (two-sided):                  0.19   Kurtosis:                         2.03
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
AIC: 383.5074127708557 | BIC: 390.2954524019796
```

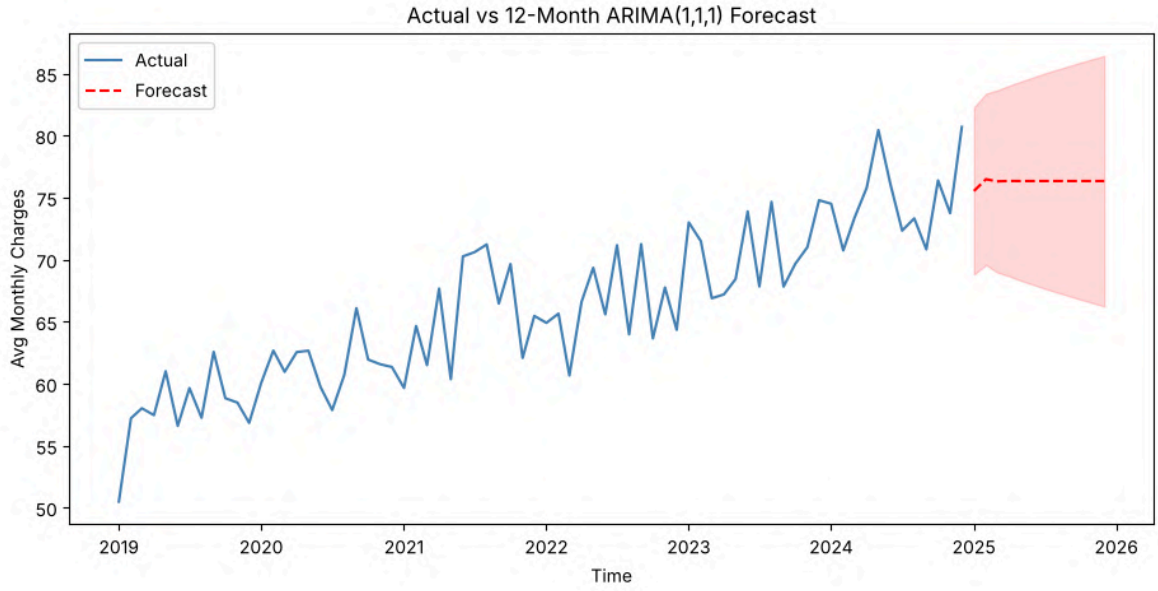
## Forecast 12 Months - PDF #6 - Question #11

```
#Forecast 12 Months - PDF #6 - Question #11
fc = model.get_forecast(steps=12)
fc_mean = fc.predicted_mean
fc_ci = fc.conf_int()
print(pd.DataFrame({"Forecast": fc_mean, "Lower": fc_ci.iloc[:, 0], "Upper": fc_ci.iloc[:, 1]}))
```

```
            Forecast      Lower      Upper
2025-01  75.527524  68.789746  82.265301
2025-02  76.480380  69.577084  83.383676
2025-03  76.304707  68.973576  83.635839
2025-04  76.337095  68.656672  84.017518
2025-05  76.331124  68.307184  84.355065
2025-06  76.332225  67.980561  84.683888
2025-07  76.332022  67.664721  84.999323
2025-08  76.332059  67.360271  85.303848
2025-09  76.332053  67.065767  85.598338
2025-10  76.332054  66.780349  85.883759
2025-11  76.332054  66.503214  86.160894
2025-12  76.332054  66.233681  86.430426
```

## Plot Actual vs Forecast - PDF #6 - Question #12

```
#Plot Actual vs Forecast - PDF #6 - Question #12
plt.figure(figsize=(11, 5))
plt.plot(ts.index.to_timestamp(), ts.values, label="Actual", color="steelblue")
plt.plot(fc_mean.index.to_timestamp(), fc_mean.values, label="Forecast", color="red", linestyle="--
")
plt.fill_between(fc_mean.index.to_timestamp(), fc_ci.iloc[:, 0], fc_ci.iloc[:, 1], color="red",
alpha=0.15)
plt.title("Actual vs 12-Month ARIMA(1,1,1) Forecast")
plt.xlabel("Time")
plt.ylabel("Avg Monthly Charges")
plt.legend()
plt.show()
```


Actual vs 12-Month ARIMA(1,1,1) Forecast
Actual
Forecast
Avg Monthly Charges
85
80
75
70
65
60
55
50
2019
2020
2021
2022
2023
2024
2025
2026
Time

# PDF 7 - 1D, 2D and ND Analysis

## Load Dataset - PDF #7 - Question #1

```
#Load Dataset - PDF #7 - Question #1
df = load_clean()
```

## Summary Statistics - PDF #7 - Question #2

```
#Summary Statistics - PDF #7 - Question #2
x = df["MonthlyCharges"]
mean, median, mn, mx = x.mean(), x.median(), x.min(), x.max()
rng, var, sd = mx - mn, x.var(), x.std()
iqr = x.quantile(0.75) - x.quantile(0.25)
skew, kurt = stats.skew(x), stats.kurtosis(x)
```

## Store in Structured Data Frame - PDF #7 - Question #3

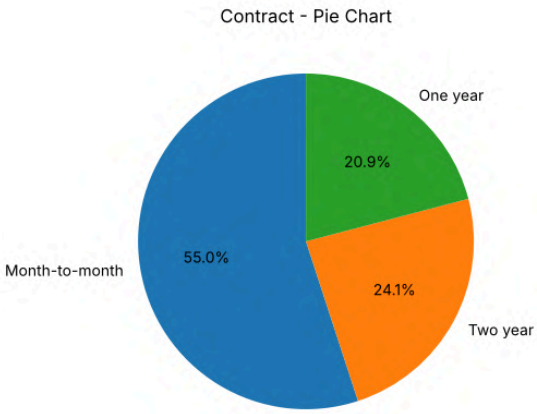
```
#Store in Structured Data Frame - PDF #7 - Question #3
summary = pd.DataFrame({"Statistic": ["Mean", "Median", "Minimum", "Maximum", "Range", "Variance",
"SD", "IQR", "Skewness", "Kurtosis"],
                        "Value": [mean, median, mn, mx, rng, var, sd, iqr, skew, kurt]}).round(4)
print(summary)
```

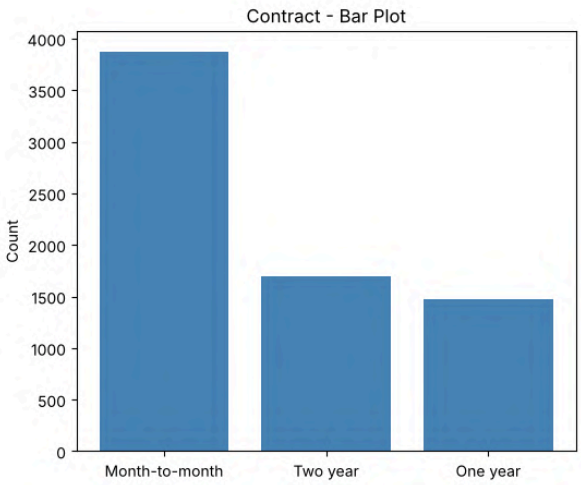
```
  Statistic     Value
0      Mean   64.7643
1    Median   70.3500
2   Minimum   18.2500
3   Maximum  118.7500
4     Range  100.5000
5  Variance  904.8783
6        SD   30.0812
7       IQR   54.3250
8  Skewness   -0.2209
9  Kurtosis   -1.2564
```

## Frequency & Relative Frequency + Pie + Bar - PDF #7 - Question #4

```
#Frequency & Relative Frequency + Pie + Bar - PDF #7 - Question #4
f = df["Contract"].value_counts()
print(pd.DataFrame({"Frequency": f, "Relative Frequency": (f / f.sum()).round(4)}))
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].pie(f.values, labels=f.index, autopct="%1.1f%%", startangle=90)
axes[0].set_title("Contract - Pie Chart")
axes[1].bar(f.index, f.values, color="steelblue")
axes[1].set_title("Contract - Bar Plot")
axes[1].set_ylabel("Count")
plt.show()
```

```
                Frequency  Relative Frequency
Contract
Month-to-month       3875              0.5502
Two year             1695              0.2407
One year             1473              0.2091
```

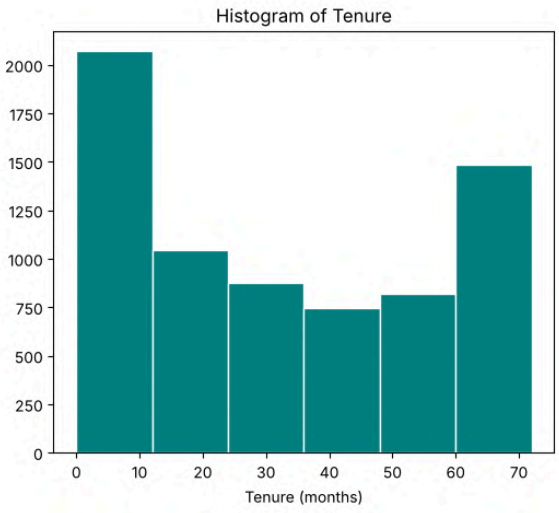


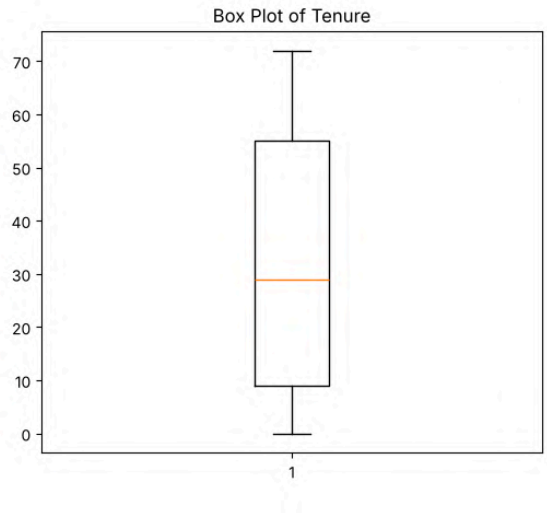




## Binned Frequency Table + Histogram + Box Plot - PDF #7 - Question #5

```
#Binned Frequency Table + Histogram + Box Plot - PDF #7 - Question #5
bins = pd.cut(df["tenure"], bins=[0, 12, 24, 36, 48, 60, 72], include_lowest=True)
print(bins.value_counts().sort_index())
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].hist(df["tenure"], bins=[0, 12, 24, 36, 48, 60, 72], color="teal", edgecolor="white")
axes[0].set_title("Histogram of Tenure")
axes[0].set_xlabel("Tenure (months)")
axes[1].boxplot(df["tenure"])
axes[1].set_title("Box Plot of Tenure")
plt.show()
```

```
tenure
(-0.001, 12.0]    2186
(12.0, 24.0]      1023
(24.0, 36.0]       833
(36.0, 48.0]       762
(48.0, 60.0]       832
(60.0, 72.0]      1407
Name: count, dtype: int64
```



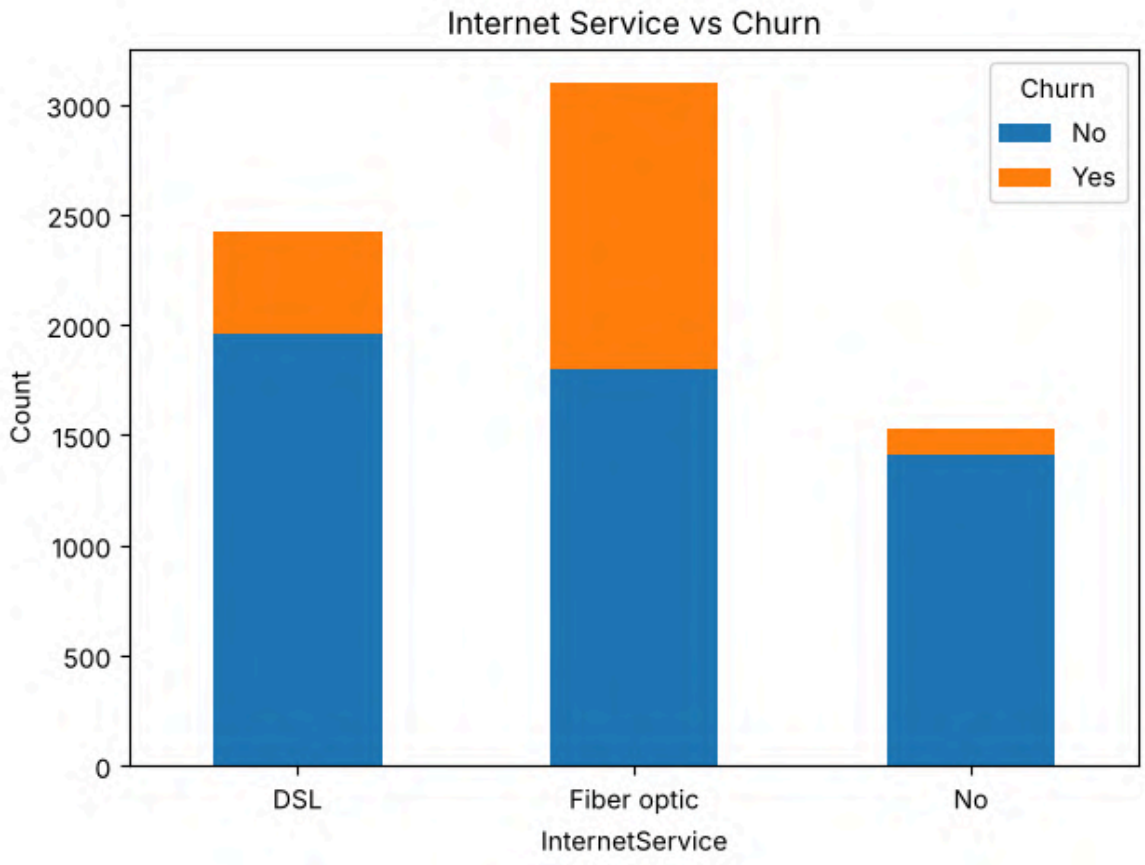

## Contingency Table + Row/Column % - PDF #7 - Question #6

```
#Contingency Table + Row/Column % - PDF #7 - Question #6
ct = pd.crosstab(df["InternetService"], df["Churn"])
print(ct)
print((ct.div(ct.sum(axis=1), axis=0) * 100).round(2))
print((ct.div(ct.sum(axis=0), axis=1) * 100).round(2))
```

```
Churn              No   Yes
InternetService
DSL              1963   458
Fiber optic      1799  1297
No               1413   113
Churn               No    Yes
InternetService
DSL              81.08  18.92
Fiber optic      58.11  41.89
No               92.60   7.40
Churn               No    Yes
InternetService
DSL              37.93  24.52
Fiber optic      34.76  69.43
No               27.30   6.05
```

## Stacked Bar Plot - PDF #7 - Question #7

```python
#Stacked Bar Plot - PDF #7 - Question #7
ct.plot(kind="bar", stacked=True, figsize=(7, 5))
plt.title("Internet Service vs Churn")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```



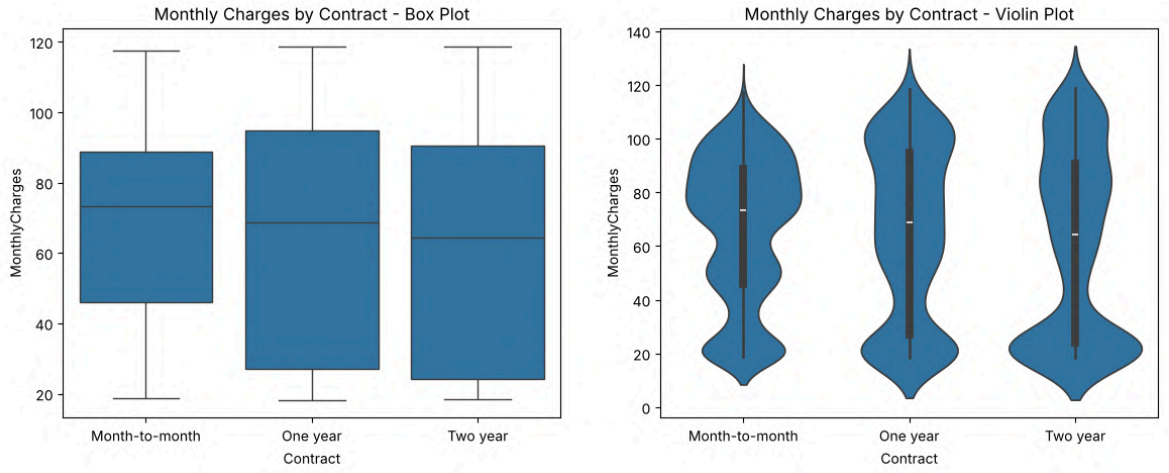

## Grouped Mean, Median, SD (aggregate) - PDF #7 - Question #8

```python
#Grouped Mean, Median, SD (aggregate) - PDF #7 - Question #8
print(df.groupby("Contract")["MonthlyCharges"].agg(["mean", "median", "std"]).round(3))
```

```
                 mean  median     std
Contract
Month-to-month  66.400   73.25  26.927
One year        65.023   68.75  31.822
Two year        60.800   64.35  34.665
```

## Side-by-Side Box + Violin - PDF #7 - Question #9

```
#Side-by-Side Box + Violin - PDF #7 - Question #9
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.boxplot(data=df, x="Contract", y="MonthlyCharges", ax=axes[0])
axes[0].set_title("Monthly Charges by Contract - Box Plot")
sns.violinplot(data=df, x="Contract", y="MonthlyCharges", ax=axes[1])
axes[1].set_title("Monthly Charges by Contract - Violin Plot")
plt.show()
```

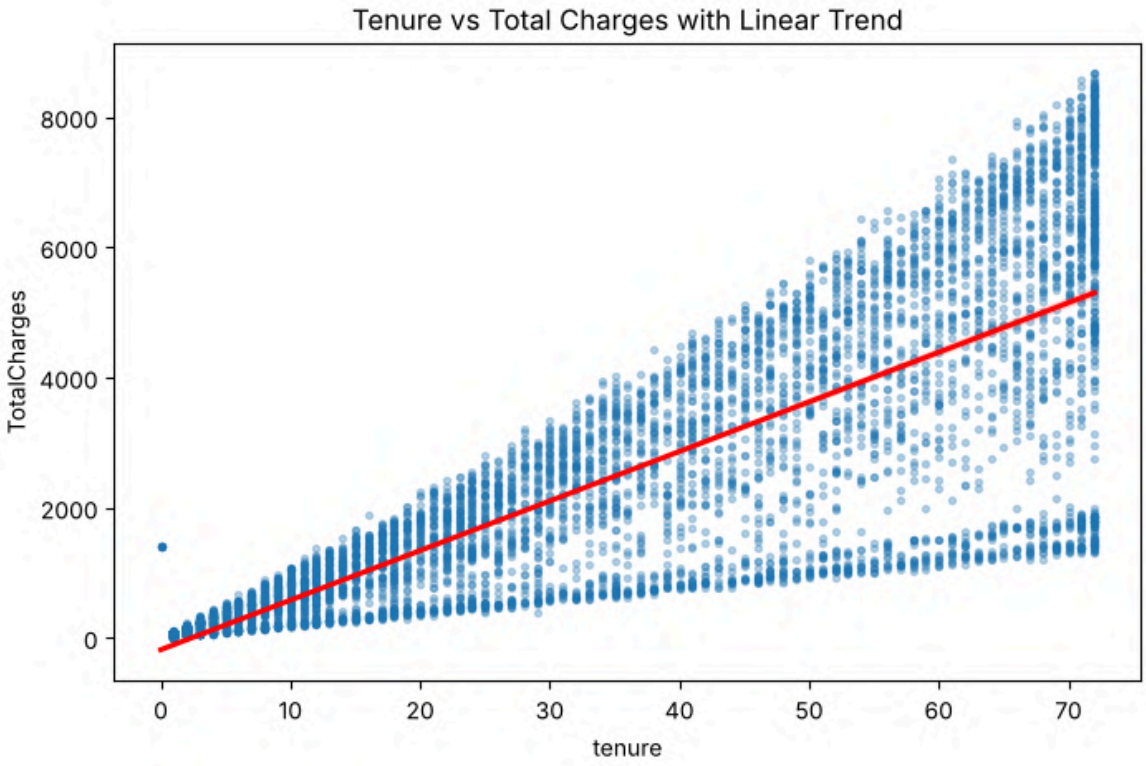



## Pearson & Spearman + Scatter with Trend Line - PDF #7 - Question #10

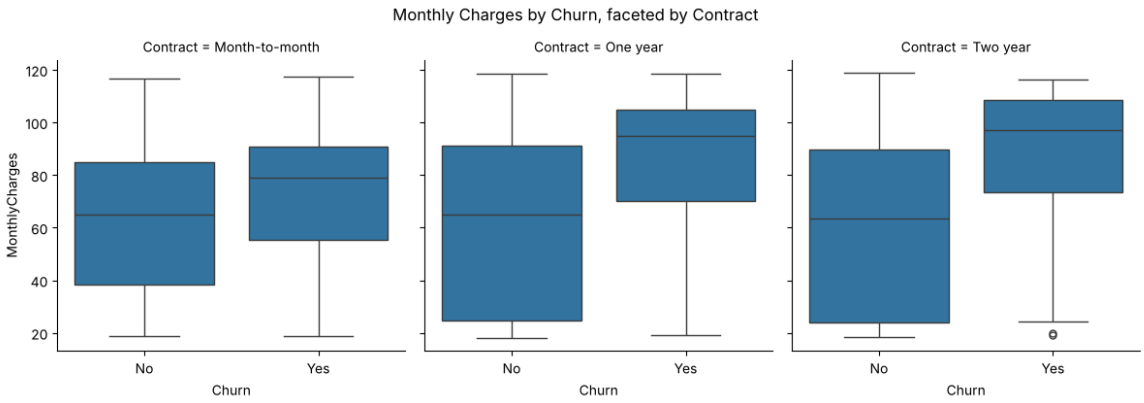
```
#Pearson & Spearman + Scatter with Trend Line - PDF #7 - Question #10
print("Pearson:", df["tenure"].corr(df["TotalCharges"], method="pearson"))
print("Spearman:", df["tenure"].corr(df["TotalCharges"], method="spearman"))
plt.figure(figsize=(8, 5))
sns.regplot(data=df, x="tenure", y="TotalCharges", scatter_kws={"s": 8, "alpha": 0.3}, line_kws=
{"color": "red"})
plt.title("Tenure vs Total Charges with Linear Trend")
plt.show()

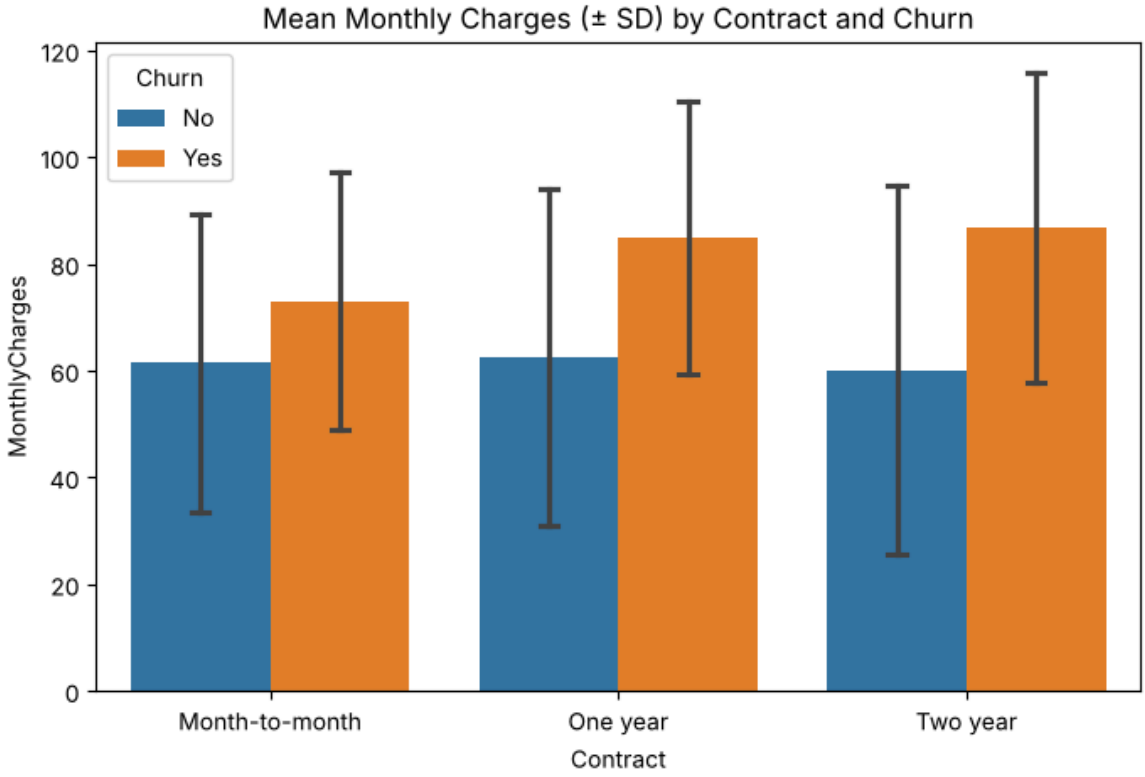
#2 Categorical + 1 Numerical Stats, Faceted Box, Grouped Bar w/ Error Bars - PDF #7 - Question #11
print(df.groupby(["Contract", "Churn"])["MonthlyCharges"].agg(["mean", "std"]).round(3))
g = sns.catplot(data=df, x="Churn", y="MonthlyCharges", col="Contract", kind="box", height=4)
g.figure.suptitle("Monthly Charges by Churn, faceted by Contract", y=1.05)
plt.show()
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x="Contract", y="MonthlyCharges", hue="Churn", errorbar="sd", capsize=0.1)
plt.title("Mean Monthly Charges (± SD) by Contract and Churn")
plt.show()
```

```
Pearson: 0.8254330756776308
Spearman: 0.8869421800059288
```




| Contract | Churn | mean | std |
|---|---|---|---|
| Month-to-month | No | 61.458 | 27.860 |
| | Yes | 73.036 | 24.082 |
| One year | No | 62.508 | 31.659 |
| | Yes | 84.827 | 25.606 |
| Two year | No | 60.043 | 34.534 |
| | Yes | 86.777 | 28.918 |






## Scatter Colored/Shaped by Category + Per-Category Correlation - PDF #7 - Question #12

```
#Scatter Colored/Shaped by Category + Per-Category Correlation - PDF #7 - Question #12
plt.figure(figsize=(9, 6))
sns.scatterplot(data=df, x="tenure", y="MonthlyCharges", hue="InternetService",
style="InternetService", s=15, alpha=0.6)
plt.title("Tenure vs Monthly Charges by Internet Service")
plt.show()
for grp, sub in df.groupby("InternetService"):
    print(grp)
    print(sub[["tenure", "MonthlyCharges"]].corr())
```

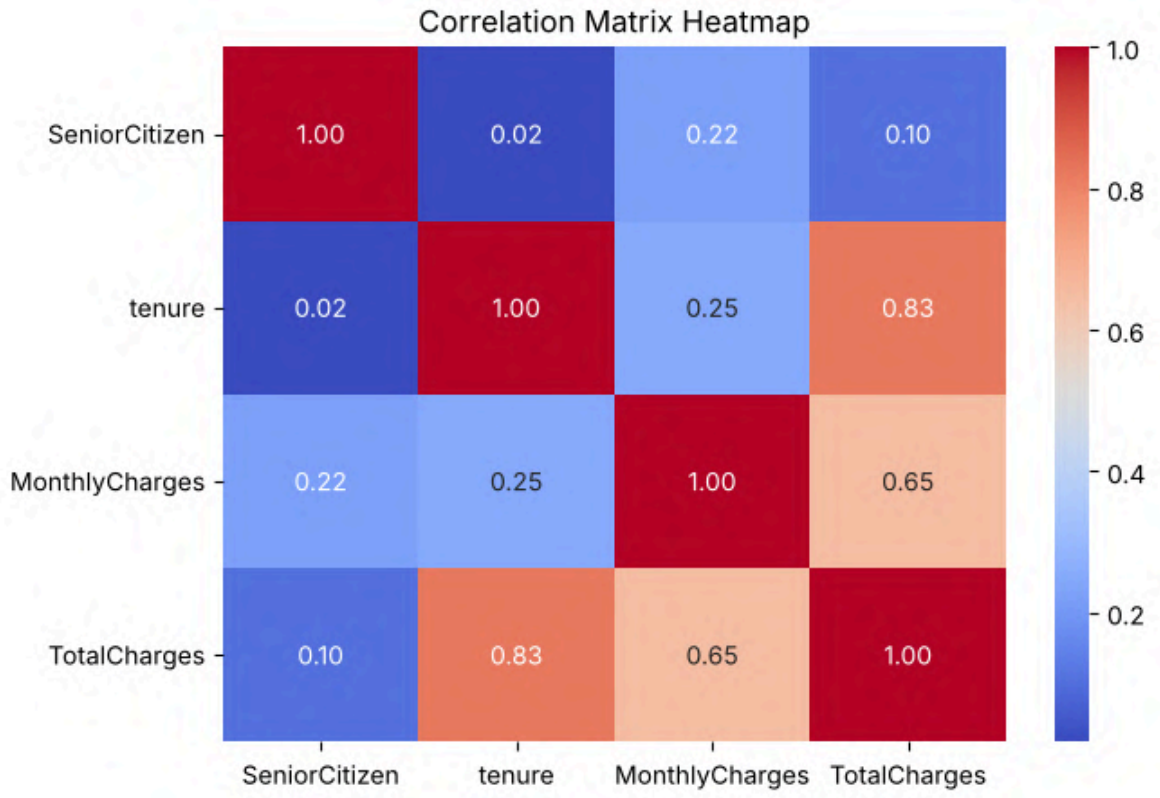



```
DSL
                 tenure  MonthlyCharges
tenure          1.000000        0.517156
MonthlyCharges  0.517156        1.000000
Fiber optic
                 tenure  MonthlyCharges
tenure          1.000000        0.623333
MonthlyCharges  0.623333        1.000000
No
                 tenure  MonthlyCharges
tenure          1.000000        0.312432
MonthlyCharges  0.312432        1.000000
```

## Correlation Matrix Heatmap (>=3 numeric) - PDF #7 - Question #13

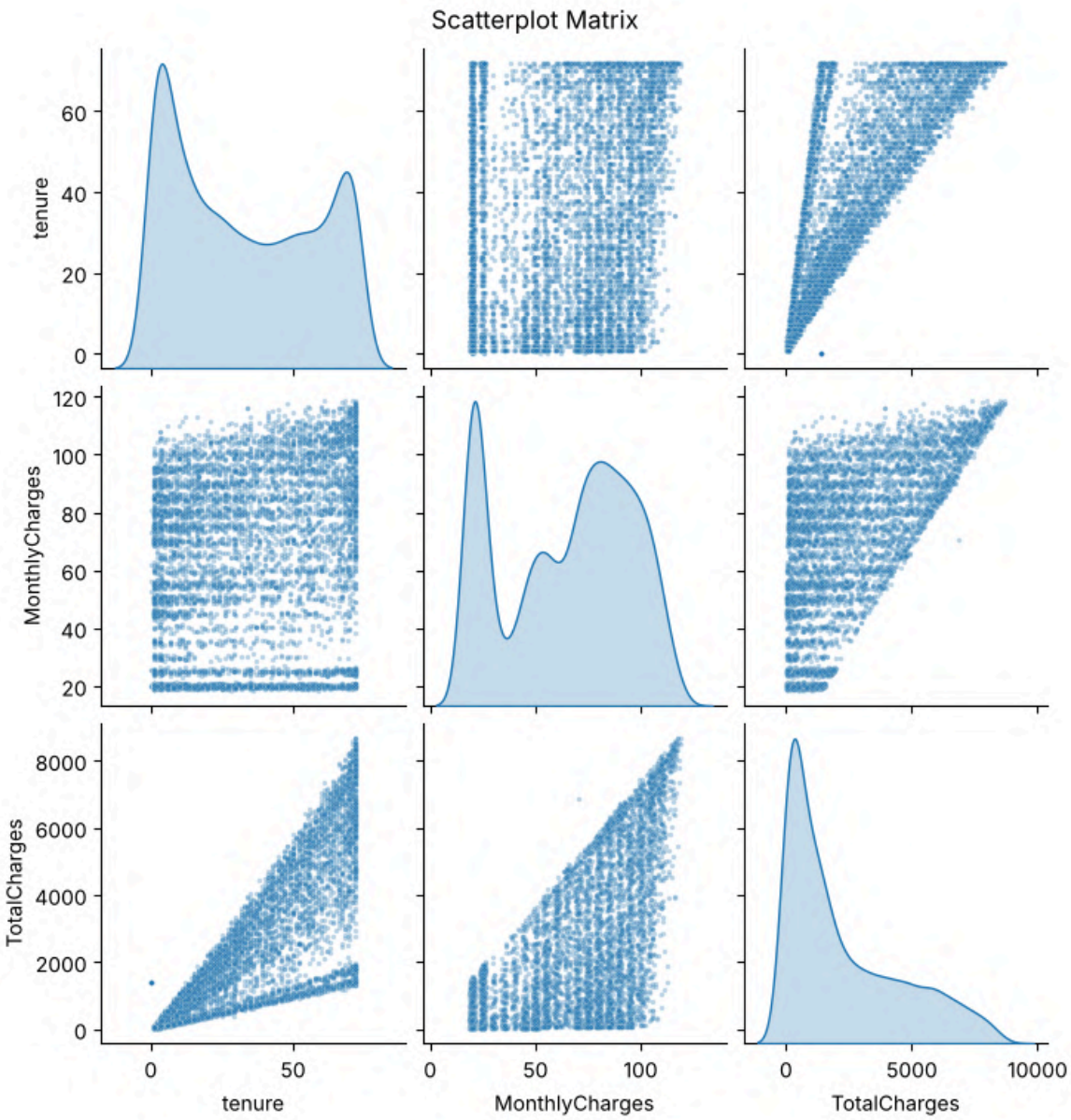
```python
#Correlation Matrix Heatmap (>=3 numeric) - PDF #7 - Question #13
cm = df[["SeniorCitizen", "tenure", "MonthlyCharges", "TotalCharges"]].corr()
print(cm)
plt.figure(figsize=(7, 5))
sns.heatmap(cm, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Matrix Heatmap")
plt.show()
```

```
                SeniorCitizen    tenure  MonthlyCharges  TotalCharges
SeniorCitizen        1.000000  0.016547        0.220200      0.102652
tenure               0.016547  1.000000        0.247916      0.825433
MonthlyCharges       0.220200  0.247916        1.000000      0.650558
TotalCharges         0.102652  0.825433        0.650558      1.000000
```




## Pairs Plot / Scatterplot Matrix - PDF #7 - Question #14

```
#Pairs Plot / Scatterplot Matrix - PDF #7 - Question #14
sns.pairplot(df[["tenure", "MonthlyCharges", "TotalCharges"]], diag_kind="kde", plot_kws={"s": 6,
"alpha": 0.3})
plt.suptitle("Scatterplot Matrix", y=1.02)
plt.show()
```


Scatterplot Matrix
tenure
MonthlyCharges
TotalCharges
0
20
40
60
20
40
60
80
100
120
0
2000
4000
6000
8000
0
50
0
50
100
0
5000
10000

# PDF 8 - K-Means Clustering

## Load Dataset - PDF #8 - K-Means - Question #1

```
#Load Dataset - PDF #8 - K-Means - Question #1
df = load_raw()
```

## View Dataset - PDF #8 - K-Means - Question #2

```
#View Dataset - PDF #8 - K-Means - Question #2
print(df.head())
df.info()
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
0  7590-VHVEG  Female            0.0     Yes         No     1.0           No  No phone service
1  5575-GNVDE    Male            0.0      No         No    34.0          Yes                No
2  3668-QPYBK    Male            0.0      No         No     2.0          Yes                No
3  7795-CFOCW    Male            0.0      No         No    45.0           No  No phone service
4  9237-HQITU  Female            0.0      No         No     2.0          Yes                No

  InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
0             DSL             No          Yes               No          No          No              No
1             DSL            Yes           No              Yes          No          No              No
2             DSL            Yes          Yes               No          No          No              No
3             DSL            Yes           No              Yes         Yes          No              No
4     Fiber optic             No           No               No          No          No              No

        Contract PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges Churn
0  Month-to-month              Yes           Electronic check           29.85         29.85    No
1        One year               No               Mailed check           56.95       1889.50    No
2  Month-to-month              Yes               Mailed check           53.85        108.15   Yes
3        One year               No  Bank transfer (automatic)           42.30       1840.75    No
4  Month-to-month              Yes           Electronic check           70.70        151.65   Yes
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7035 non-null   float64
 3   Partner           7038 non-null   str
 4   Dependents        7041 non-null   str
 5   tenure            7041 non-null   float64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7042 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7042 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7040 non-null   float64
 19  TotalCharges      7032 non-null   float64
 20  Churn             7040 non-null   str
dtypes: float64(4), str(17)
memory usage: 1.1 MB
```

## Check Missing Values - PDF #8 - K-Means - Question #3

```
#Check Missing Values - PDF #8 - K-Means - Question #3
print(df.isna().sum())
```

```
customerID          0
gender              0
SeniorCitizen       8
Partner             5
Dependents          2
tenure              2
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         1
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    1
PaymentMethod       0
MonthlyCharges      3
TotalCharges       11
Churn               3
dtype: int64
```

## Dependent Variable Distribution (table) - PDF #8 - K-Means - Question #4

```
#Dependent Variable Distribution (table) - PDF #8 - K-Means - Question #4
print(df["Churn"].value_counts(dropna=False))
```

```
Churn
No     5172
Yes    1868
NaN       3
Name: count, dtype: int64
```

## Select Numerical Features - PDF #8 - K-Means - Question #5

```
#Select Numerical Features - PDF #8 - K-Means - Question #5
km_data = df.select_dtypes(include="number").dropna()
print(km_data.head())
```

```
   SeniorCitizen  tenure  MonthlyCharges  TotalCharges
0            0.0     1.0           29.85         29.85
1            0.0    34.0           56.95       1889.50
2            0.0     2.0           53.85        108.15
3            0.0    45.0           42.30       1840.75
4            0.0     2.0           70.70        151.65
```

## Standardize (scale) - PDF #8 - K-Means - Question #6

```
#Standardize (scale) - PDF #8 - K-Means - Question #6
km_scaled = r_scale(km_data)
print(km_scaled.head())
```

```
   SeniorCitizen    tenure  MonthlyCharges  TotalCharges
0      -0.440782 -1.279515       -1.162064     -0.994168
1      -0.440782  0.064720       -0.261030     -0.173506
2      -0.440782 -1.238781       -0.364101     -0.959614
3      -0.440782  0.512799       -0.748120     -0.195019
4      -0.440782 -1.238781        0.196136     -0.940418
```

## K-Means with k=2 - PDF #8 - K-Means - Question #7

```
#K-Means with k=2 - PDF #8 - K-Means - Question #7
km2 = KMeans(n_clusters=2, n_init=25, random_state=42).fit(km_scaled)
```

## Display K-Means Results - PDF #8 - K-Means - Question #8

```
#Display K-Means Results - PDF #8 - K-Means - Question #8
print("Cluster sizes:", np.bincount(km2.labels_))
print("Cluster labels (first 20):", km2.labels_[:20])
tss = ((km_scaled - km_scaled.mean()) ** 2).sum().sum()
print("Total WCSS:", km2.inertia_, "| Between_SS / Total_SS =", round((tss - km2.inertia_) / tss *
100, 2), "%")
```

```
Cluster sizes: [4620 2399]
Cluster labels (first 20): [0 0 0 0 0 0 0 0 1 1 0 0 1 1 1 1 1 0 0 0]
Total WCSS: 16591.939219150347 | Between_SS / Total_SS = 40.9 %
```

## Cluster Centers - PDF #8 - K-Means - Question #9

```
#Cluster Centers - PDF #8 - K-Means - Question #9
print(pd.DataFrame(km2.cluster_centers_, columns=km_data.columns))
```

```
   SeniorCitizen    tenure  MonthlyCharges  TotalCharges
0      -0.112845 -0.506470       -0.428978     -0.630153
1       0.217042  0.974127        0.825081      1.212015
```

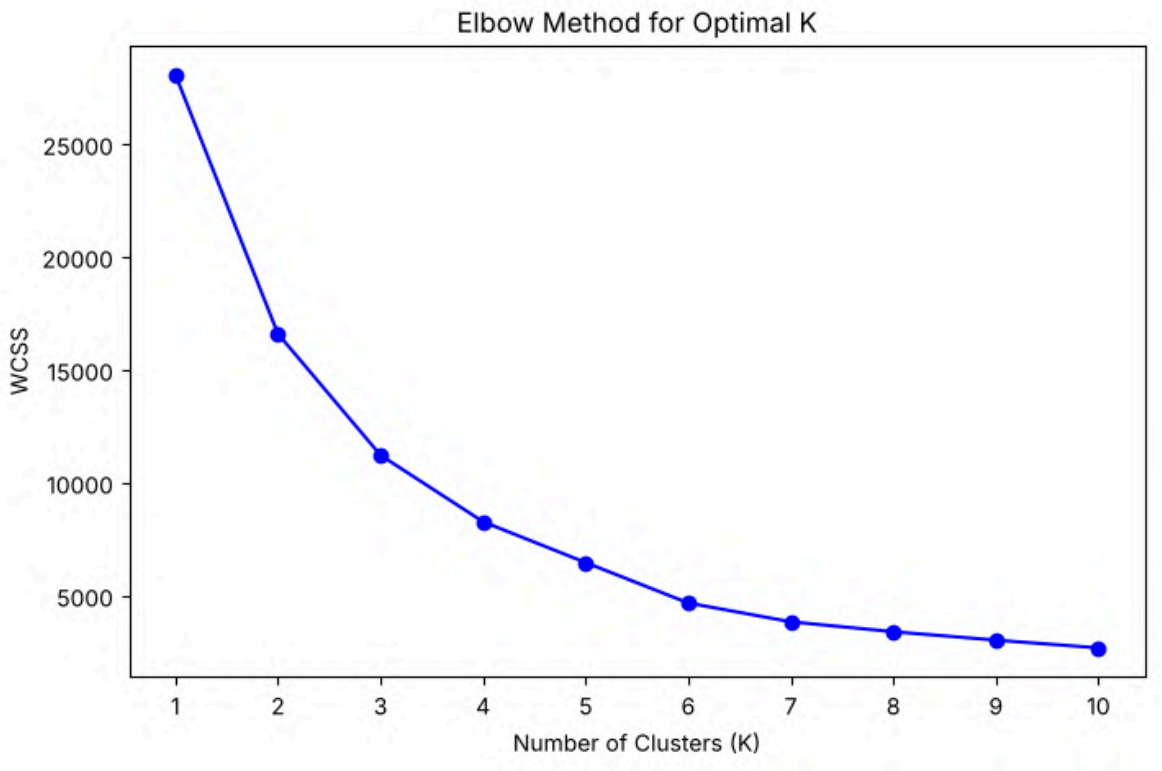
## WCSS for K = 1 to 10 - PDF #8 - K-Means - Question #10

```
#WCSS for K = 1 to 10 - PDF #8 - K-Means - Question #10
wcss = [KMeans(n_clusters=k, n_init=25, random_state=42).fit(km_scaled).inertia_ for k in range(1,
11)]
print(pd.DataFrame({"K": range(1, 11), "WCSS": wcss}))
```

```
    K          WCSS
0   1  28072.000000
1   2  16591.939219
2   3  11212.859426
3   4   8265.424751
4   5   6456.925166
5   6   4666.217315
6   7   3833.652692
7   8   3393.496517
8   9   3020.576773
9  10   2677.997871
```

## Elbow Plot - PDF #8 - K-Means - Question #11

```
#Elbow Plot - PDF #8 - K-Means - Question #11
plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), wcss, "bo-")
plt.title("Elbow Method for Optimal K")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.xticks(range(1, 11))
plt.show()
```




## K-Means with Elbow-Recommended K - PDF #8 - K-Means - Question #12

```
#K-Means with Elbow-Recommended K - PDF #8 - K-Means - Question #12
kx = (np.arange(1, 11) - 1) / 9
wy = (np.array(wcss) - min(wcss)) / (max(wcss) - min(wcss))
best_k = int(np.argmax(1 - kx - wy) + 1)
print("Elbow K:", best_k)
km_best = KMeans(n_clusters=best_k, n_init=25, random_state=42).fit(km_scaled)
print("Cluster sizes:", np.bincount(km_best.labels_))
print(pd.DataFrame(km_best.cluster_centers_, columns=km_data.columns))
```

```
Elbow K: 4
Cluster sizes: [1957 1746 1142 2174]
   SeniorCitizen    tenure  MonthlyCharges  TotalCharges
0      -0.440782 -0.114266       -1.302056     -0.659434
1      -0.440782  1.079717        0.793447      1.294881
2       2.268370  0.036042        0.499374      0.232915
3      -0.440782 -0.781511        0.273009     -0.566979
```

## Load Dataset - PDF #8 - Hierarchical - Question #1

```
#Load Dataset - PDF #8 - Hierarchical - Question #1
df = load_raw()
```

## View Dataset - PDF #8 - Hierarchical - Question #2

```
#View Dataset - PDF #8 - Hierarchical - Question #2
print(df.head())
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService     MultipleLines  \
0  7590-VHVEG  Female            0.0     Yes         No     1.0           No  No phone service
1  5575-GNVDE    Male            0.0      No         No    34.0          Yes                No
2  3668-QPYBK    Male            0.0      No         No     2.0          Yes                No
3  7795-CFOCW    Male            0.0      No         No    45.0           No  No phone service
4  9237-HQITU  Female            0.0      No         No     2.0          Yes                No

  InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies  \
0             DSL             No          Yes               No          No          No              No
1             DSL            Yes           No              Yes          No          No              No
2             DSL            Yes          Yes               No          No          No              No
3             DSL            Yes           No              Yes         Yes          No              No
4     Fiber optic             No           No               No          No          No              No

         Contract PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges Churn
0  Month-to-month              Yes           Electronic check           29.85         29.85    No
1        One year               No               Mailed check           56.95       1889.50    No
2  Month-to-month              Yes               Mailed check           53.85        108.15   Yes
3        One year               No  Bank transfer (automatic)           42.30       1840.75    No
4  Month-to-month              Yes           Electronic check           70.70        151.65   Yes
```

## Check Missing Values - PDF #8 - Hierarchical - Question #3

```
#Check Missing Values - PDF #8 - Hierarchical - Question #3
print(df.isna().sum())
```

```
customerID          0
gender              0
SeniorCitizen       8
Partner             5
Dependents          2
tenure              2
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         1
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    1
PaymentMethod       0
MonthlyCharges      3
TotalCharges       11
Churn               3
dtype: int64
```

## Select Numerical Attributes - PDF #8 - Hierarchical - Question #4

```
#Select Numerical Attributes - PDF #8 - Hierarchical - Question #4
hc_data = df.select_dtypes(include="number").dropna()
```

## Standardize (scale) - PDF #8 - Hierarchical - Question #5

```
#Standardize (scale) - PDF #8 - Hierarchical - Question #5
hc_scaled = r_scale(hc_data)
```

## Euclidean Distance (dist) - PDF #8 - Hierarchical - Question #6

```
#Euclidean Distance (dist) - PDF #8 - Hierarchical - Question #6
d = pdist(hc_scaled.values, metric="euclidean")
print("Number of pairwise distances:", len(d))
```

```
Number of pairwise distances: 24629671
```

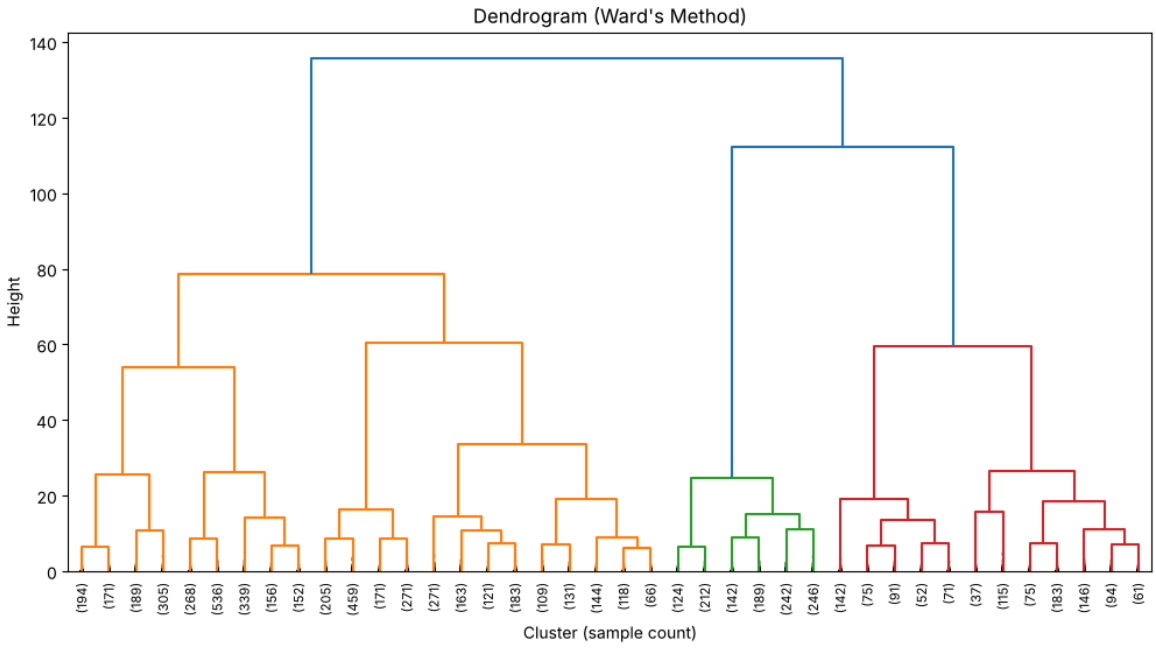
## Ward's Method (hclust) - PDF #8 - Hierarchical - Question #7

```
#Ward's Method (hclust) - PDF #8 - Hierarchical - Question #7
hc = linkage(d, method="ward")
```

## Dendrogram - PDF #8 - Hierarchical - Question #8

```
#Dendrogram - PDF #8 - Hierarchical - Question #8
plt.figure(figsize=(12, 6))
dendrogram(hc, truncate_mode="lastp", p=40, leaf_rotation=90, show_contracted=True)
plt.title("Dendrogram (Ward's Method)")
plt.xlabel("Cluster (sample count)")
plt.ylabel("Height")
plt.show()

#3 Clusters + Assignments (cutree) - PDF #8 - Hierarchical - Question #9
clusters = fcluster(hc, t=3, criterion="maxclust")
print(clusters[:30])
```




```
[1 1 1 1 1 1 1 1 1 1 1 2 2 1 2 2 1 1 3 1 1 1 1 1 2 1 2 1 3]
```

## Samples per Cluster (table) - PDF #8 - Hierarchical - Question #10

```
#Samples per Cluster (table) - PDF #8 - Hierarchical - Question #10
print(pd.Series(clusters).value_counts().sort_index())
```

```
1    4722
2    1155
3    1142
Name: count, dtype: int64
```

# PDF 9 - Principal Component Analysis

## Load Dataset + Verify Packages - PDF #9 - Question #1

```
#Load Dataset + Verify Packages - PDF #9 - Question #1
df = load_raw()
import sklearn, matplotlib, statsmodels
print("pandas", pd.__version__, "| sklearn", sklearn.__version__, "| matplotlib",
matplotlib.__version__,
      "| seaborn", sns.__version__, "| statsmodels", statsmodels.__version__)
```

```
pandas 3.0.5 | sklearn 1.9.1 | matplotlib 3.11.2 | seaborn 0.13.2 | statsmodels 0.15.0
```

## Rows and Columns - PDF #9 - Question #2

```
#Rows and Columns - PDF #9 - Question #2
print("Rows:", df.shape[0], "Columns:", df.shape[1])
```

```
Rows: 7043 Columns: 21
```

## Numeric Columns, Clean, Scale → data_scaled - PDF #9 - Question #3

```
#Numeric Columns, Clean, Scale -> data_scaled - PDF #9 - Question #3
num = df.select_dtypes(include="number").dropna().drop_duplicates()
data_scaled = r_scale(num)
print(data_scaled.head())
```

```
   SeniorCitizen    tenure  MonthlyCharges  TotalCharges
0      -0.447299 -1.335373       -1.196198     -1.026105
1      -0.447299  0.023973       -0.290238     -0.205285
2      -0.447299 -1.294181       -0.393872     -0.991544
3      -0.447299  0.477088       -0.779991     -0.226803
4      -0.447299 -1.294181        0.169428     -0.972344
```

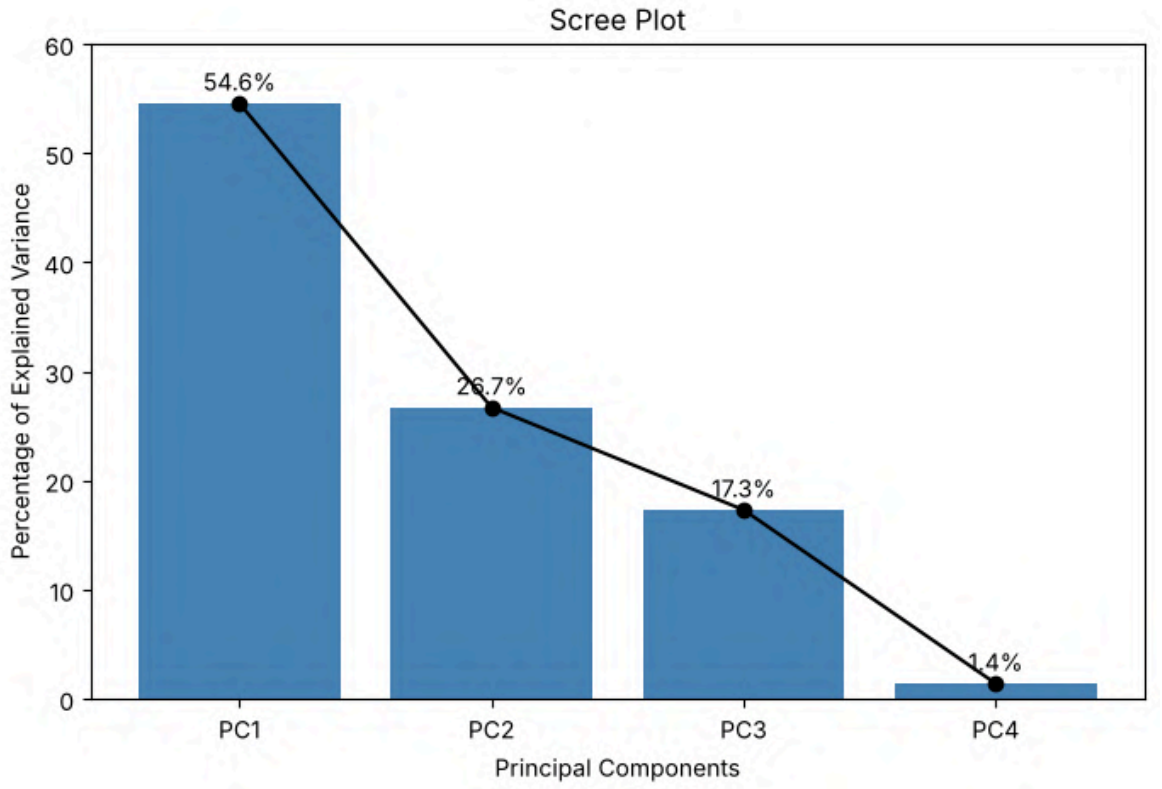
## PCA + Summary - PDF #9 - Question #4

```
#PCA + Summary - PDF #9 - Question #4
pca = PCA().fit(data_scaled)
pcs = [f"PC{i+1}" for i in range(pca.n_components_)]
print(pd.DataFrame({"Standard deviation": np.sqrt(pca.explained_variance_),
                    "Proportion of Variance": pca.explained_variance_ratio_,
                    "Cumulative Proportion": np.cumsum(pca.explained_variance_ratio_)},
index=pcs).T.round(4))
print(pd.DataFrame(pca.components_.T, index=num.columns, columns=pcs).round(4))
```

```
                           PC1     PC2     PC3     PC4
Standard deviation      1.4774  1.0336  0.8323  0.2372
Proportion of Variance  0.5457  0.2671  0.1732  0.0141
Cumulative Proportion   0.5457  0.8128  0.9859  1.0000
                    PC1     PC2     PC3     PC4
SeniorCitizen    0.1420  0.8594 -0.4910  0.0138
tenure           0.5503 -0.3698 -0.5036 -0.5539
MonthlyCharges   0.4903  0.3312  0.7108 -0.3803
TotalCharges     0.6608 -0.1225 -0.0025  0.7405
```

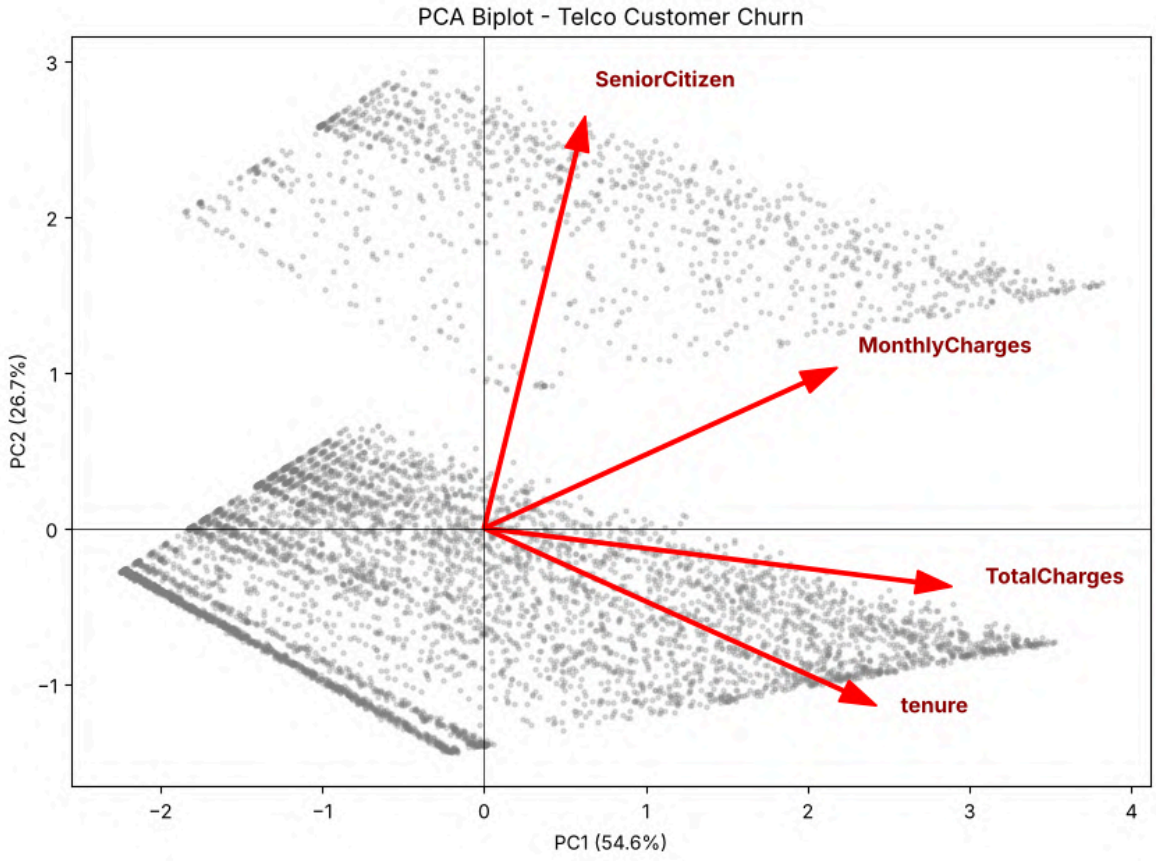
## Scree Plot (labels, ylim 0-60) - PDF #9 - Question #5

```
#Scree Plot (labels, ylim 0-60) - PDF #9 - Question #5
var_pct = pca.explained_variance_ratio_ * 100
plt.figure(figsize=(8, 5))
bars = plt.bar(pcs, var_pct, color="steelblue")
plt.plot(pcs, var_pct, "ko-")
plt.bar_label(bars, fmt="%.1f%%", padding=3)
plt.ylim(0, 60)
plt.title("Scree Plot")
plt.xlabel("Principal Components")
plt.ylabel("Percentage of Explained Variance")
plt.show()
```



## PCA Biplot (non-overlapping labels) - PDF #9 - Question #6

```
#PCA Biplot (non-overlapping labels) - PDF #9 - Question #6
scores = pca.transform(data_scaled)
loadings = pca.components_.T * np.sqrt(pca.explained_variance_)
scale = np.abs(scores[:, :2]).max() / np.abs(loadings[:, :2]).max() * 0.7
plt.figure(figsize=(10, 7))
plt.scatter(scores[:, 0], scores[:, 1], s=5, alpha=0.25, color="grey")
texts = []
for i, var in enumerate(num.columns):
    plt.arrow(0, 0, loadings[i, 0] * scale, loadings[i, 1] * scale, color="red", width=0.02,
head_width=0.15)
    texts.append(plt.text(loadings[i, 0] * scale * 1.15, loadings[i, 1] * scale * 1.15, var,
color="darkred", fontsize=11, weight="bold"))
try:
    from adjustText import adjust_text
    adjust_text(texts, arrowprops=dict(arrowstyle="-", color="darkred"))
except ImportError:
    pass
plt.axhline(0, color="black", lw=0.5)
plt.axvline(0, color="black", lw=0.5)
plt.title("PCA Biplot - Telco Customer Churn")
plt.xlabel(f"PC1 ({var_pct[0]:.1f}%)")
plt.ylabel(f"PC2 ({var_pct[1]:.1f}%)")
plt.show()
```




## PCA Scores Data Frame (first 6) - PDF #9 - Question #7

```
#PCA Scores Data Frame (first 6) - PDF #9 - Question #7
pca_scores = pd.DataFrame(scores, columns=pcs, index=num.index)
print(pca_scores.head(6))
```

```
        PC1       PC2       PC3       PC4
0 -2.062872 -0.161041  0.044395  0.428640
1 -0.328290 -0.464248  0.001748 -0.061055
2 -1.624000  0.085203  0.593885  0.126260
3 -0.333289 -0.791373 -0.574529 -0.141698
4 -1.335136  0.269401  0.994250 -0.073767
5 -0.529555  0.462354  1.556995 -0.360149
```

## PC1 vs PC2 Scatter (blue circles, reg no.) - PDF #9 - Question #8

```python
#PC1 vs PC2 Scatter (blue circles, reg no.) - PDF #9 - Question #8
plt.figure(figsize=(8, 6))
plt.scatter(pca_scores["PC1"], pca_scores["PC2"], color="blue", marker="o", s=12, alpha=0.5)
plt.title(f"PC1 vs PC2 - {REG_NO}")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.show()
```

